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Theoretical Knowledge

TASK 1. SOC Fundamentals and Operations

1. Explanation

A **Security Operations Center (SOC)** is a centralized team responsible for **monitoring, detecting, analyzing, and responding to cybersecurity incidents** in an organization.

Think of a SOC like a **cybersecurity command center**.

The SOC team continuously monitors:

- Network traffic
- Security logs
- User activity
- Endpoint behavior
- Threat intelligence feeds

The main goal is to **detect threats early and stop attacks before damage occurs**.

Example

If a hacker tries to log in multiple times to a company server:

1. Logs are generated
2. SIEM detects suspicious activity
3. SOC analyst investigates
4. Incident is contained

This process happens **24/7 in a SOC environment**.

2. Purpose of a SOC

A SOC exists to **protect an organization's IT infrastructure from cyber threats**.

Main Objectives

1. Continuous Monitoring

SOC teams monitor systems **24/7** for suspicious activity.



Example:

- Multiple failed login attempts
- Malware alerts
- Unusual network traffic

2. Threat Detection

Security tools analyze logs and identify potential attacks.

Example:

- Brute force login
- Data exfiltration
- Malware infection

3. Incident Response

When a threat is detected:

SOC analysts investigate and respond.

Example actions:

- Block attacker IP
- Isolate infected system
- Remove malware

4. Threat Intelligence

SOC teams use threat intelligence sources to identify known attacker indicators.

Examples:

- Malicious IP addresses
- Malware signatures
- Command & control servers

5. Log Analysis

Logs from different systems are analyzed to detect anomalies.

Sources include:

- Firewalls
- Servers
- Applications
- Endpoints



3. SOC Team Structure

A SOC typically has **multiple analyst tiers**.

Tier 1 Analyst (SOC Analyst Level 1)

First line of defense.

Responsibilities:

- Monitor SIEM alerts
- Perform initial investigation
- Escalate serious incidents

Example tasks:

- Investigate login failures
- Validate alerts
- Check suspicious IP activity

Tools used:

- SIEM (Splunk, ELK)
- Threat intelligence platforms

Tier 2 Analyst (Incident Responder)

Handles **advanced investigations**.

Responsibilities:

- Deep analysis of alerts
- Malware investigation
- Network traffic analysis

Tools used:

- Wireshark
- Endpoint detection tools
- Forensic tools

Tier 3 Analyst (Threat Hunter)

Proactively searches for hidden threats.

Responsibilities:

- Threat hunting
- Attack pattern analysis
- Developing detection rules



Tools used:

- MITRE ATT&CK
- Advanced SIEM queries
- Threat intelligence platforms

SOC Manager

Responsible for managing the SOC team.

Responsibilities:

- SOC strategy
- Incident management
- Compliance reporting
- Team coordination

4. Key SOC Functions

1. Log Monitoring

SOC teams collect logs from multiple systems.

Examples:

- Windows event logs
- Firewall logs
- DNS logs
- Authentication logs

Logs are centralized using **SIEM platforms**.

Examples of SIEM tools:

- Splunk
- ELK Stack
- IBM QRadar
- Microsoft Sentinel

2. Alert Triage

Alert triage means **evaluating security alerts to determine if they are real threats**.

Typical triage workflow:

1. Alert generated
2. Validate alert



3. Check severity
4. Investigate source
5. Escalate if needed

3. Threat Intelligence Integration

Threat intelligence provides information about known cyber threats.

Examples of threat intelligence sources:

- AlienVault OTX
- VirusTotal
- AbuseIPDB

SOC analysts compare alerts with threat intelligence data.

5. SOC Frameworks

SOC operations are based on cybersecurity frameworks.

NIST Cybersecurity Framework

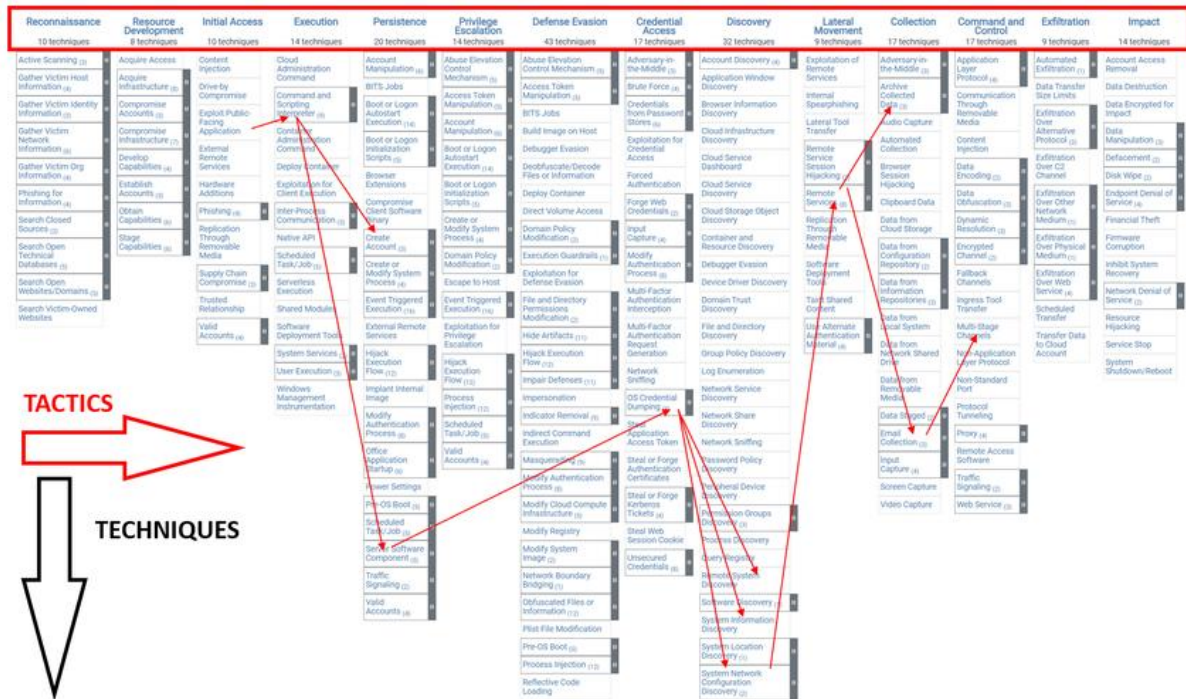
Key functions:

1. Identify
2. Protect
3. Detect
4. Respond
5. Recover

SOC teams mainly focus on:

- Detect
- Respond

MITRE ATT&CK Framework

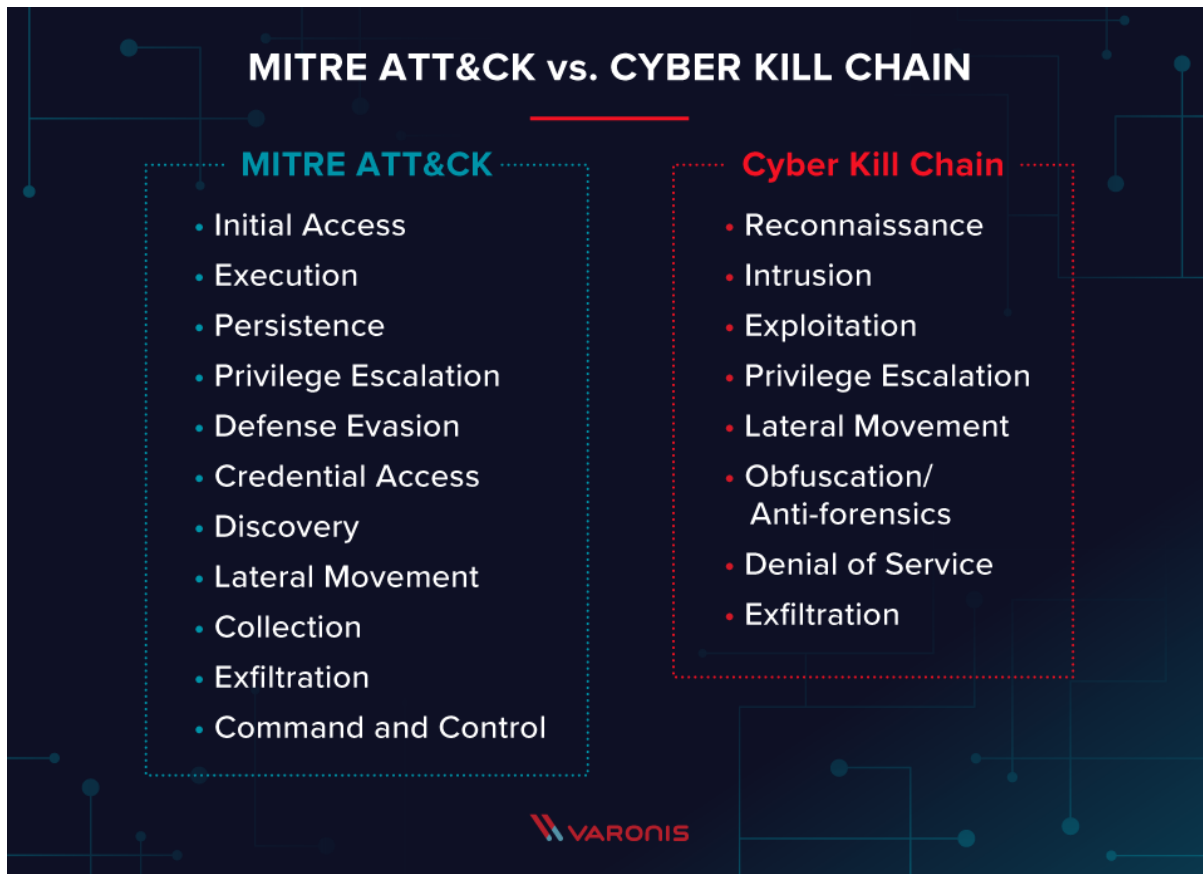


Software Execution x +													
selection controls layer controls technique controls													
Initial Access	Execution	Persistence	Privilege Escalation	Defense Evasion	Credential Access	Discovery	Lateral Movement	Collection					
9 techniques	10 techniques	18 techniques	12 techniques	37 techniques	14 techniques	25 techniques	9 techniques	17 techniques					
Replication Through Removable Media	Native API	BITS Jobs	Process Injection	Obfuscated Files or Information	Credentials from Password Stores	System Information Discovery	Replication Through Removable Media	Screen Capture					
Drive-by Compromise	Windows Management Instrumentation	Hijack Execution Flow	Access Token Manipulation	Deobfuscate/Decode Files or Information	Network Sniffing	File and Directory Discovery	Lateral Tool Transfer	Data from Local System					
Valid Accounts	Command and Scripting Interpreter	Traffic Signaling	Exploitation for Privilege Escalation	Modify Registry	OS Credential Dumping	Process Discovery	Exploitation of Remote Services	Audio Capture					
Exploit Public-Facing Application	Exploitation for Client Execution	Valid Accounts	Hijack Execution Flow	Rootkit	Brute Force	System Network Configuration Discovery	Taint Shared Content	Archive Collected Data					
External Remote Services	Shared Modules	Account Manipulation	Valid Accounts	Indicator Removal on Host	Steal Web Session Cookie	System Owner/User Discovery	Remote Service Hijacking	Clipboard Data					
Hardware Additions	Scheduled Task/Job	Browser Extensions	Boot or Logon Autostart Execution	Access Token Manipulation	Two-Factor Authentication Interception	Query Registry	Remote Services	Video Capture					
Phishing	Software Deployment Tools	Boot or Logon Autostart Execution	Group Policy Modification	Virtualization/Sandbox Evasion	Unsecured Credentials	System Network Connections Discovery	Software Deployment Tools	Automated Collection					
Supply Chain Compromise	Inter-Process Communication	Compromise Client Software Binary	Scheduled Task/Job	BITS Jobs	Exploitation for Credential Access	System Time Discovery	Internal Spearphishing	Data from Removable Media					
Trusted Relationship	System Services	External Remote Services	Abuse Elevation Control Mechanism	Hijack Execution Flow	Forced Authentication	System Service Discovery	Remote Service Session Hijacking	Data from Network Shared Drive					
	User Execution	Scheduled Task/Job	Boot or Logon Initialization Scripts	Masquerading	Input Capture	Peripheral Device Discovery	Use Alternate Authentication Material	Data from Cloud Storage Object					
		Boot or Logon Initialization Scripts	Create or Modify System Process	Valid Accounts	Man-in-the-Middle	Remote System Discovery		Data from Configuration Repository					
		Create or Modify System Process	Event Triggered Execution	Indirect Command Execution	Modify Authentication Process	Application Window Discovery		Data from Information Repositories					
		Event Triggered Execution	Implant Container Image	Rogue Domain Controller	Steal Application Access Token	Network Service Scanning		Data Staged					
				XSL Script Processing	Steal or Forge Kerberos Tickets	Network Share Discovery		Email Collection					
				Abuse Elevation Control Mechanism		Software Discovery		Input Capture					
				Direct Volume Access		Domain Trust							



CYBER KILL CHAIN VS. MITRE ATT&CK





MITRE ATT&CK is a **knowledge base of attacker techniques and tactics**.

It helps SOC analysts:

- Understand attacker behavior
- Detect attack patterns
- Improve threat detection

Example attack stages:

- Initial Access
- Privilege Escalation
- Lateral Movement
- Data Exfiltration

6. SOC Tools Used in Real Environments

Common SOC tools include:

SIEM

- Splunk
- Elastic SIEM
- IBM QRadar



Purpose:

Collect and analyze security logs.

EDR (Endpoint Detection & Response)

Examples:

- CrowdStrike
- Microsoft Defender
- SentinelOne

Purpose:

Detect endpoint attacks.

Network Monitoring Tools

Examples:

- Wireshark
- Zeek
- Suricata

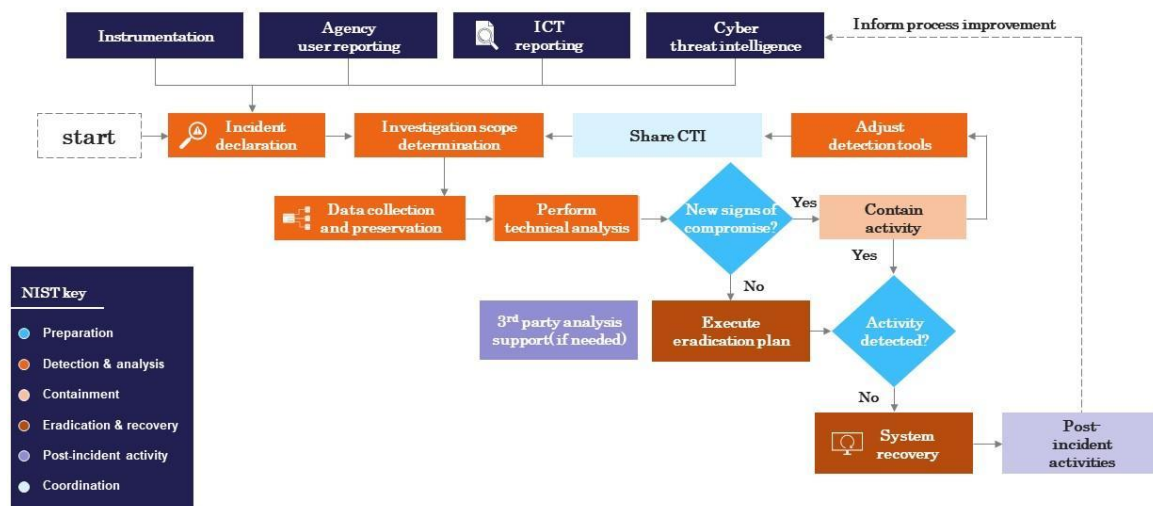
Purpose:

Analyze network traffic.

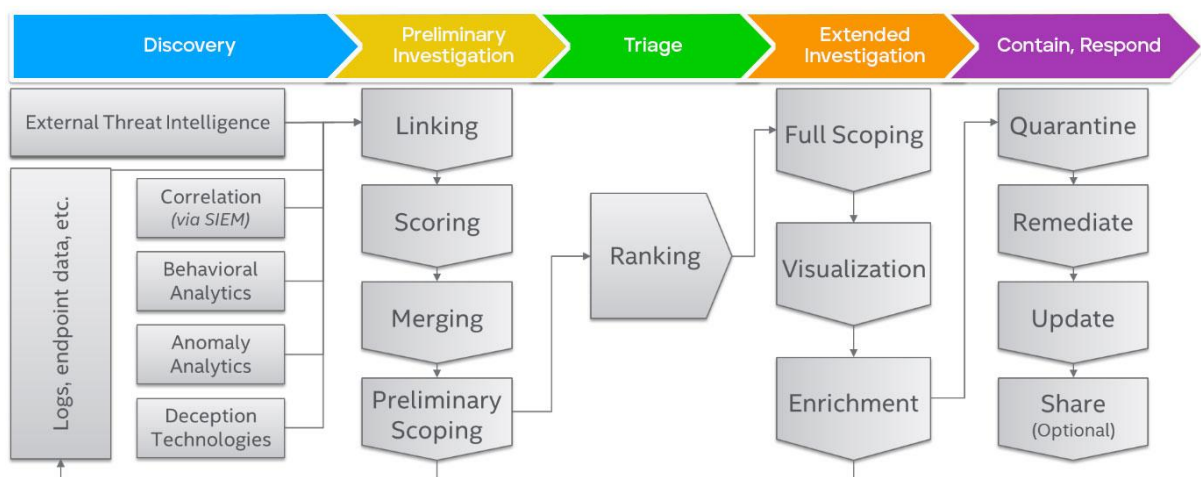
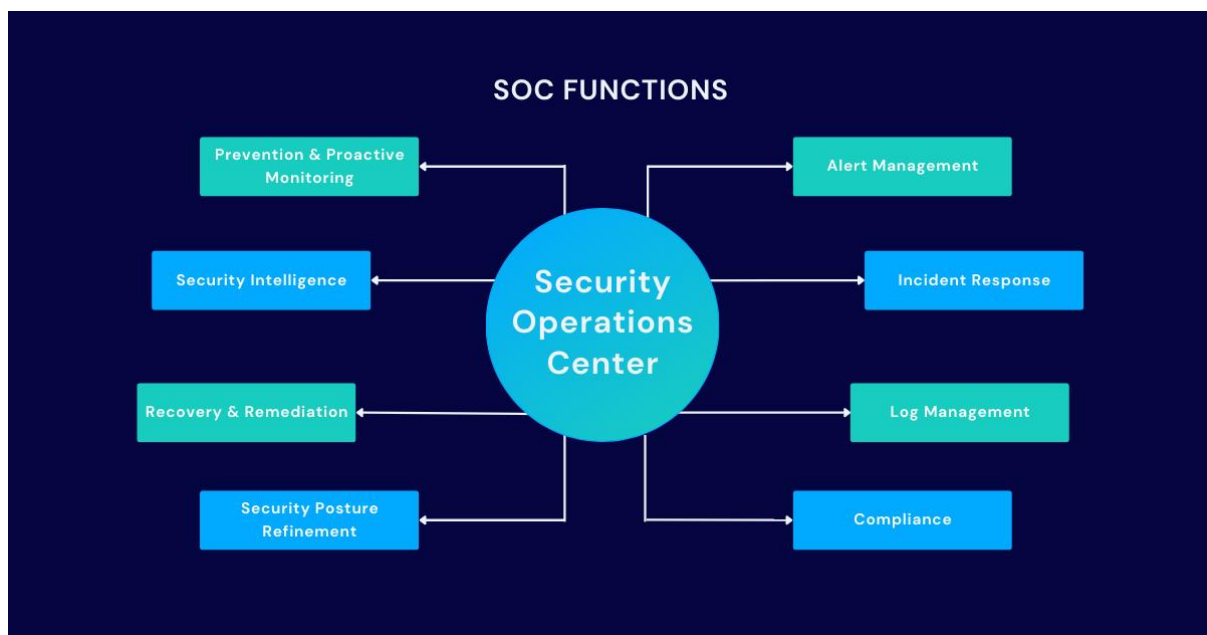
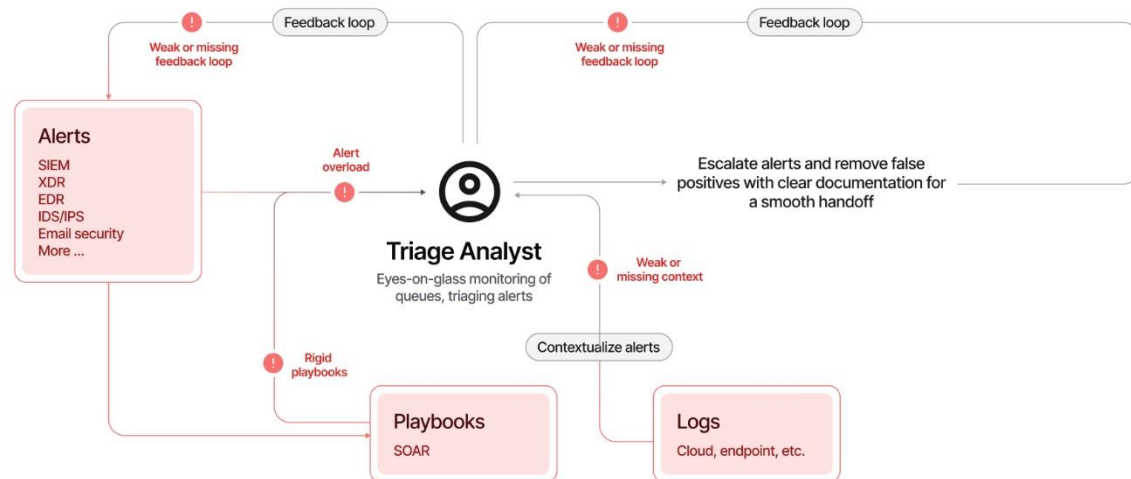
7. SOC Workflow (Real SOC Environment)

Cyber security incident response process flow chart

This slide represents the flow chart representing the detection and reaction to cyber security incidents, determination of their scope and risk and reduction of likelihood of incident from reoccurring. It starts with incident declaration and ends with system recovery.



This slide is 100% editable. Adapt it to your needs and capture your audience's attention.



Typical SOC workflow:



Step 1 — Log Collection

Logs collected from:

- Servers
- Firewalls
- Applications
- Endpoints

Step 2 — SIEM Analysis

SIEM correlates logs and generates alerts.

Example:

Multiple login failures detected.

Step 3 — Alert Triage

Tier 1 analyst:

- Reviews alert
- Checks severity
- Verifies activity

Step 4 — Investigation

Tier 2 analyst investigates further:

- Check logs
- Analyze network traffic
- Identify attacker behavior

Step 5 — Incident Response

Actions taken:

- Block attacker IP
- Disable compromised account
- Isolate infected device

Step 6 — Documentation

SOC analysts document:

- Incident details
- Timeline
- Response actions



8. Conclusion

A Security Operations Center plays a critical role in protecting organizations from cyber threats. SOC analysts continuously monitor logs, analyze alerts, investigate incidents, and respond to attacks using specialized tools and frameworks like NIST and MITRE ATT&CK. Understanding SOC operations is essential for building strong cybersecurity defense capabilities.



TASK 2. Security Monitoring Basics:

1. Explanation

Security Monitoring is the **continuous observation of system, network, and application activity** to detect security threats.

A **SOC analyst watches logs and network traffic** to find signs of:

- Unauthorized login attempts
- Malware activity
- Suspicious network traffic
- Policy violations
- Data exfiltration

SOC analysts mainly rely on **logs and network traffic**.

Two major monitoring sources:

1. Log Monitoring

Logs come from:

- Operating systems
- Firewalls
- Servers
- Applications
- Security tools

These logs are collected into a **SIEM system**.

Example tools:

- Splunk
- Elastic SIEM

The SIEM analyzes logs and generates **alerts**.

2. Network Traffic Monitoring

Network monitoring helps detect:

- Suspicious connections
- Malware communication
- Data exfiltration
- Port scanning

Common tool:

- Wireshark

Wireshark allows analysts to **capture and analyze packets**.



2. Key Security Monitoring Objectives

Detect Anomalies

Example:

Normal login location → India

Suddenly login from → Russia

SOC should detect this anomaly.

Detect Unauthorized Access

Example:

Multiple failed SSH login attempts

Possible **Brute-force attack**

Detect Policy Violations

Example:

Employee downloading company data to external storage.

3. Key Metrics Used in SOC

Mean Time to Detect (MTTD)

Time taken to detect an attack.

Example:

Attack starts → 10:00

SOC detects → 10:05

MTTD = **5 minutes**

Lower MTTD = **Better SOC performance**

False Positives

Alert triggered but **not a real attack**.

Example:

Security tool flags admin login as suspicious.

False Negatives

Real attack happened but **SOC missed it**.

Example:

Malware communication not detected.



False negatives are **very dangerous**.

4. Tools Required for Lab

For your project, we will use **open-source SOC tools**.

Tool	Purpose
Elastic SIEM	Log monitoring
Wireshark	Network traffic analysis
Kali Linux / Ubuntu VM	Lab environment
Boss of the SOC dataset	Attack log dataset

5. Lab Environment Setup

Recommended environment:

SOC LAB

Attacker Machine
(Kali Linux)



Target Machine
(Ubuntu)



Monitoring System
Elastic SIEM
Wireshark

Lab environment used:

Operating System: Ubuntu / Kali Linux VM

Monitoring Tool: Wireshark

Network Activity: Web browsing and ping commands

The lab environment allowed capturing and analyzing real-time network traffic.



6.Installation of Wireshark

Wireshark was installed using the Linux package manager.

Command Used

```
sudo apt update
```

```
sudo apt install wireshark
```

After installation, the version was verified.

```
wireshark --version
```

```
kali@kali: ~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task-2-Security-Monitoring-Basics
$ ls
README.md  screenshots

(kali@kali)~[~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task-2-Security-Monitoring-Basics]
$ wireshark --version
Wireshark 4.6.2.

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Licensed under the terms of the GNU General Public License (version 2 or later).
This is free software; see the file named COPYING in the distribution. There is
NO WARRANTY; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

Compile-time info:
  Bit width: 64-bit
  Compiler: GCC 15.2.0
  GLib: 2.86.2
  With:
    +brotli
    +Gcrypt 1.11.3
    +GnuTLS 3.8.11 and PKCS#11
    +Kerberos (MIT)
    +libnl 3
    +libpcap
    +libsmb 0.4.8
    +libxml2 2.15.1
    +Lua 5.4.8
    +LZ4 1.10.0
    +MaxMind
    +Minizip 1.3.1
    +nghttp2 1.64.0
    +nghttp3 1.12.0
    +PCRE2 10.46 2025-08-27
    +POSIX capabilities (Linux)
    +Qt 6.9.2
    +QtDBus
    +QtMultimedia
    +Snappy 1.2.2
    +xxhash 0.8.3
    +zlib 1.3.1
    +Zstandard 1.5.7
  Without:
    -automatic updates  -zlib-ng

Runtime info:
  OS: Linux 6.18.3+kali1-amd64
  CPU: 13th Gen Intel(R) Core(TM) i5-13420H (with SSE4.2)
  Memory: 10794 MB of physical memory
  GLib: 2.86.3
  Locale: LC_TYPE=en_US.UTF-8
```

Figure 1: Wireshark Installation on Linux System



7.Launching Wireshark

Wireshark was launched using the following command.

wireshark

The application displays available network interfaces that can be used for packet capture.

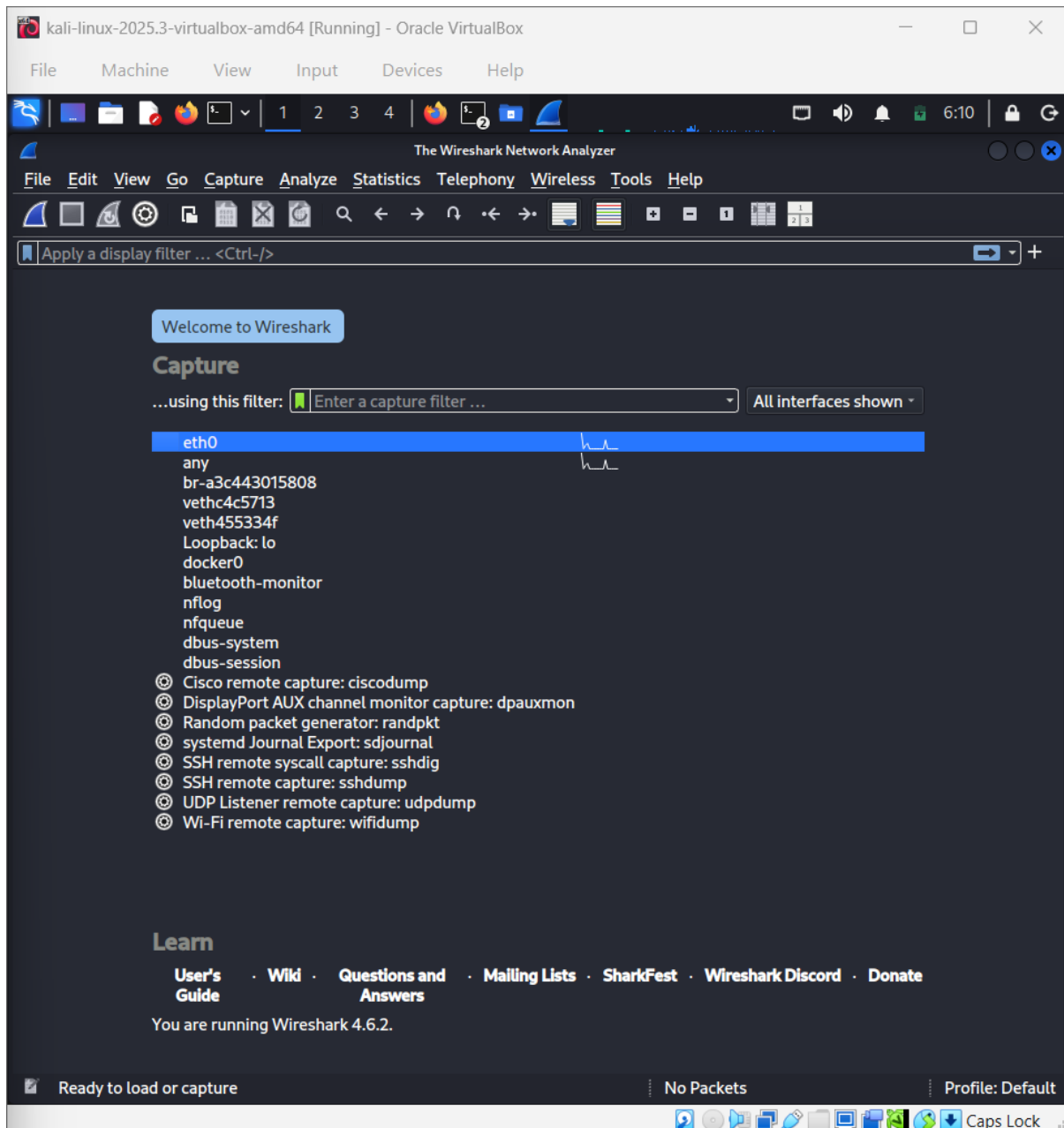


Figure 2: Available Network Interfaces in Wireshark



8. Capturing Network Traffic

To capture network traffic, the active network interface was selected and packet capture was started.

Wireshark immediately began displaying live network packets.

The packet capture interface shows several columns such as:

- Packet number
- Time
- Source IP address
- Destination IP address
- Protocol
- Packet length
- Packet information

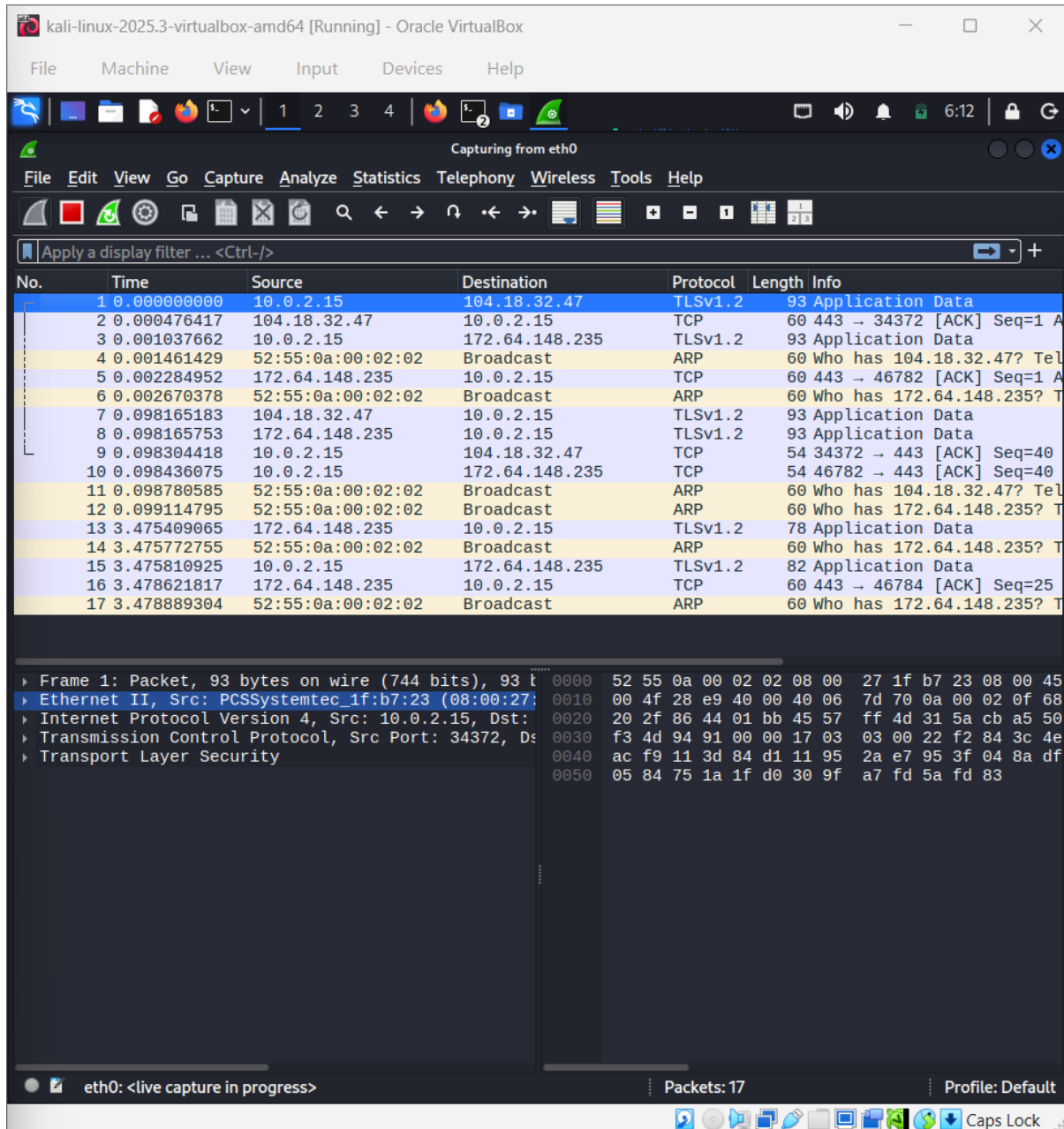


Figure 3 : Live Packet Capture in Wireshark



9. Generating Network Traffic

Network activity was generated to produce packets for analysis.

Traffic was generated by:

- Visiting websites such as Google and GitHub
- Sending ICMP ping requests

Command used:

ping google.com

This generated DNS queries and ICMP packets visible in Wireshark.

```
kali@kali: ~  
$ ping google.com  
PING google.com (142.250.66.14) 56(84) bytes of data:  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=1 ttl=255 time=64.8 ms  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=1 ttl=255 time=64.8 ms (DUP!)  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=2 ttl=255 time=41.5 ms  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=2 ttl=255 time=41.5 ms (DUP!)  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=3 ttl=255 time=45.0 ms  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=3 ttl=255 time=45.0 ms (DUP!)  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=4 ttl=255 time=57.1 ms  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=4 ttl=255 time=57.1 ms (DUP!)  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=5 ttl=255 time=52.9 ms  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=5 ttl=255 time=52.9 ms (DUP!)  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=6 ttl=255 time=56.8 ms  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=6 ttl=255 time=56.8 ms (DUP!)  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=7 ttl=255 time=45.0 ms  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=7 ttl=255 time=45.0 ms (DUP!)  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=8 ttl=255 time=50.4 ms  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=8 ttl=255 time=50.4 ms (DUP!)  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=9 ttl=255 time=46.8 ms  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=9 ttl=255 time=46.8 ms (DUP!)  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=10 ttl=255 time=53.8 ms  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=10 ttl=255 time=53.8 ms (DUP!)  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=11 ttl=255 time=54.1 ms  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=11 ttl=255 time=54.1 ms (DUP!)  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=12 ttl=255 time=50.3 ms  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=12 ttl=255 time=50.3 ms (DUP!)  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=13 ttl=255 time=38.4 ms  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=13 ttl=255 time=38.4 ms (DUP!)  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=14 ttl=255 time=51.6 ms  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=14 ttl=255 time=51.6 ms (DUP!)  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=15 ttl=255 time=36.3 ms  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=15 ttl=255 time=36.3 ms (DUP!)  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=16 ttl=255 time=44.7 ms  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=16 ttl=255 time=44.7 ms (DUP!)  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=17 ttl=255 time=43.2 ms  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=17 ttl=255 time=43.2 ms (DUP!)  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=18 ttl=255 time=41.8 ms  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=18 ttl=255 time=41.8 ms (DUP!)  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=19 ttl=255 time=47.3 ms  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=19 ttl=255 time=47.3 ms (DUP!)  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=20 ttl=255 time=35.6 ms  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=20 ttl=255 time=35.6 ms (DUP!)  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=21 ttl=255 time=52.8 ms  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=21 ttl=255 time=52.8 ms (DUP!)  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=22 ttl=255 time=52.1 ms  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=22 ttl=255 time=52.1 ms (DUP!)  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=23 ttl=255 time=60.0 ms  
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=23 ttl=255 time=60.0 ms (DUP!)
```

Figure 4 : Generated Network Traffic for Monitoring



10. Applying Packet Filters

Packet filters help analysts isolate specific traffic types.

Several filters were applied during analysis.

DNS Filter

dns

Displays DNS queries and responses.

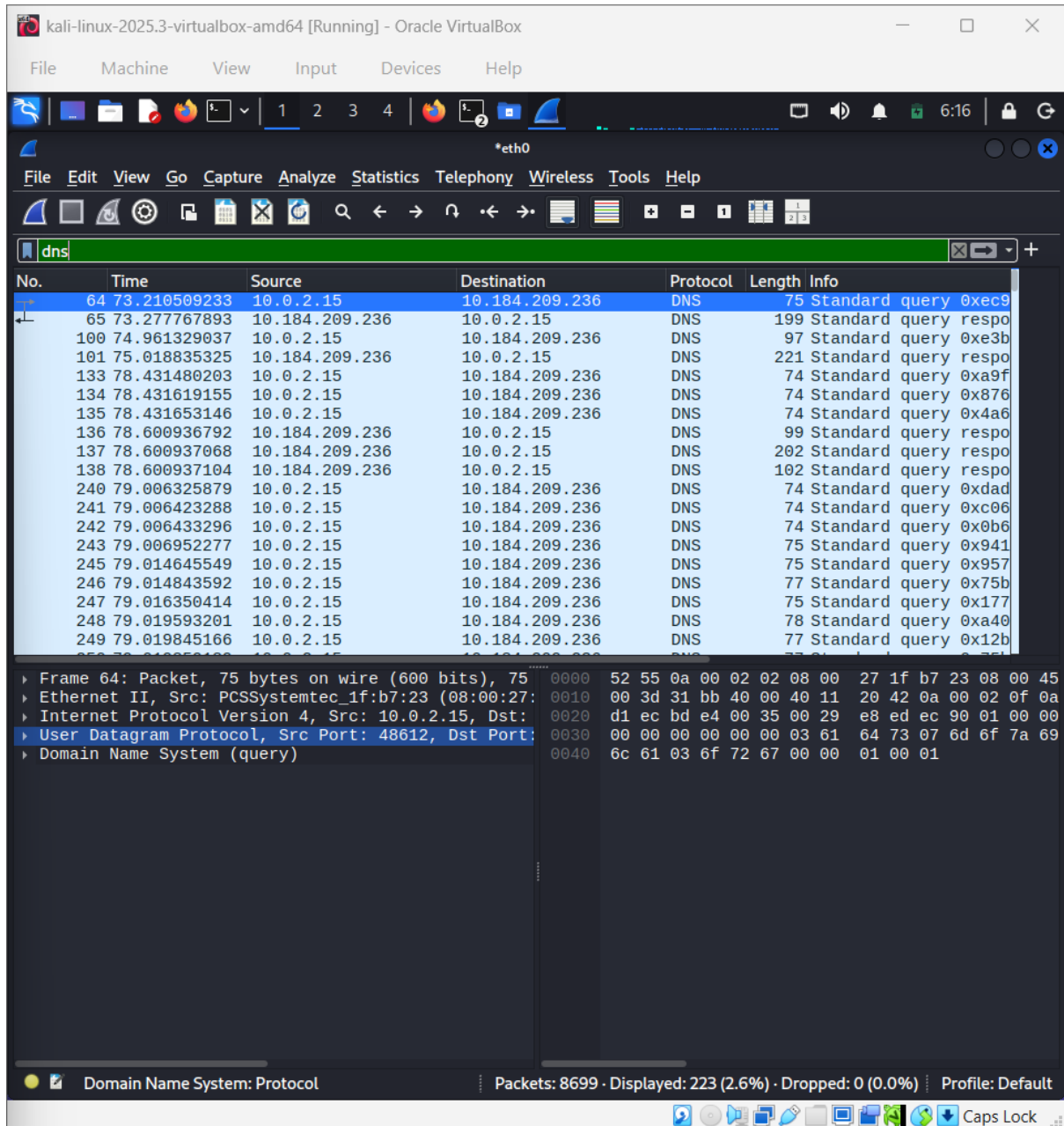


Figure 5: DNS traffic Analysis



11.HTTP Filter

http

Displays HTTP communication.

The screenshot shows the Wireshark network protocol analyzer interface. The top menu bar includes File, Machine, View, Input, Devices, and Help. The toolbar contains various icons for file operations, capture, and analysis. The main window is titled "http" and displays a list of captured packets. The packet list shows the following data:

No.	Time	Source	Destination	Protocol	Length	Info
875	80.248399448	10.0.2.15	142.250.206.67	OCSP	482	Request
912	80.352602367	10.0.2.15	142.250.206.67	OCSP	482	Request
914	80.353183811	10.0.2.15	142.250.206.67	OCSP	482	Request
922	80.361506592	142.250.206.67	10.0.2.15	OCSP	966	Response
1026	80.407512415	10.0.2.15	142.250.206.67	OCSP	482	Request
1046	80.430563225	142.250.206.67	10.0.2.15	OCSP	966	Response
1060	80.448752581	142.250.206.67	10.0.2.15	OCSP	966	Response
1168	80.593652189	142.250.206.67	10.0.2.15	OCSP	966	Response
1642	81.275074411	10.0.2.15	142.250.206.67	OCSP	481	Request
1650	81.287814869	10.0.2.15	142.250.206.67	OCSP	481	Request
1656	81.366512853	142.250.206.67	10.0.2.15	OCSP	1156	Response
1659	81.372585005	142.250.206.67	10.0.2.15	OCSP	1156	Response
2034	83.911932796	10.0.2.15	142.250.206.67	OCSP	481	Request
2165	84.054857782	142.250.206.67	10.0.2.15	OCSP	964	Response
2380	84.229143426	10.0.2.15	142.250.206.67	OCSP	481	Request
2522	84.329865000	142.250.206.67	10.0.2.15	OCSP	964	Response
5669	94.837215737	10.0.2.15	142.250.206.67	OCSP	482	Request
5763	94.915430300	142.250.206.67	10.0.2.15	OCSP	1157	Response
5893	95.077785107	10.0.2.15	142.250.206.67	OCSP	482	Request

The bottom pane shows the details of the selected packet (Frame 875), including Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, Hypertext Transfer Protocol, and Online Certificate Status Protocol. The status bar at the bottom indicates "Packets: 8699 · Displayed: 28 (0.3%) · Dropped: 0 (0.0%) · Profile: Default".

Figure 6 : Http monitoring

Displays ping requests and replies.





13.Packet Inspection

Individual packets were inspected to analyze detailed protocol information.

Wireshark provides layered packet analysis including:

- Ethernet layer
- IP layer
- TCP/UDP layer
- Application layer

Packet inspection helps analysts identify:

- Source IP
- Destination IP
- Communication protocol
- Requested domain names

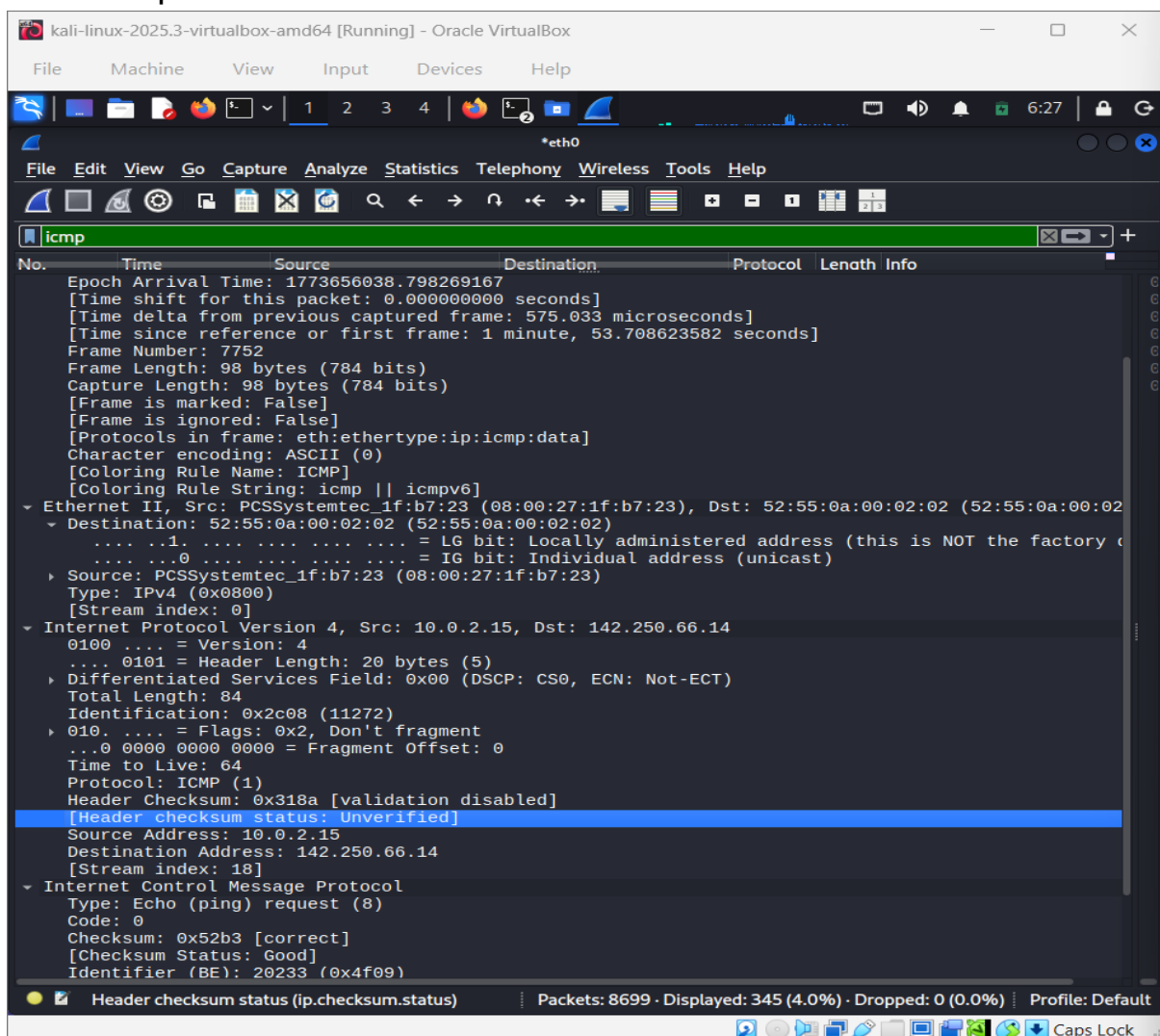


Figure 8: Analyzing random Packet



14.Observations

During packet capture, multiple types of network traffic were observed including DNS queries, ICMP packets, and encrypted HTTPS communication.

The analysis showed how network monitoring tools can reveal communication patterns between systems.

Packet filtering and inspection allowed identification of source and destination addresses as well as protocol information.

15. Challenges Faced

Some challenges encountered during this task included:

- Identifying the correct network interface for packet capture
- Understanding packet protocol details
- Filtering relevant packets from large traffic volumes

These challenges were resolved by applying protocol filters and analyzing packet structures.

16. Conclusion

Security monitoring is a critical capability in cybersecurity operations. SOC analysts rely on monitoring tools to detect suspicious activity and respond to potential threats.

Through this task, the concepts of security monitoring were studied and practical packet analysis was performed using Wireshark. The exercise demonstrated how network traffic monitoring can help identify unusual activity and support incident investigation



TASK 3 : Log Management Fundamentals:

1. Introduction

Log management is a critical function in a Security Operations Center (SOC). Logs provide visibility into system activities and are used to detect security incidents such as unauthorized access, brute-force attacks, and malware activity.

This task demonstrates how logs are collected, processed, and analyzed using industry-standard tools.

2. Objective

- Understand log lifecycle (collection, normalization, storage, analysis)
- Configure a log collection pipeline using Fluentd
- Generate and verify logs
- Simulate real-world SOC log ingestion

3. Log Lifecycle

The log lifecycle consists of:

- **Collection** → Gathering logs from systems
- **Normalization** → Converting logs into structured format
- **Storage** → Storing logs in SIEM systems
- **Retention** → Keeping logs for analysis/compliance
- **Analysis** → Detecting suspicious activity

4. Tools Used

- Fluentd (Log Collector)
- Syslog (Log Source)
- Ubuntu Linux

5. Lab 1 Log Collection Pipeline

Description: In this lab, Fluentd was configured to collect syslog messages from the local system. Logs were generated using network-based syslog simulation and verified in Fluentd output.



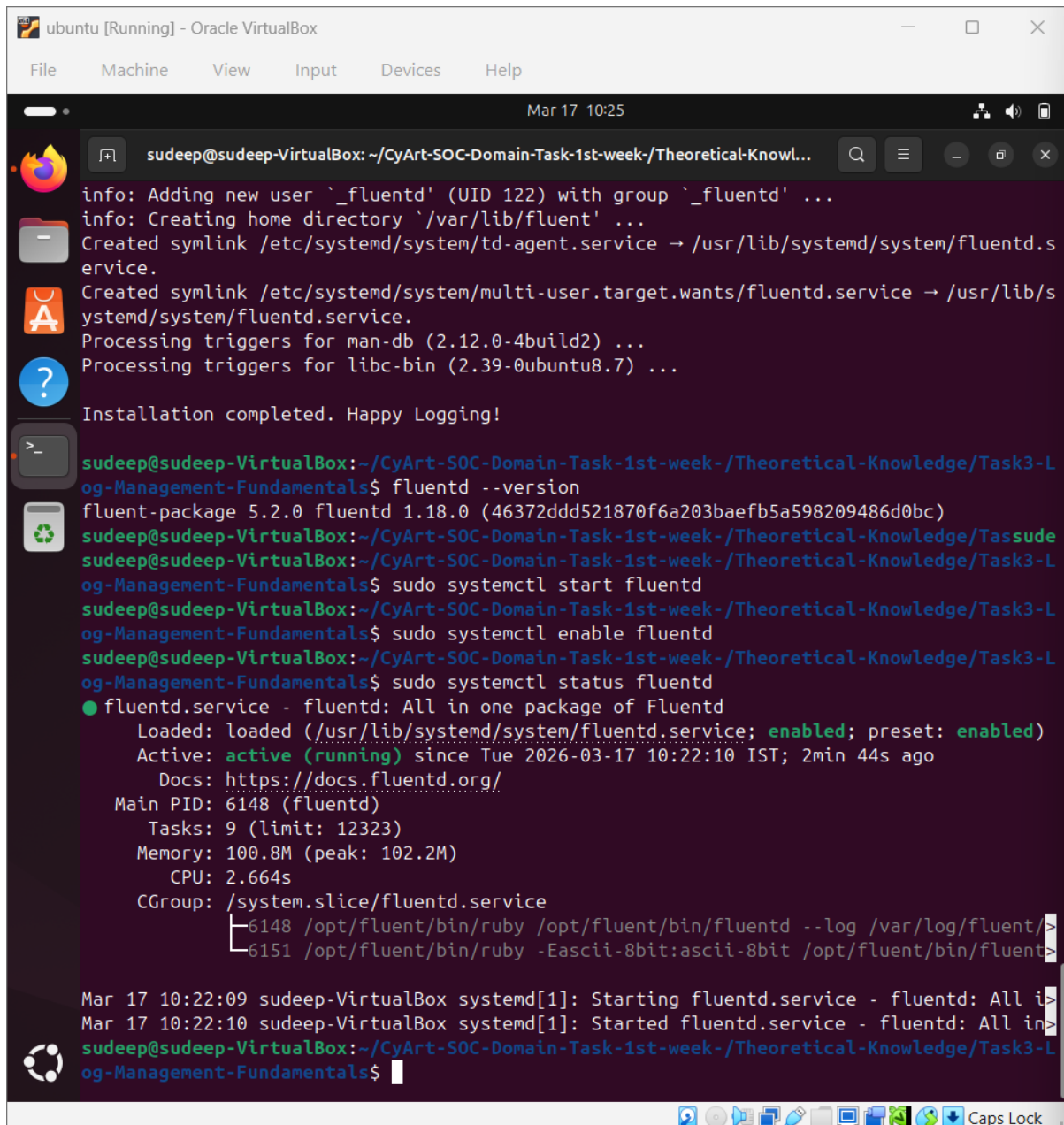
5.1 Installation & Setup

System updated and Fluentd installed using official package.

```
sudo apt update && sudo apt upgrade -y  
curl -fsSL https://toolbelt.treasuredata.com/sh/install-ubuntu-jammy-  
fluent-package5.sh | sh
```

```
ubuntu [Running] - Oracle VirtualBox  
File Machine View Input Devices Help  
Mar 17 10:23  
sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowl...  
The following NEW packages will be installed:  
  fluent-package  
0 upgraded, 1 newly installed, 0 to remove and 1 not upgraded.  
Need to get 17.5 MB of archives.  
After this operation, 82.7 MB of additional disk space will be used.  
Get:1 https://packages.treasuredata.com/5/ubuntu/jammy jammy/contrib amd64 fluent-packag  
e amd64 5.2.0-1 [17.5 MB]  
Fetched 17.5 MB in 13s (1,345 kB/s)  
  
Selecting previously unselected package fluent-package.  
(Reading database ... 153239 files and directories currently installed.)  
Preparing to unpack ../fluent-package_5.2.0-1_amd64.deb ...  
Unpacking fluent-package (5.2.0-1) ...  
Setting up fluent-package (5.2.0-1) ...  
info: Selecting UID from range 100 to 999 ...  
  
info: Selecting GID from range 100 to 999 ...  
info: Adding system user `_fluentd' (UID 122) ...  
info: Adding new group `_fluentd' (GID 124) ...  
info: Adding new user `_fluentd' (UID 122) with group `_fluentd' ...  
info: Creating home directory `/var/lib/fluent' ...  
Created symlink /etc/systemd/system/td-agent.service → /usr/lib/systemd/system/fluentd.s  
ervice.  
Created symlink /etc/systemd/system/multi-user.target.wants/fluentd.service → /usr/lib/s  
ystemd/system/fluentd.service.  
Processing triggers for man-db (2.12.0-4build2) ...  
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...  
  
Installation completed. Happy Logging!  
  
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-L  
og-Management-Fundamentals$ fluentd --version  
fluent-package 5.2.0 fluentd 1.18.0 (46372ddd521870f6a203baefb5a598209486d0bc)  
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Tassude  
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-L  
og-Management-Fundamentals$
```

Figure 1: Fluentd Installation and version checking



```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Mar 17 10:25
sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowl...
info: Adding new user '_fluentd' (UID 122) with group '_fluentd' ...
info: Creating home directory '/var/lib/fluentd' ...
Created symlink /etc/systemd/system/td-agent.service → /usr/lib/systemd/system/fluentd.s
service.
Created symlink /etc/systemd/system/multi-user.target.wants/fluentd.service → /usr/lib/s
ystemd/system/fluentd.service.
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Installation completed. Happy Logging!
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-L
og-Management-Fundamentals$ fluentd --version
fluent-package 5.2.0 fluentd 1.18.0 (46372ddd521870f6a203baefb5a598209486d0bc)
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Tassude
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-L
og-Management-Fundamentals$ sudo systemctl start fluentd
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-L
og-Management-Fundamentals$ sudo systemctl enable fluentd
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-L
og-Management-Fundamentals$ sudo systemctl status fluentd
● fluentd.service - fluentd: All in one package of Fluentd
   Loaded: loaded (/usr/lib/systemd/system/fluentd.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-03-17 10:22:10 IST; 2min 44s ago
     Docs: https://docs.fluentd.org/
    Main PID: 6148 (fluentd)
       Tasks: 9 (limit: 12323)
      Memory: 100.8M (peak: 102.2M)
         CPU: 2.664s
        CGroup: /system.slice/fluentd.service
                └─6148 /opt/fluent/bin/ruby /opt/fluent/bin/fluentd --log /var/log/fluent/
                  └─6151 /opt/fluent/bin/ruby -Eascii-8bit:ascii-8bit /opt/fluent/bin/fluent>
Mar 17 10:22:09 sudeep-VirtualBox systemd[1]: Starting fluentd.service - fluentd: All i>
Mar 17 10:22:10 sudeep-VirtualBox systemd[1]: Started fluentd.service - fluentd: All in>
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-L
og-Management-Fundamentals$
```

Figure 2: Fluentd Active running

5.2 Fluentd Configuration

Fluentd was configured to listen for syslog messages.

```
<source>
  @type syslog
  port 5140
  bind 0.0.0.0
  tag syslog
```




```
</source>
```

```
<match syslog.**>
```

```
  @type stdout
```

```
</match>
```

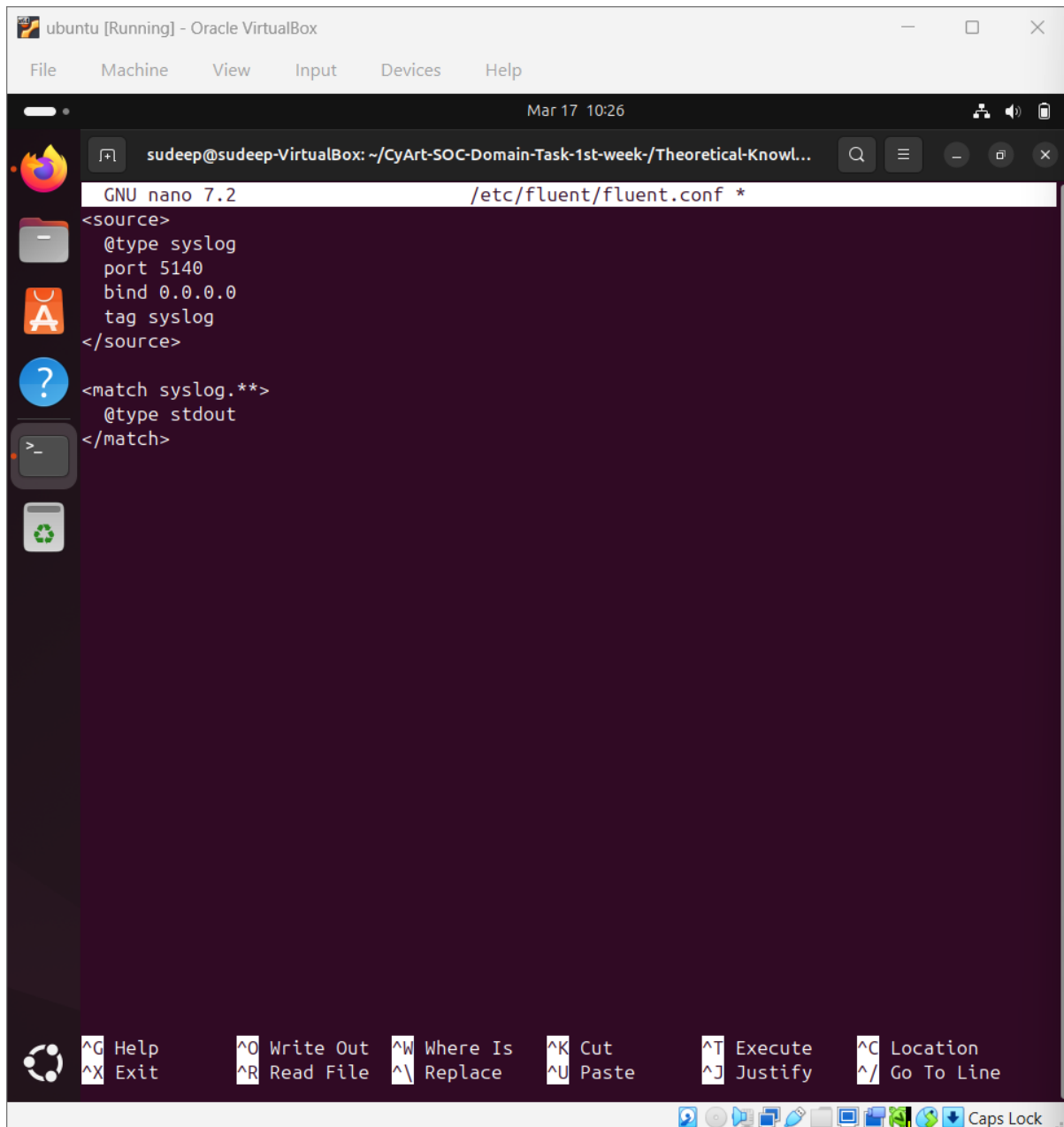


Figure 3:Flentd Syslog configuration

5.3 Running Fluentd

Fluentd was executed manually to ensure correct configuration loading.

```
sudo fluentd -c /etc/fluent/fluent.conf -vv
```



The screenshot shows a terminal window titled 'ubuntu [Running] - Oracle VirtualBox'. The terminal output displays the command `sudo fluentd -c /etc/fluent/fluent.conf -vv` and its output. The output shows the Fluentd version '1.18.0' and lists various plugins and their versions, including 'fluent-plugin-elasticsearch', 'fluent-plugin-flowcounter-simple', 'fluent-plugin-kafka', 'fluent-plugin-metrics-cmectrics', 'fluent-plugin-opensearch', 'fluent-plugin-prometheus', 'fluent-plugin-prometheus-pushgateway', 'fluent-plugin-record-modifier', 'fluent-plugin-rewrite-tag-filter', 'fluent-plugin-s3', 'fluent-plugin-sd-dns', 'fluent-plugin-systemd', 'fluent-plugin-td', and 'fluent-plugin-utmpx'.

```
sudeep@udeep-VirtualBox: ~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowl...
sudeep@udeep-VirtualBox: ~/CyArt-SOC-Domain-T... x sudeep@udeep-VirtualBox: ~/CyArt-SOC-Domain-T... x
og-Management-Fundamentals$ sudo fluentd -c /etc/fluent/fluent.conf -vv
2026-03-17 10:39:57 +0530 [info]: fluent/log.rb:362:info: init supervisor logger path=nil
rotate_age=nil rotate_size=nil
2026-03-17 10:39:57 +0530 [info]: fluent/log.rb:362:info: parsing config file is succeed
ed path="/etc/fluent/fluent.conf"
2026-03-17 10:39:57 +0530 [info]: fluent/log.rb:362:info: gem 'fluentd' version '1.18.0'
2026-03-17 10:39:57 +0530 [info]: fluent/log.rb:362:info: gem 'fluent-plugin-calyptia-mo
nitoring' version '0.1.3'
2026-03-17 10:39:57 +0530 [info]: fluent/log.rb:362:info: gem 'fluent-plugin-elasticsearch'
version '5.4.3'
2026-03-17 10:39:57 +0530 [info]: fluent/log.rb:362:info: gem 'fluent-plugin-flowcounter
-simple' version '0.1.0'
2026-03-17 10:39:57 +0530 [info]: fluent/log.rb:362:info: gem 'fluent-plugin-kafka' vers
ion '0.19.3'
2026-03-17 10:39:57 +0530 [info]: fluent/log.rb:362:info: gem 'fluent-plugin-metrics-cme
trics' version '0.1.2'
2026-03-17 10:39:57 +0530 [info]: fluent/log.rb:362:info: gem 'fluent-plugin-opensearch'
version '1.1.4'
2026-03-17 10:39:57 +0530 [info]: fluent/log.rb:362:info: gem 'fluent-plugin-prometheus'
version '2.1.0'
2026-03-17 10:39:57 +0530 [info]: fluent/log.rb:362:info: gem 'fluent-plugin-prometheus_
pushgateway' version '0.1.1'
2026-03-17 10:39:57 +0530 [info]: fluent/log.rb:362:info: gem 'fluent-plugin-record-modi
fier' version '2.2.0'
2026-03-17 10:39:57 +0530 [info]: fluent/log.rb:362:info: gem 'fluent-plugin-rewrite-tag
-filter' version '2.4.0'
2026-03-17 10:39:57 +0530 [info]: fluent/log.rb:362:info: gem 'fluent-plugin-s3' version
'1.8.1'
2026-03-17 10:39:57 +0530 [info]: fluent/log.rb:362:info: gem 'fluent-plugin-sd-dns' ver
sion '0.1.0'
2026-03-17 10:39:57 +0530 [info]: fluent/log.rb:362:info: gem 'fluent-plugin-systemd' ve
rsion '1.1.0'
2026-03-17 10:39:57 +0530 [info]: fluent/log.rb:362:info: gem 'fluent-plugin-td' version
'1.2.0'
2026-03-17 10:39:57 +0530 [info]: fluent/log.rb:362:info: gem 'fluent-plugin-utmpx' vers
ion '0.5.0'
```

Figure 4: Fluentd running with syslog listener

5.4 Log Generation

A test syslog message was sent using netcat.

```
echo "<13>Mar 17 testhost SOC MANUAL TEST" | nc -u localhost 5140
```

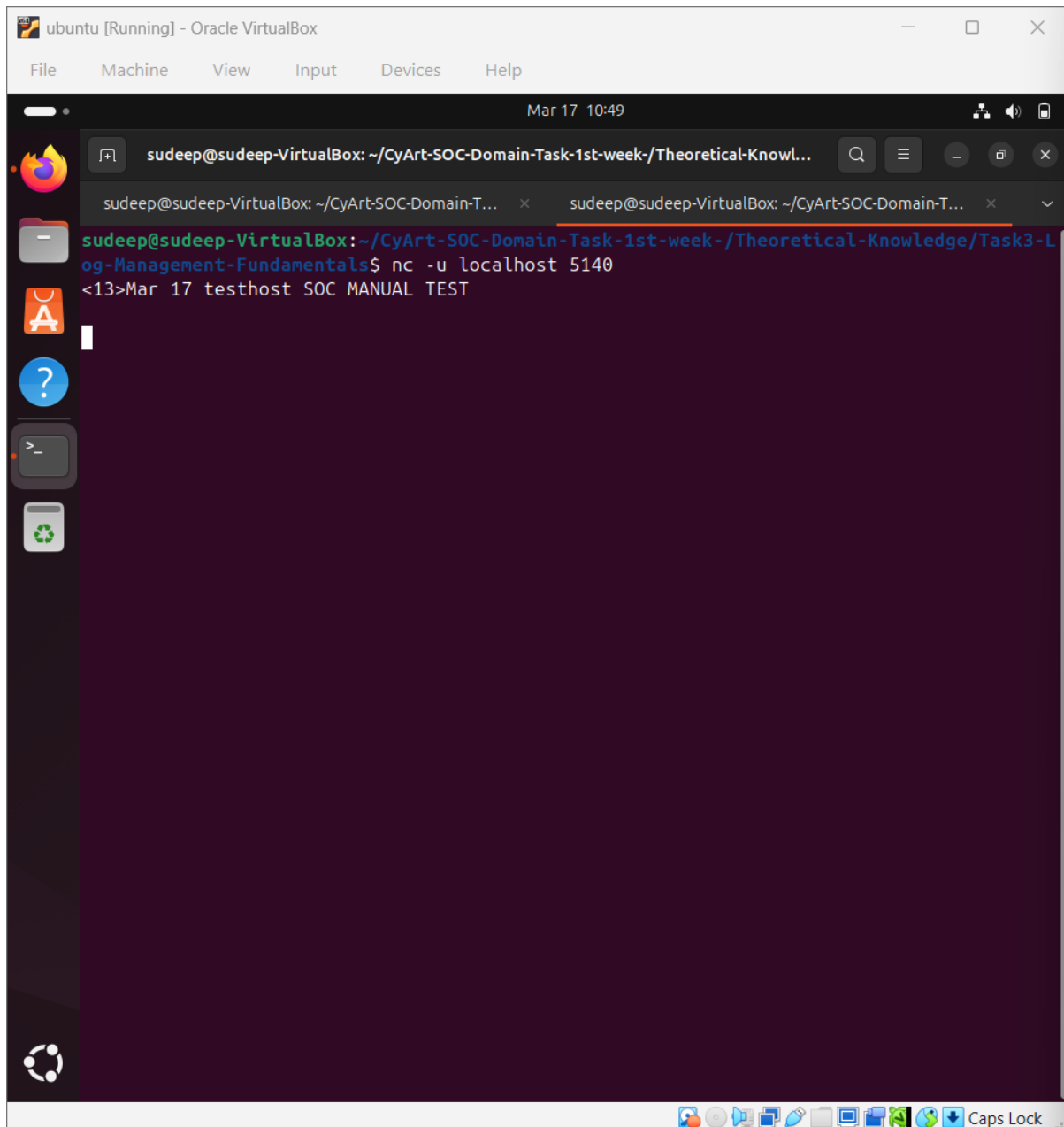


Figure 5: Fluentd configuration for Syslog

5.5 Log Verification

The message was successfully received in Fluentd output.



```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Mar 17 10:51
sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-Task-1st-week-Theoretical-Knowl...
'syslog'
2026-03-17 10:44:13 +0530 [debug]: #0 fluent/log.rb:341:debug: No fluent logger for internal event
2026-03-17 10:44:13 +0530 [info]: #0 fluent/log.rb:362:info: starting fluentd worker pid=7479 ppid=7472 worker=0
2026-03-17 10:44:13 +0530 [info]: #0 fluent/log.rb:362:info: listening syslog socket on 0.0.0.0:5140 with udp
2026-03-17 10:44:13 +0530 [info]: #0 fluent/log.rb:362:info: fluentd worker is now running worker=0
2026-03-17 10:46:04 +0530 [error]: #0 fluent/log.rb:404:error: invalid input data="<13>Mar 17 testhost SOC MANUAL TEST" error_class=Fluent::TimeParser::TimeParseError error="invalid time format: value = Mar 17 testhost, error_class = ArgumentError, error = string doesn't match"
2026-03-17 10:46:04 +0530 [error]: #0 plugin_helper/server.rb:579:on_readable_with_socket: /opt/fluent/lib/ruby/gems/3.2.0/gems/fluentd-1.18.0/lib/fluent/time.rb:280:in 'rescue in parse'
2026-03-17 10:46:04 +0530 [error]: #0 plugin_helper/server.rb:579:on_readable_with_socket: /opt/fluent/lib/ruby/gems/3.2.0/gems/fluentd-1.18.0/lib/fluent/time.rb:277:in 'parse'
2026-03-17 10:46:04 +0530 [error]: #0 plugin_helper/server.rb:579:on_readable_with_socket: /opt/fluent/lib/ruby/gems/3.2.0/gems/fluentd-1.18.0/lib/fluent/plugin/parser_syslog.rb:193:in 'block in parse_rfc3164_regexp'
2026-03-17 10:46:04 +0530 [error]: #0 plugin_helper/server.rb:579:on_readable_with_socket: /opt/fluent/lib/ruby/gems/3.2.0/gems/fluentd-1.18.0/lib/fluent/plugin/parser_syslog.rb:193:in 'synchronize'
2026-03-17 10:46:04 +0530 [error]: #0 plugin_helper/server.rb:579:on_readable_with_socket: /opt/fluent/lib/ruby/gems/3.2.0/gems/fluentd-1.18.0/lib/fluent/plugin/parser_syslog.rb:193:in 'parse_rfc3164_regexp'
2026-03-17 10:46:04 +0530 [error]: #0 plugin_helper/server.rb:579:on_readable_with_socket: /opt/fluent/lib/ruby/gems/3.2.0/gems/fluentd-1.18.0/lib/fluent/plugin/in_syslog.rb:245:in 'message_handler'
2026-03-17 10:46:04 +0530 [error]: #0 plugin_helper/server.rb:579:on_readable_with_socket: /opt/fluent/lib/ruby/gems/3.2.0/gems/fluentd-1.18.0/lib/fluent/plugin/in_syslog.rb:178:in 'block in start_udp_server'
2026-03-17 10:46:04 +0530 [error]: #0 plugin_helper/server.rb:579:on_readable_with_socket: /opt/fluent/lib/ruby/gems/3.2.0/gems/fluentd-1.18.0/lib/fluent/plugin_helper/server.rb:579:in 'on_readable_with_socket'
```

Figure 6: Successful log ingestion in Fluentd

6. SOC Workflow Explanation

The implemented pipeline represents a real SOC architecture:

Log Source → Network → Fluentd → Output

In real environments:

Servers → Log Collector → SIEM → Analyst Investigation

SOC analysts monitor logs to detect:

- Failed login attempts



- Suspicious IP activity
- Unauthorized access

7. Challenges Faced

- Fluentd service was not using the correct configuration file
- Logs were not received initially due to pipeline misconfiguration
- Port binding issues with default syslog port

8. Solution

- Ran Fluentd manually with configuration file
- Used custom port (5140) for testing
- Verified log ingestion using netcat

9. Results

- Fluentd successfully collected syslog messages
- Log ingestion pipeline was validated
- System logs were captured and displayed correctly

10. Conclusion

This task demonstrated the implementation of a log collection pipeline using Fluentd. The lab simulated real-world SOC log ingestion, where logs are collected, processed, and analyzed for security monitoring. Troubleshooting steps provided practical experience in resolving log pipeline issues.



11.LAB 2 Log Querying & Detection:

11.1 Introduction

This lab focuses on **log analysis and threat detection** using both Linux-based tools and Elastic SIEM (Kibana).

The objective is to identify **failed login attempts (Event ID 4625)** and analyze them to detect potential brute force attacks.

11.2 Tools Used

Tool	Purpose
Linux CLI (grep, awk)	Log filtering and analysis
Elasticsearch	Log storage
Kibana	SIEM visualization and querying
Fluentd	Log collection

11.3 Linux-Based Log Analysis

Step 1 — Creating Sample Logs

A JSON file containing login events was created.

```
cat security_logs.json
```

Example logs:

```
{"EventID":4625,"SourceIP":"192.168.1.10","User":"admin"}
{"EventID":4625,"SourceIP":"192.168.1.10","User":"admin"}
{"EventID":4625,"SourceIP":"192.168.1.20","User":"root"}
{"EventID":4624,"SourceIP":"192.168.1.15","User":"user1"}
```



The screenshot shows a terminal window titled 'ubuntu [Running] - Oracle VirtualBox'. The terminal is running the 'nano' text editor, editing a file named 'security_logs.json'. The file contains five JSON objects representing login events. The first four objects have an 'EventID' of 4625, indicating successful logins for 'admin' (from 192.168.1.10) and 'root' (from 192.168.1.20). The fifth object has an 'EventID' of 4624, indicating a failed login attempt for 'user1' (from 192.168.1.15). The terminal window also shows a taskbar with various application icons and a system tray at the bottom.

```
GNU nano 7.2 security_logs.json *
{"EventID":4625,"SourceIP":"192.168.1.10","User":"admin"}
{"EventID":4625,"SourceIP":"192.168.1.10","User":"admin"}
{"EventID":4625,"SourceIP":"192.168.1.20","User":"root"}
{"EventID":4624,"SourceIP":"192.168.1.15","User":"user1"}
{"EventID":4625,"SourceIP":"192.168.1.10","User":"admin"}
```

Figure 7: Sample security logs containing login events

Step 2 — Filtering Failed Logins

Command used:

```
grep 4625 security_logs.json
```

This filters only failed login attempts



```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Mar 17 11:17
sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowl...
sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-T... x sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-T... x
2026-03-17 11:14:24 +0530 [debug]: #0 fluent/log.rb:341:debug: preparing shutdown input
plugin type=:syslog plugin_id="object:988"
2026-03-17 11:14:24 +0530 [info]: #0 fluent/log.rb:362:info: shutting down input plugin
type=:syslog plugin_id="object:988"
2026-03-17 11:14:24 +0530 [debug]: #0 fluent/log.rb:341:debug: preparing shutdown output
plugin type=:stdout plugin_id="object:960"
2026-03-17 11:14:24 +0530 [info]: #0 fluent/log.rb:362:info: shutting down output plugin
type=:stdout plugin_id="object:960"
2026-03-17 11:14:25 +0530 [debug]: #0 fluent/log.rb:341:debug: calling after_shutdown on
input plugin type=:syslog plugin_id="object:988"
2026-03-17 11:14:25 +0530 [debug]: #0 fluent/log.rb:341:debug: calling after_shutdown on
output plugin type=:stdout plugin_id="object:960"
2026-03-17 11:14:25 +0530 [debug]: #0 fluent/log.rb:341:debug: closing input plugin type
=:syslog plugin_id="object:988"
2026-03-17 11:14:25 +0530 [debug]: #0 fluent/log.rb:341:debug: closing output plugin typ
e=:stdout plugin_id="object:960"
2026-03-17 11:14:25 +0530 [debug]: #0 fluent/log.rb:341:debug: calling terminate on inpu
t plugin type=:syslog plugin_id="object:988"
2026-03-17 11:14:25 +0530 [debug]: #0 fluent/log.rb:341:debug: calling terminate on outp
ut plugin type=:stdout plugin_id="object:960"
2026-03-17 11:14:25 +0530 [info]: fluent/log.rb:362:info: Worker 0 finished with status
0
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-L
og-Management-Fundamentals$ ls
fluent-apt-source.deb README.md screenshots
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-L
og-Management-Fundamentals$ nano security_logs.json
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-L
og-Management-Fundamentals$ grep 4625 security_logs.json
{"EventID":4625,"SourceIP":"192.168.1.10","User":"admin"}
{"EventID":4625,"SourceIP":"192.168.1.10","User":"admin"}
{"EventID":4625,"SourceIP":"192.168.1.20","User":"root"}
{"EventID":4625,"SourceIP":"192.168.1.10","User":"admin"}
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-L
og-Management-Fundamentals$
```

Figure 8: Filtering failed login events using grep

Step 3 — Grouping by Source IP

Command used:

```
grep 4625 security_logs.json | awk -F'"' '{print $6}' | sort | uniq -c
```

Output:

```
3 192.168.1.10
1 192.168.1.20
```



```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Mar 17 11:21
sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-KnowL...
sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-T... x sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-T... x
plugin type=:stdout plugin_id="object:960"
2026-03-17 11:14:24 +0530 [info]: #0 fluent/log.rb:362:info: shutting down output plugin
type=:stdout plugin_id="object:960"
2026-03-17 11:14:24 +0530 [debug]: #0 fluent/log.rb:341:debug: calling after_shutdown on
input plugin type=:syslog plugin_id="object:988"
2026-03-17 11:14:25 +0530 [debug]: #0 fluent/log.rb:341:debug: calling after_shutdown on
output plugin type=:stdout plugin_id="object:960"
2026-03-17 11:14:25 +0530 [debug]: #0 fluent/log.rb:341:debug: closing input plugin type
=:syslog plugin_id="object:988"
2026-03-17 11:14:25 +0530 [debug]: #0 fluent/log.rb:341:debug: closing output plugin typ
e=:stdout plugin_id="object:960"
2026-03-17 11:14:25 +0530 [debug]: #0 fluent/log.rb:341:debug: calling terminate on inpu
t plugin type=:syslog plugin_id="object:988"
2026-03-17 11:14:25 +0530 [debug]: #0 fluent/log.rb:341:debug: calling terminate on outp
ut plugin type=:stdout plugin_id="object:960"
2026-03-17 11:14:25 +0530 [info]: fluent/log.rb:362:info: Worker 0 finished with status
0
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-L
og-Management-Fundamentals$ ls
fluent-apt-source.deb README.md screenshots
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-L
og-Management-Fundamentals$ nano security_logs.json
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-L
og-Management-Fundamentals$ grep 4625 security_logs.json
{"EventID":4625,"SourceIP":"192.168.1.10","User":"admin"}
{"EventID":4625,"SourceIP":"192.168.1.10","User":"admin"}
{"EventID":4625,"SourceIP":"192.168.1.20","User":"root"}
{"EventID":4625,"SourceIP":"192.168.1.10","User":"admin"}
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-L
og-Management-Fundamentals$ grep 4625 security_logs.json | awk -F'"' '{print $6}' | sort
| uniq -c
  3 192.168.1.10
  1 192.168.1.20
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-L
og-Management-Fundamentals$
```

Figure 9: Failed login attempts grouped by source IP

Analysis (Linux)

- IP 192.168.1.10 generated multiple failed attempts
- Indicates possible **brute force attack**



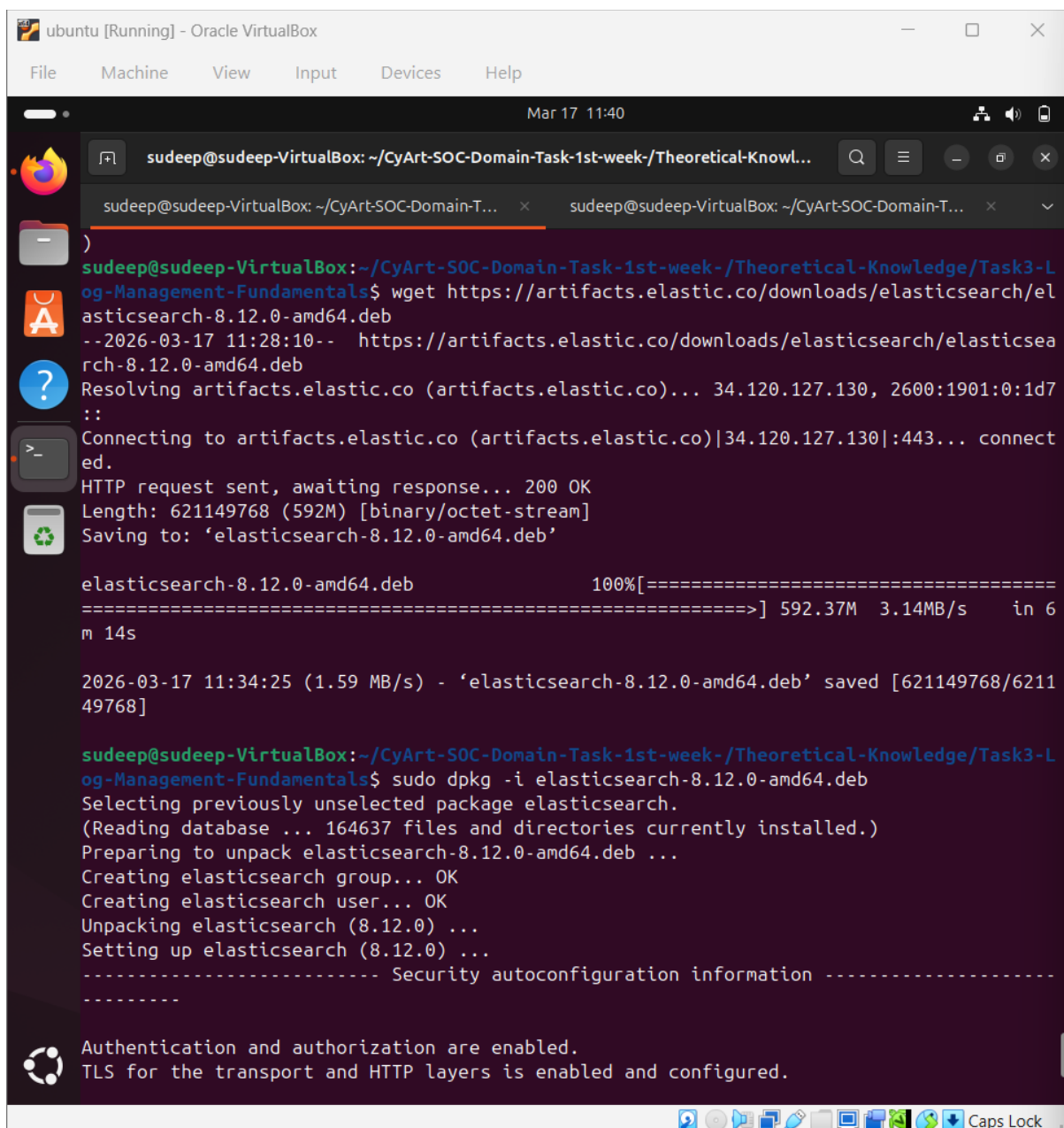
11.4 Elastic SIEM (Kibana) Analysis

Step 1- Installing Elastic search

command used

wget <https://artifacts.elastic.co/downloads/elasticsearch/elasticsearch-8.12.0-amd64.deb>

sudo dpkg -i elasticsearch-8.12.0-amd64.deb



```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Mar 17 11:40
sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowl...
sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-T... x sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-T... x
)
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-L
og-Management-Fundamentals$ wget https://artifacts.elastic.co/downloads/elasticsearch/el
asticsearch-8.12.0-amd64.deb
--2026-03-17 11:28:10-- https://artifacts.elastic.co/downloads/elasticsearch/elasticsea
rch-8.12.0-amd64.deb
Resolving artifacts.elastic.co (artifacts.elastic.co)... 34.120.127.130, 2600:1901:0:1d7
::
Connecting to artifacts.elastic.co (artifacts.elastic.co)|34.120.127.130|:443... connect
ed.
HTTP request sent, awaiting response... 200 OK
Length: 621149768 (592M) [binary/octet-stream]
Saving to: 'elasticsearch-8.12.0-amd64.deb'

elasticsearch-8.12.0-amd64.deb      100%[=====] 592.37M  3.14MB/s   in 6
m 14s

2026-03-17 11:34:25 (1.59 MB/s) - 'elasticsearch-8.12.0-amd64.deb' saved [621149768/6211
49768]

sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-L
og-Management-Fundamentals$ sudo dpkg -i elasticsearch-8.12.0-amd64.deb
Selecting previously unselected package elasticsearch.
(Reading database ... 164637 files and directories currently installed.)
Preparing to unpack elasticsearch-8.12.0-amd64.deb ...
Creating elasticsearch group... OK
Creating elasticsearch user... OK
Unpacking elasticsearch (8.12.0) ...
Setting up elasticsearch (8.12.0) ...
----- Security autoconfiguration information -----
-----
Authentication and authorization are enabled.
TLS for the transport and HTTP layers is enabled and configured.
```



Figure 10: Elastic Installation

Step 2-Enabling Elastic search

Command used

```
sudo systemctl daemon-reexec
```

```
sudo systemctl enable elasticsearch
```

```
sudo systemctl start elasticsearch
```

```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Mar 17 11:42
sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowl...
sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-T... x sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-T... x
lasticsearch service to start automatically using systemd
sudo systemctl daemon-reload
sudo systemctl enable elasticsearch.service
### You can start elasticsearch service by executing
sudo systemctl start elasticsearch.service
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-L
og-Management-Fundamentals$ sudo systemctl daemon-reexec
sudo systemctl enable elasticsearch
sudo systemctl start elasticsearch
Created symlink /etc/systemd/system/multi-user.target.wants/elasticsearch.service → /usr
/lib/systemd/system/elasticsearch.service.
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Tassusu
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-L
og-Management-Fundamentals$ sudo systemctl status elasticsearch
● elasticsearch.service - Elasticsearch
   Loaded: loaded (/usr/lib/systemd/system/elasticsearch.service; enabled; preset: en>
   Active: active (running) since Tue 2026-03-17 11:36:22 IST; 6min ago
     Docs: https://www.elastic.co
    Main PID: 9833 (java)
      Tasks: 105 (limit: 12323)
     Memory: 5.6G (peak: 5.6G)
        CPU: 1min 46.166s
      CGroup: /system.slice/elasticsearch.service
              └─9833 /usr/share/elasticsearch/jdk/bin/java -Xms4m -Xmx64m -XX:+UseSerial>
                9903 /usr/share/elasticsearch/jdk/bin/java -Des.networkaddress.cache.ttl>
                9932 /usr/share/elasticsearch/modules/x-pack-ml/platform/linux-x86_64/bi>

Mar 17 11:35:53 sudeep-VirtualBox systemd[1]: Starting elasticsearch.service - Elasticse>
Mar 17 11:36:02 sudeep-VirtualBox systemd-entrypoint[9903]: CompileCommand: exclude org>
Mar 17 11:36:02 sudeep-VirtualBox systemd-entrypoint[9903]: CompileCommand: exclude org>
Mar 17 11:36:03 sudeep-VirtualBox systemd-entrypoint[9833]: Mar 17, 2026 11:36:03 AM su>
Mar 17 11:36:03 sudeep-VirtualBox systemd-entrypoint[9833]: WARNING: COMPAT locale prov>
Mar 17 11:36:22 sudeep-VirtualBox systemd[1]: Started elasticsearch.service - Elasticse>
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-L
og-Management-Fundamentals$
```

Figure 11: Elasticsearch service running



STEP 3 — Verify Elasticsearch

command used:

`curl -k https://localhost:9200`

```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Mar 17 11:44

sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowl...
sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-T... x sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-T... x
Memory: 5.6G (peak: 5.6G)
CPU: 1min 46.166s
CGroup: /system.slice/elasticsearch.service
├─9833 /usr/share/elasticsearch/jdk/bin/java -Xms4m -Xmx64m -XX:+UseSerial>
├─9903 /usr/share/elasticsearch/jdk/bin/java -Des.networkaddress.cache.ttl>
└─9932 /usr/share/elasticsearch/modules/x-pack-ml/platform/linux-x86_64/bi>

Mar 17 11:35:53 sudeep-VirtualBox systemd[1]: Starting elasticsearch.service - Elasticse>
Mar 17 11:36:02 sudeep-VirtualBox systemd-entrypoint[9903]: CompileCommand: exclude org>
Mar 17 11:36:02 sudeep-VirtualBox systemd-entrypoint[9903]: CompileCommand: exclude org>
Mar 17 11:36:03 sudeep-VirtualBox systemd-entrypoint[9833]: Mar 17, 2026 11:36:03 AM su>
Mar 17 11:36:03 sudeep-VirtualBox systemd-entrypoint[9833]: WARNING: COMPAT locale prov>
Mar 17 11:36:22 sudeep-VirtualBox systemd[1]: Started elasticsearch.service - Elasticse>
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-L
og-Management-Fundamentals$ curl -k -u elastic https://localhost:9200
Enter host password for user 'elastic':
{
  "name" : "sudeep-VirtualBox",
  "cluster_name" : "elasticsearch",
  "cluster_uuid" : "K3SeuW1YRhOSMXemUPeaWg",
  "version" : {
    "number" : "8.12.0",
    "build_flavor" : "default",
    "build_type" : "deb",
    "build_hash" : "1665f706fd9354802c02146c1e6b5c0fbcdddfbc9",
    "build_date" : "2024-01-11T10:05:27.953830042Z",
    "build_snapshot" : false,
    "lucene_version" : "9.9.1",
    "minimum_wire_compatibility_version" : "7.17.0",
    "minimum_index_compatibility_version" : "7.0.0"
  },
  "tagline" : "You Know, for Search"
}
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-L
og-Management-Fundamentals$
```

Figure 12: Elasticsearch running and responding on port 9200



Step 5: Install Kibana (SIEM Dashboard)

wget <https://artifacts.elastic.co/downloads/kibana/kibana-8.12.0-amd64.deb>

sudo dpkg -i kibana-8.12.0-amd64.deb

```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Mar 17 11:57
sudeep@sudeep-VirtualBox: ~

sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-Task-1st-week-$ cd ..
sudeep@sudeep-VirtualBox:~$ wget https://artifacts.elastic.co/downloads/kibana/kibana-8.12.0-amd64.deb
--2026-03-17 11:50:42-- https://artifacts.elastic.co/downloads/kibana/kibana-8.12.0-amd64.deb
Resolving artifacts.elastic.co (artifacts.elastic.co)... 34.120.127.130, 2600:1901:0:1d7::
Connecting to artifacts.elastic.co (artifacts.elastic.co)|34.120.127.130|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 317649798 (303M) [binary/octet-stream]
Saving to: 'kibana-8.12.0-amd64.deb'

kibana-8.12.0-amd64.d 100%[=====] 302.93M 2.02MB/s in 4m 23s

2026-03-17 11:55:07 (1.15 MB/s) - 'kibana-8.12.0-amd64.deb' saved [317649798/317649798]

sudeep@sudeep-VirtualBox:~$ sudo dpkg -i kibana-8.12.0-amd64.deb
[sudo] password for sudeep:
Selecting previously unselected package kibana.
(Reading database ... 165981 files and directories currently installed.)
Preparing to unpack kibana-8.12.0-amd64.deb ...
Unpacking kibana (8.12.0) ...
Setting up kibana (8.12.0) ...
Creating kibana group... OK
Creating kibana user... OK
Kibana is currently running with legacy OpenSSL providers enabled! For details and instructions on how to disable see https://www.elastic.co/guide/en/kibana/8.12/production.html#openssl-legacy-provider
Created Kibana keystore in /etc/kibana/kibana.keystore
sudeep@sudeep-VirtualBox:~$ sudo systemctl enable kibana
sudo systemctl start kibana
Created symlink /etc/systemd/system/multi-user.target.wants/kibana.service → /usr/lib/systemd/system/kibana.service.
sudeep@sudeep-VirtualBox:~$
```

Figure 13: Kibana installation



STEP 6 — Start Kibana

```
sudo systemctl enable kibana
```

```
sudo systemctl start kibana
```

Check:

```
sudo systemctl status kibana
```

The screenshot shows a terminal window titled 'ubuntu [Running] - Oracle VirtualBox'. The user is logged in as 'sudeep@sudeep-VirtualBox'. The terminal output shows the following steps:

```
Unpacking kibana (8.12.0) ...
Setting up kibana (8.12.0) ...
Creating kibana group... OK
Creating kibana user... OK
Kibana is currently running with legacy OpenSSL providers enabled! For details and instructions on how to disable see https://www.elastic.co/guide/en/kibana/8.12/production.html#openssl-legacy-provider
Created Kibana keystore in /etc/kibana/kibana.keystore
sudeep@sudeep-VirtualBox:~$ sudo systemctl enable kibana
sudeep@sudeep-VirtualBox:~$ sudo systemctl start kibana
Created symlink /etc/systemd/system/multi-user.target.wants/kibana.service → /usr/lib/systemd/system/kibana.service.
sudeep@sudeep-VirtualBox:~$ sudo systemctl status kibana
● kibana.service - Kibana
   Loaded: loaded (/usr/lib/systemd/system/kibana.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-03-17 11:57:34 IST; 1min 41s ago
     Docs: https://www.elastic.co
    Main PID: 10873 (node)
      Tasks: 11 (limit: 12323)
     Memory: 302.0M (peak: 393.3M)
        CPU: 17.532s
     CGroup: /system.slice/kibana.service
            └─10873 /usr/share/kibana/bin/./node/bin/node /usr/share/kibana/bin/./sr>

Mar 17 11:57:53 sudeep-VirtualBox kibana[10873]: [2026-03-17T11:57:52.932+05:30][INFO ]>
Mar 17 11:57:53 sudeep-VirtualBox kibana[10873]: [2026-03-17T11:57:52.932+05:30][INFO ]>
Mar 17 11:57:53 sudeep-VirtualBox kibana[10873]: [2026-03-17T11:57:52.932+05:30][INFO ]>
Mar 17 11:57:53 sudeep-VirtualBox kibana[10873]: [2026-03-17T11:57:52.932+05:30][INFO ]>
Mar 17 11:57:53 sudeep-VirtualBox kibana[10873]: [2026-03-17T11:57:53.152+05:30][INFO ]>
Mar 17 11:57:53 sudeep-VirtualBox kibana[10873]: [2026-03-17T11:57:53.525+05:30][INFO ]>
Mar 17 11:57:53 sudeep-VirtualBox kibana[10873]: [2026-03-17T11:57:53.583+05:30][INFO ]>
Mar 17 11:57:53 sudeep-VirtualBox kibana[10873]: [2026-03-17T11:57:53.632+05:30][INFO ]>
Mar 17 11:57:53 sudeep-VirtualBox kibana[10873]: i Kibana has not been configured.
Mar 17 11:57:53 sudeep-VirtualBox kibana[10873]: Go to http://localhost:5601/?code=5131>
sudeep@sudeep-VirtualBox:~$
```

Figure 14: Kibana service running



STEP 7 — Open Browser - Access Kibana Dashboard

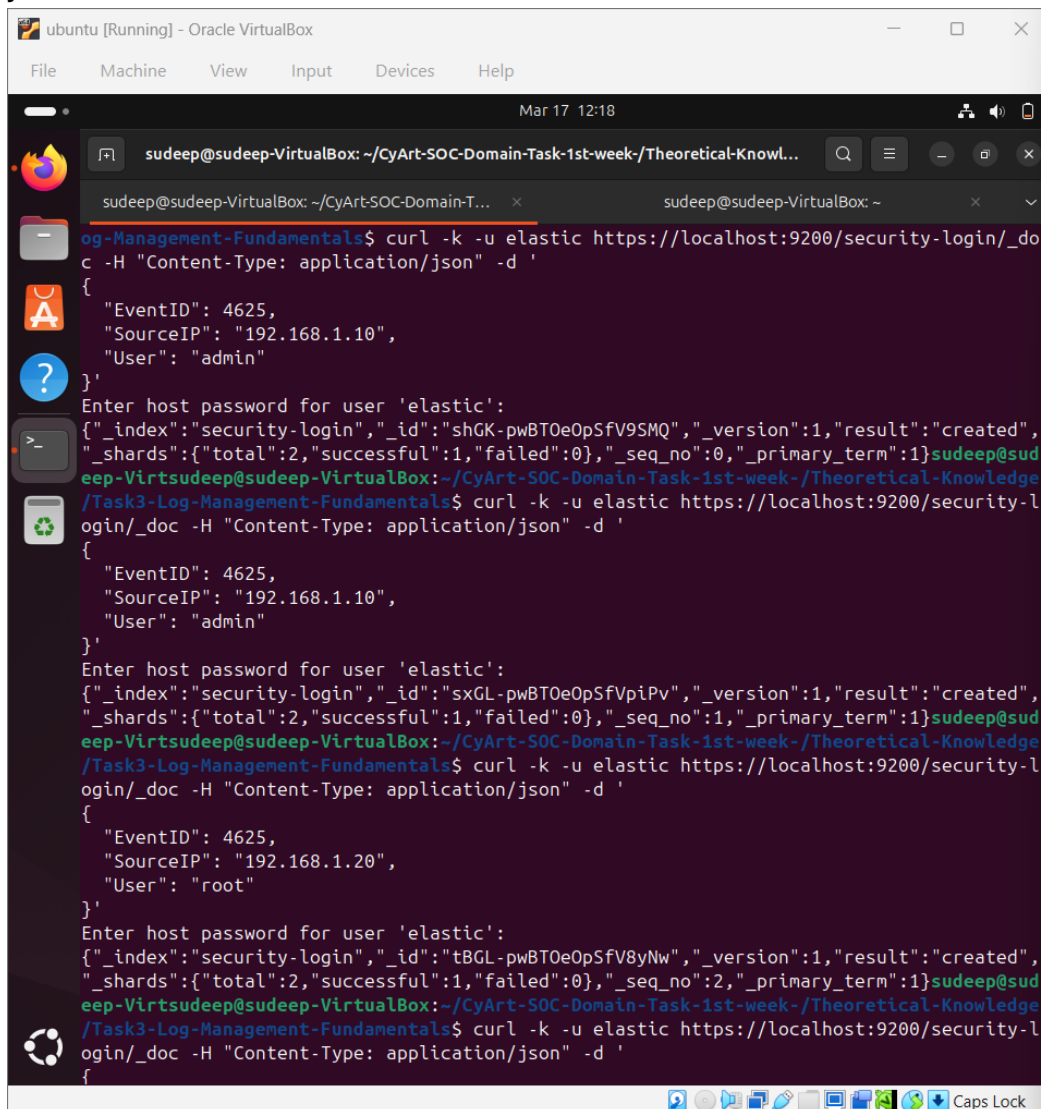
<http://localhost:5601>

Step 8 — Inserting Logs into Elasticsearch

Logs were inserted using curl:

```
curl -k -u elastic https://localhost:9200/security-login/_doc -H "Content-Type: application/json" -d '{
```

```
{
  "EventID": 4625,
  "SourceIP": "192.168.1.10",
  "User": "admin"
}'
```



```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Mar 17 12:18
sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-Task-1st-week/Theoretical-Knowl...
sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-T... sudeep@sudeep-VirtualBox: ~
og-Management-Fundamentals$ curl -k -u elastic https://localhost:9200/security-login/_doc -H "Content-Type: application/json" -d '{
{
  "EventID": 4625,
  "SourceIP": "192.168.1.10",
  "User": "admin"
}'
Enter host password for user 'elastic':
{"_index": "security-login", "_id": "shGK-pwBT0eOpSfV9SMQ", "_version": 1, "result": "created",
 "_shards": {"total": 2, "successful": 1, "failed": 0}, "_seq_no": 0, "_primary_term": 1}sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-Task-1st-week/Theoretical-Knowledge
/Task3-Log-Management-Fundamentals$ curl -k -u elastic https://localhost:9200/security-l
ogin/_doc -H "Content-Type: application/json" -d '{
{
  "EventID": 4625,
  "SourceIP": "192.168.1.10",
  "User": "admin"
}'
Enter host password for user 'elastic':
{"_index": "security-login", "_id": "sxGL-pwBT0eOpSfVpiPv", "_version": 1, "result": "created",
 "_shards": {"total": 2, "successful": 1, "failed": 0}, "_seq_no": 1, "_primary_term": 1}sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-Task-1st-week/Theoretical-Knowledge
/Task3-Log-Management-Fundamentals$ curl -k -u elastic https://localhost:9200/security-l
ogin/_doc -H "Content-Type: application/json" -d '{
{
  "EventID": 4625,
  "SourceIP": "192.168.1.20",
  "User": "root"
}'
Enter host password for user 'elastic':
{"_index": "security-login", "_id": "tBGL-pwBT0eOpSfV8yNw", "_version": 1, "result": "created",
 "_shards": {"total": 2, "successful": 1, "failed": 0}, "_seq_no": 2, "_primary_term": 1}sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-Task-1st-week/Theoretical-Knowledge
/Task3-Log-Management-Fundamentals$ curl -k -u elastic https://localhost:9200/security-l
ogin/_doc -H "Content-Type: application/json" -d '{
{
```

Figure 15: Inserting logs into Elasticsearch index



Step 9— Creating Data View in Kibana

- Opened Kibana → Discover
- Created index pattern:

security-login*

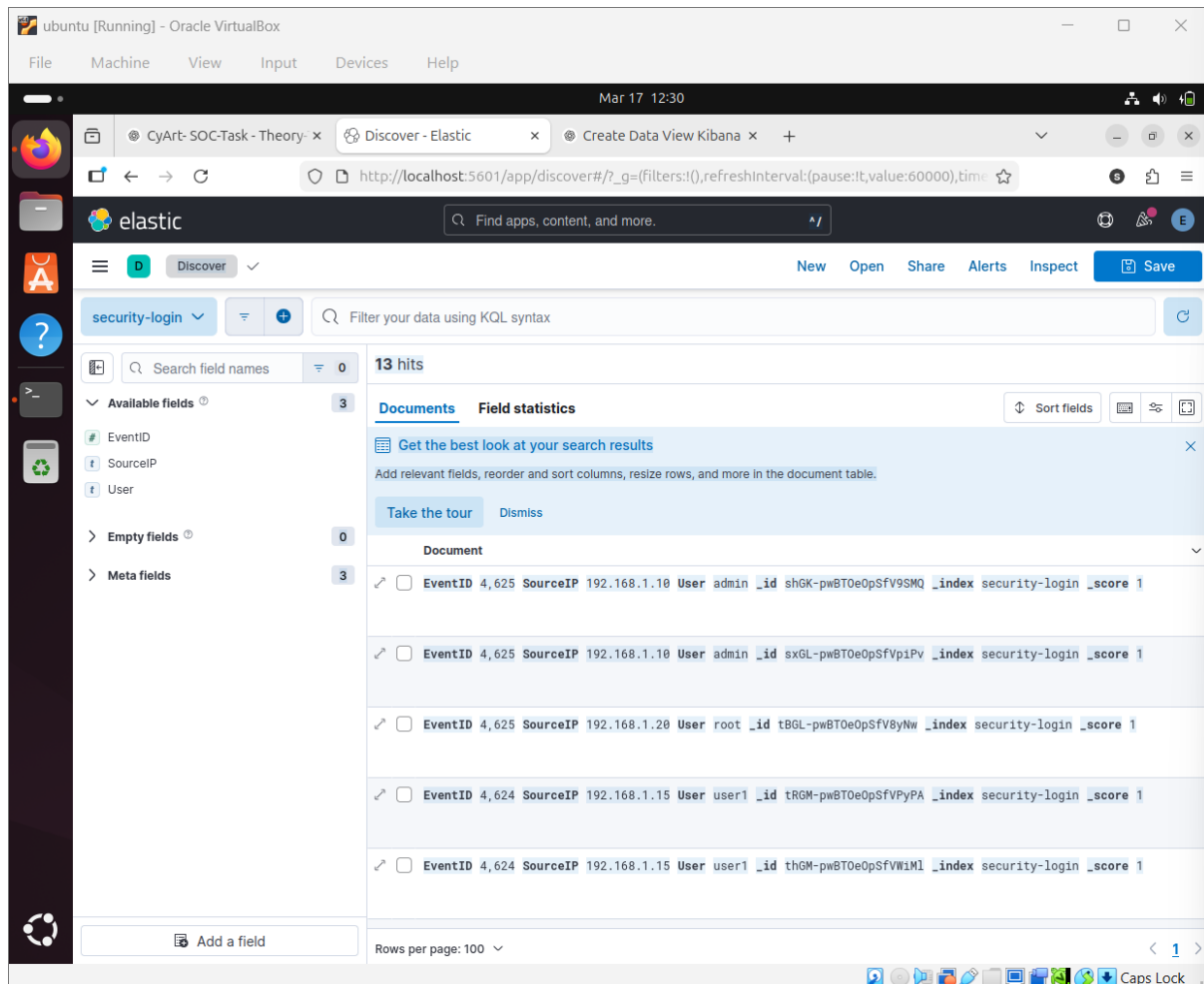


Figure 16: Creating data view in Kibana

Step 10 — Running KQL Query

Query used:

EventID: 4625

Displays failed login events.

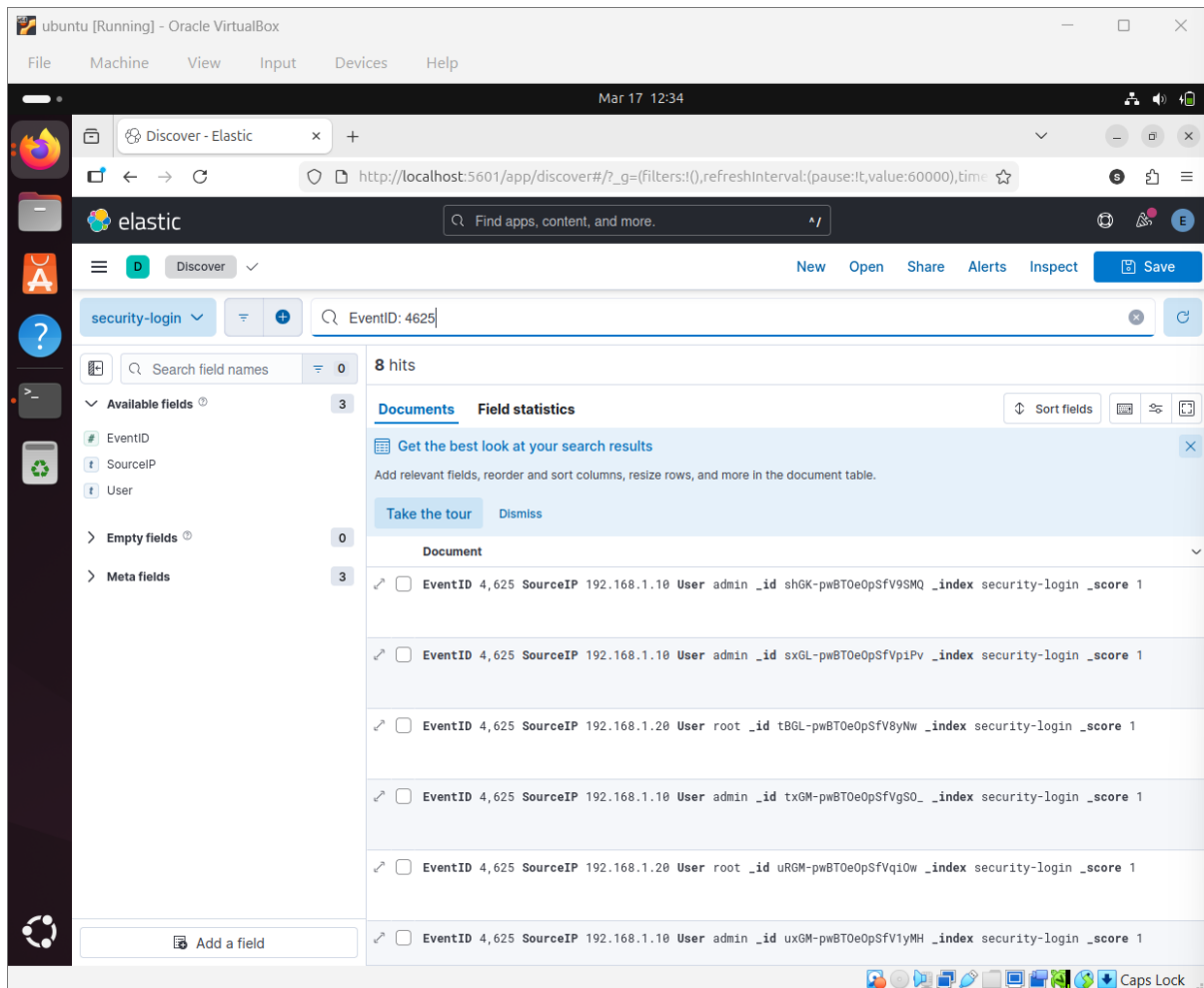


Figure 17: KQL query showing failed login events

Step 11 — Analyzing by Source IP

Logs were analyzed by grouping Source IP.

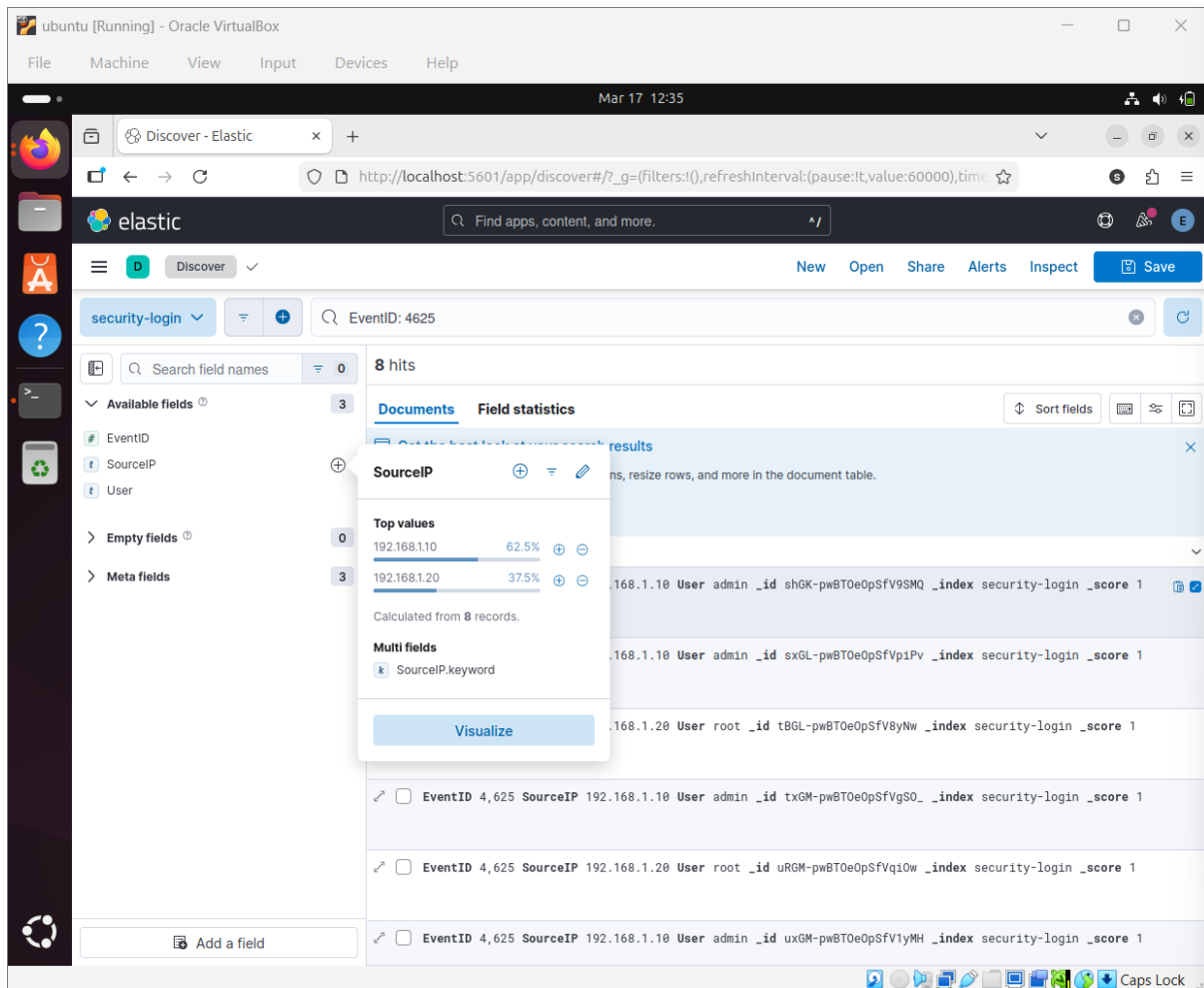


Figure 18: Failed login attempts grouped by Source IP in Kibana

11.5 Analysis (Kibana)

- Multiple failed logins observed from:
 - 192.168.1.10
- Indicates potential:
 - Brute force attack
 - Credential guessing activity

11.6 SOC Analyst Workflow

Log Collection → Log Storage → Query Execution → Detection → Investigation

Steps performed:

1. Logs collected and stored in Elasticsearch
2. Queries executed using KQL



3. Suspicious patterns identified
4. Analyst investigates source IP

11.7 Conclusion

This lab demonstrated how SOC analysts detect suspicious login activity using both:

- Linux command-line tools
- SIEM platforms like Elasticsearch and Kibana

The analysis successfully identified repeated failed login attempts from a single IP, indicating a potential brute force attack.

12.LAB 3 Log Normalization using Logstash:

12.1 Introduction:

In a Security Operations Center (SOC), logs come from multiple sources in different formats.

Normalization is the process of converting these logs into a standard structured format such as JSON.

This allows:

- Easier searching and querying
- Correlation of events across systems
- Faster incident detection

Logstash is commonly used in SOC environments to parse and transform logs using Grok patterns.

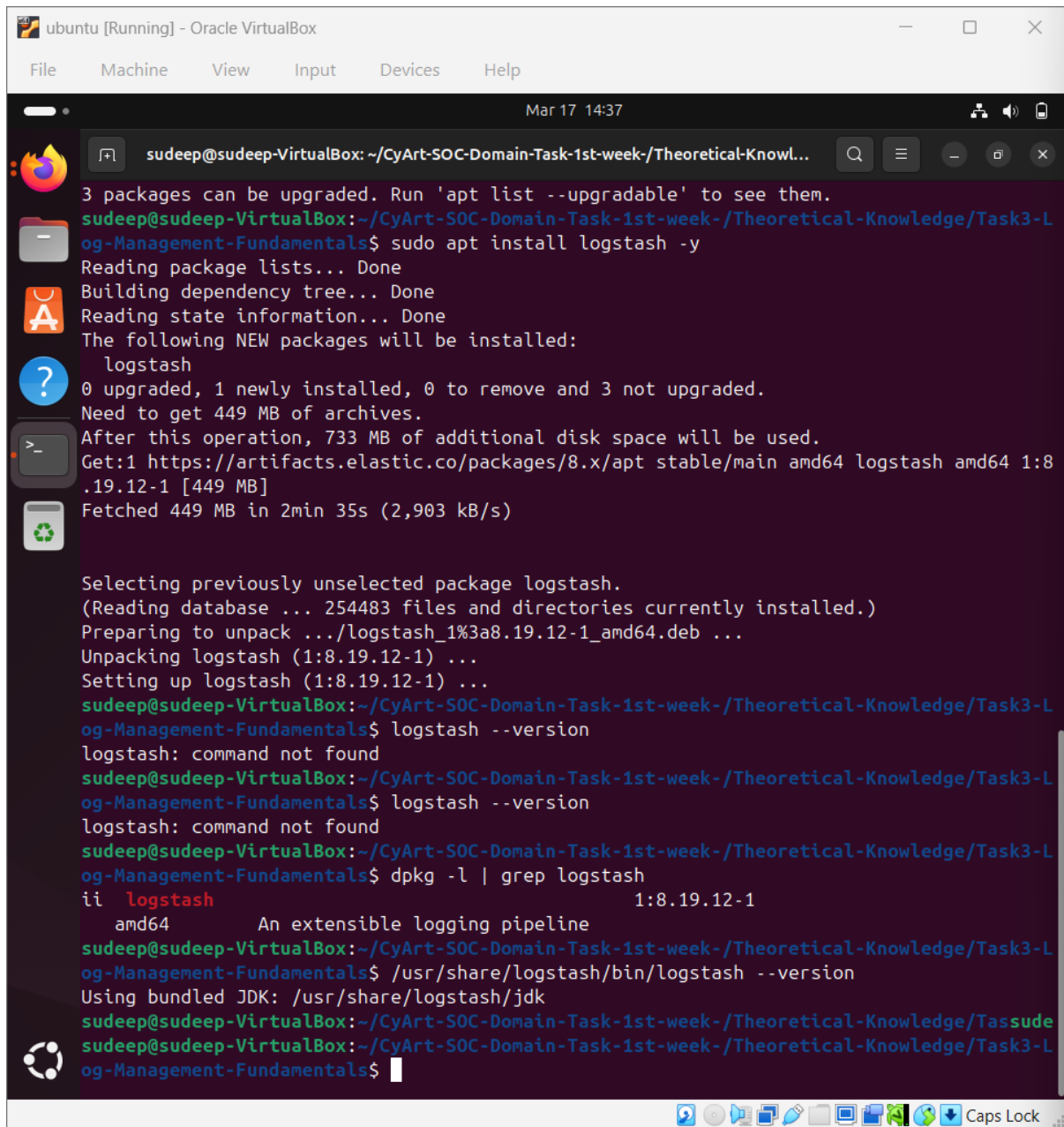
12.2 Installation & Setup:



Install Logstash

```
sudo apt update
```

```
sudo apt install logstash -y
```



```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Mar 17 14:37
sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowl...
3 packages can be upgraded. Run 'apt list --upgradable' to see them.
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-L
og-Management-Fundamentals$ sudo apt install logstash -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  logstash
0 upgraded, 1 newly installed, 0 to remove and 3 not upgraded.
Need to get 449 MB of archives.
After this operation, 733 MB of additional disk space will be used.
Get:1 https://artifacts.elastic.co/packages/8.x/apt stable/main amd64 logstash amd64 1:8
.19.12-1 [449 MB]
Fetched 449 MB in 2min 35s (2,903 kB/s)

Selecting previously unselected package logstash.
(Reading database ... 254483 files and directories currently installed.)
Preparing to unpack .../logstash_1%3a8.19.12-1_amd64.deb ...
Unpacking logstash (1:8.19.12-1) ...
Setting up logstash (1:8.19.12-1) ...
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-L
og-Management-Fundamentals$ logstash --version
logstash: command not found
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-L
og-Management-Fundamentals$ logstash --version
logstash: command not found
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-L
og-Management-Fundamentals$ dpkg -l | grep logstash
ii  logstash                                1:8.19.12-1
    amd64 An extensible logging pipeline
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-L
og-Management-Fundamentals$ /usr/share/logstash/bin/logstash --version
Using bundled JDK: /usr/share/logstash/jdk
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-L
og-Management-Fundamentals$
```

Figure 19: Logstash installation and verification on Ubuntu

12.3 Log File Preparation

Created sample Apache log file:



```
nano apache.log
192.168.1.10 - - [17/Mar/2026:10:00:00] "GET /index.html HTTP/1.1"
200 1043
192.168.1.20 - - [17/Mar/2026:10:01:00] "POST /login HTTP/1.1" 401
560
192.168.1.10 - - [17/Mar/2026:10:02:00] "GET /admin HTTP/1.1" 403
720
```

The screenshot shows a virtual machine window titled 'ubuntu [Running] - Oracle VirtualBox'. Inside the VM, a terminal window is open with the nano 7.2 text editor editing a file named 'apache.log'. The terminal output shows the following log entries:

```
GNU nano 7.2      apache.log *
192.168.1.10 - - [17/Mar/2026:10:00:00] "GET /index.html HTTP/1.1" 200 1043
192.168.1.20 - - [17/Mar/2026:10:01:00] "POST /login HTTP/1.1" 401 560
192.168.1.10 - - [17/Mar/2026:10:02:00] "GET /admin HTTP/1.1" 403 720
```

The terminal window has a dark background and a light-colored text. The nano editor's status bar at the bottom shows various keyboard shortcuts like ^G Help, ^X Exit, ^O Write Out, ^R Read File, ^W Where Is, ^_ Replace, ^K Cut, ^U Paste, ^T Execute, ^J Justify, ^C Location, and ^_ Go To Line. The terminal window is titled 'sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowl...'. The window's title bar shows 'Mar 17 14:39' and standard window controls.

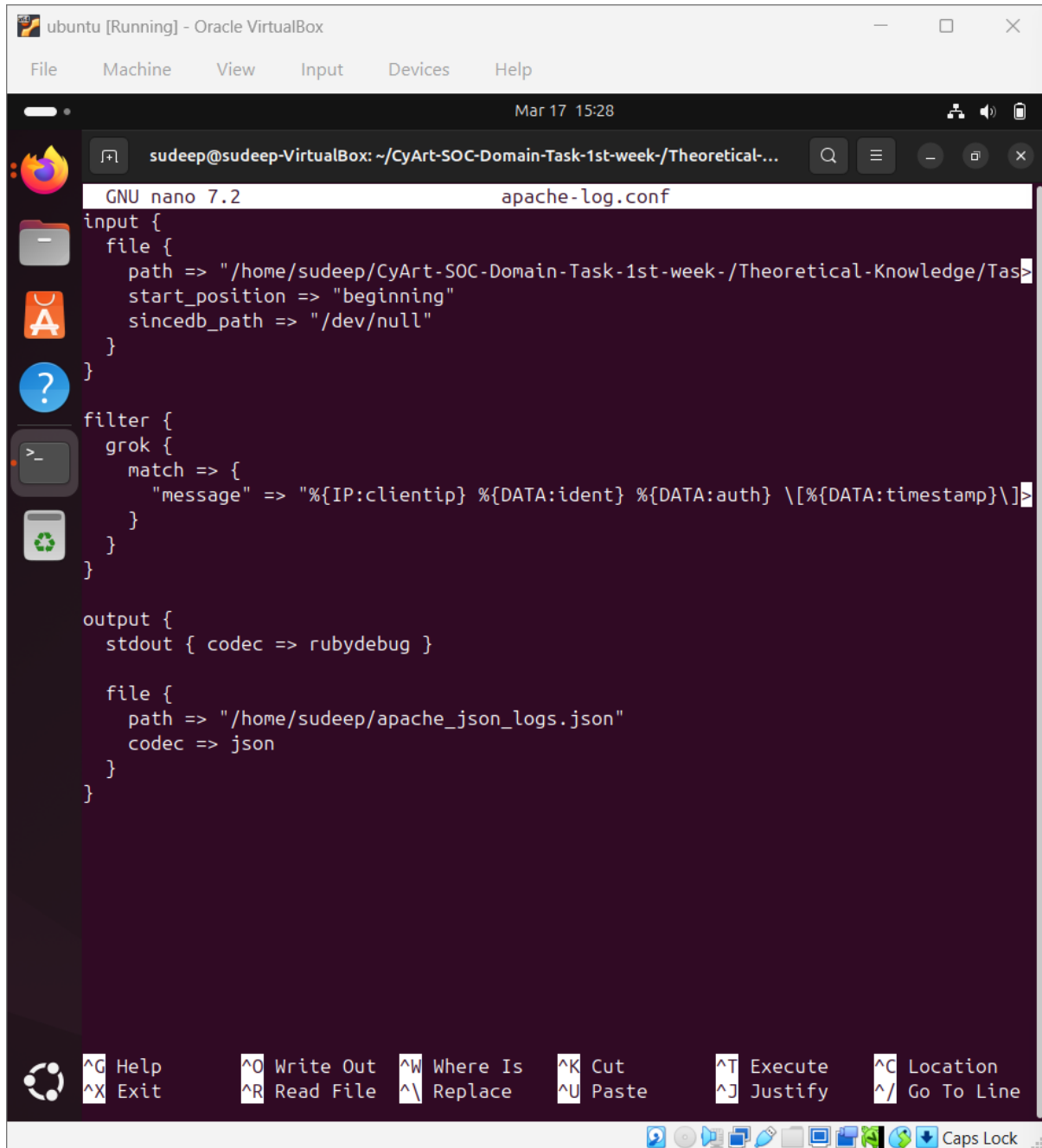
Figure 20 : Apache log file used as input



12.4 Logstash Configuration

Created config file:

nano apache-log.conf



```
GNU nano 7.2 apache-log.conf
input {
  file {
    path => "/home/sudeep/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Tas
    start_position => "beginning"
    sincedb_path => "/dev/null"
  }
}

filter {
  grok {
    match => {
      "message" => "%{IP:clientip} %{DATA:ident} %{DATA:auth} \[%{DATA:timestamp}\]>
    }
  }
}

output {
  stdout { codec => rubydebug }

  file {
    path => "/home/sudeep/apache_json_logs.json"
    codec => json
  }
}
```

Figure 21: Logstash configuration for Apache log normalization



12.5 Execution

Run Logstash:

```
sudo /usr/share/logstash/bin/logstash -f apache-log.conf
```

The screenshot shows a terminal window titled 'ubuntu [Running] - Oracle VirtualBox'. The terminal output for the command `sudo /usr/share/logstash/bin/logstash -f /home/sudeep/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-Log-Management-Fundamentals/apache-log.conf` is as follows:

```
k3-Log-Management-Fundamentals$ sudo /usr/share/logstash/bin/logstash -f /home/sudeep/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-Log-Management-Fundamentals/apache-log.conf
Using bundled JDK: /usr/share/logstash/jdk
WARNING: Could not find logstash.yml which is typically located in $LS_HOME/config or /etc/logstash. You can specify the path using --path.settings. Continuing using the defaults
Could not find log4j2 configuration at path /usr/share/logstash/config/log4j2.properties. Using default config which logs errors to the console
[WARN ] 2026-03-17 15:22:16.459 [main] runner - Starting from version 9.0, running with superuser privileges is not permitted unless you explicitly set 'allow_superuser' to true, thereby acknowledging the possible security risks
[WARN ] 2026-03-17 15:22:16.469 [main] runner - NOTICE: Running Logstash as a superuser is strongly discouraged as it poses a security risk. Set 'allow_superuser' to false for better security.
[WARN ] 2026-03-17 15:22:16.481 [main] runner - 'pipeline.buffer.type' setting is not explicitly defined. Before moving to 9.x set it to 'heap' and tune heap size upward, or set it to 'direct' to maintain existing behavior.
[INFO ] 2026-03-17 15:22:16.485 [main] runner - Starting Logstash {"logstash.version"=>"8.19.12", "jruby.version"=>"jruby 9.4.9.0 (3.1.4) 2024-11-04 547c6b150e OpenJDK 64-Bit Server VM 21.0.9+10-LTS on 21.0.9+10-LTS +indy +jit [x86_64-linux]}"
[INFO ] 2026-03-17 15:22:16.490 [main] runner - JVM bootstrap flags: [-Xms1g, -Xmx1g, -Djava.awt.headless=true, -Dfile.encoding=UTF-8, -Djruby.compile.invokedynamic=true, -XX:+HeapDumpOnOutOfMemoryError, -Djava.security.egd=file:/dev/urandom, -Dlog4j2.isThreadContextMapInheritable=true, -Djruby.regex.interruptible=true, -Djdk.io.File.enableADS=true, --add-exports=jdk.compiler/com.sun.tools.javac.api=ALL-UNNAMED, --add-exports=jdk.compiler/com.sun.tools.javac.file=ALL-UNNAMED, --add-exports=jdk.compiler/com.sun.tools.javac.parser=ALL-UNNAMED, --add-exports=jdk.compiler/com.sun.tools.javac.tree=ALL-UNNAMED, --add-exports=jdk.compiler/com.sun.tools.javac.util=ALL-UNNAMED, --add-opens=java.base/java.security=ALL-UNNAMED, --add-opens=java.base/java.io=ALL-UNNAMED, --add-opens=java.base/java.nio.channels=ALL-UNNAMED, --add-opens=java.base/sun.nio.ch=ALL-UNNAMED, --add-opens=java.management/sun.management=ALL-UNNAMED, -Dio.netty allocator.maxOrder=11]
[INFO ] 2026-03-17 15:22:16.705 [main] StreamReadConstraintsUtil - Jackson default value override `logstash.jackson.stream-read-constraints.max-string-length` configured to `2000000000` (logstash default)
[INFO ] 2026-03-17 15:22:16.706 [main] StreamReadConstraintsUtil - Jackson default value override `logstash.jackson.stream-read-constraints.max-number-length` configured to `2000000000` (logstash default)
```

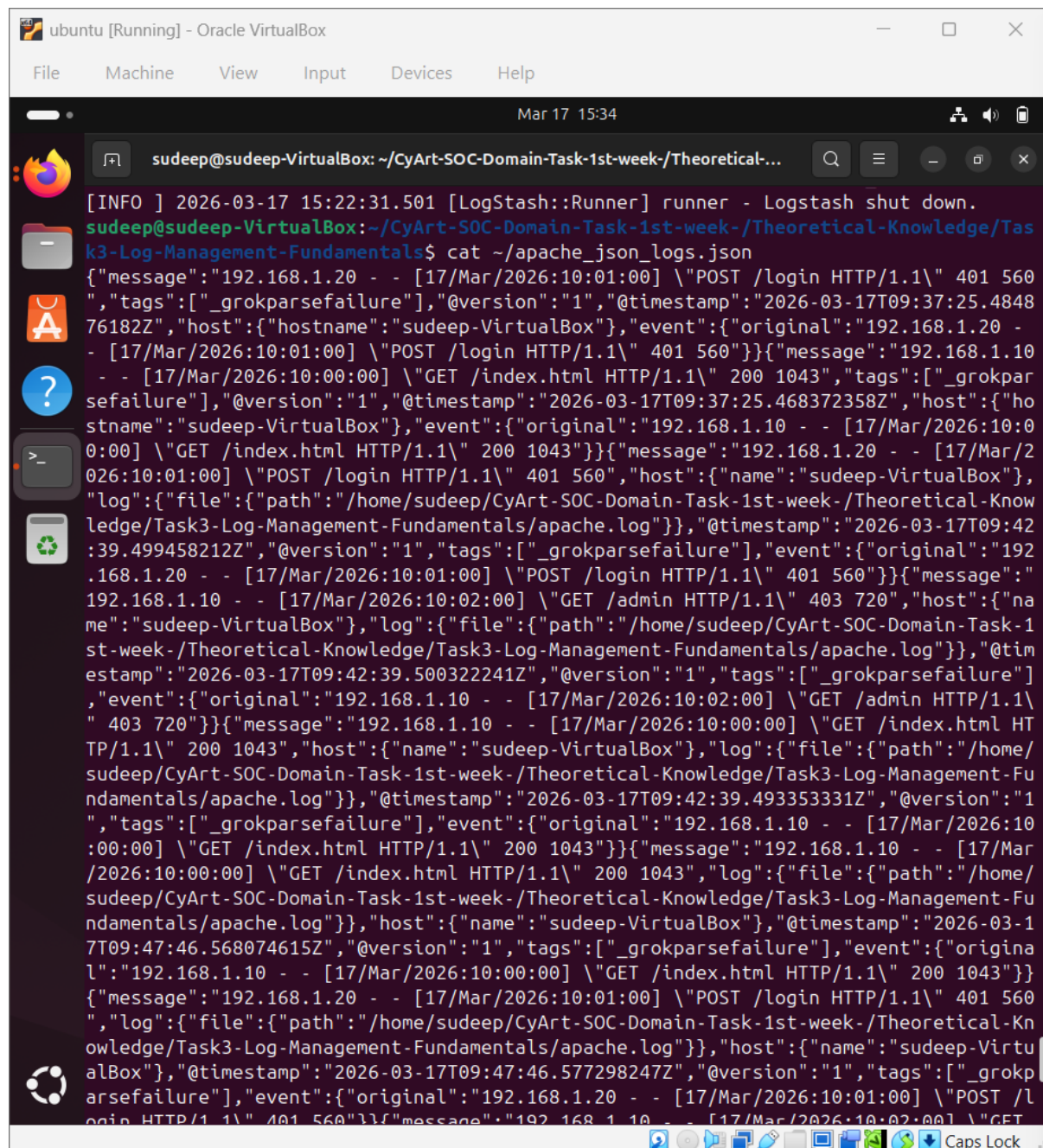
Figure 22: Logstash processing Apache logs



12.6 Output Verification

Check output:

```
cat ~/apache_json_logs.json
```



```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Mar 17 15:34
sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-...
[INFO ] 2026-03-17 15:22:31.501 [LogStash::Runner] runner - Logstash shut down.
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Tas
k3-Log-Management-Fundamentals$ cat ~/apache_json_logs.json
{"message":"192.168.1.20 - - [17/Mar/2026:10:01:00] \"POST /login HTTP/1.1\" 401 560
\", \"tags\": [\"_grokparsefailure\"], \"@version\": \"1\", \"@timestamp\": \"2026-03-17T09:37:25.4848
76182Z\", \"host\": {\"hostname\": \"sudeep-VirtualBox\"}, \"event\": {\"original\": \"192.168.1.20 -
- [17/Mar/2026:10:01:00] \"POST /login HTTP/1.1\" 401 560\"}} {\"message\": \"192.168.1.10
- - [17/Mar/2026:10:00:00] \"GET /index.html HTTP/1.1\" 200 1043\", \"tags\": [\"_grokpar
sefailure\"], \"@version\": \"1\", \"@timestamp\": \"2026-03-17T09:37:25.468372358Z\", \"host\": {\"ho
stname\": \"sudeep-VirtualBox\"}, \"event\": {\"original\": \"192.168.1.10 - - [17/Mar/2026:10:0
0:00] \"GET /index.html HTTP/1.1\" 200 1043\"}} {\"message\": \"192.168.1.20 - - [17/Mar/20
26:10:01:00] \"POST /login HTTP/1.1\" 401 560\", \"host\": {\"name\": \"sudeep-VirtualBox\"},
\"log\": {\"file\": {\"path\": \"/home/sudeep/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Know
ledge/Task3-Log-Management-Fundamentals/apache.log\"}}, \"@timestamp\": \"2026-03-17T09:42
:39.499458212Z\", \"@version\": \"1\", \"tags\": [\"_grokparsefailure\"], \"event\": {\"original\": \"192
.168.1.20 - - [17/Mar/2026:10:01:00] \"POST /login HTTP/1.1\" 401 560\"}} {\"message\": \"
192.168.1.10 - - [17/Mar/2026:10:02:00] \"GET /admin HTTP/1.1\" 403 720\", \"host\": {\"na
me\": \"sudeep-VirtualBox\"}, \"log\": {\"file\": {\"path\": \"/home/sudeep/CyArt-SOC-Domain-Task-1
st-week-/Theoretical-Knowledge/Task3-Log-Management-Fundamentals/apache.log\"}}, \"@tim
estamp\": \"2026-03-17T09:42:39.500322241Z\", \"@version\": \"1\", \"tags\": [\"_grokparsefailure\"
], \"event\": {\"original\": \"192.168.1.10 - - [17/Mar/2026:10:02:00] \"GET /admin HTTP/1.1\"
403 720\"}} {\"message\": \"192.168.1.10 - - [17/Mar/2026:10:00:00] \"GET /index.html HT
TP/1.1\" 200 1043\", \"host\": {\"name\": \"sudeep-VirtualBox\"}, \"log\": {\"file\": {\"path\": \"/home/
sudeep/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-Log-Management-Fu
ndamentals/apache.log\"}}, \"@timestamp\": \"2026-03-17T09:42:39.493353331Z\", \"@version\": \"1
\", \"tags\": [\"_grokparsefailure\"], \"event\": {\"original\": \"192.168.1.10 - - [17/Mar/2026:10
:00:00] \"GET /index.html HTTP/1.1\" 200 1043\"}} {\"message\": \"192.168.1.10 - - [17/Mar/20
26:10:00:00] \"GET /index.html HTTP/1.1\" 200 1043\", \"log\": {\"file\": {\"path\": \"/home/
sudeep/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Task3-Log-Management-Fu
ndamentals/apache.log\"}}, \"host\": {\"name\": \"sudeep-VirtualBox\"}, \"@timestamp\": \"2026-03-1
7T09:47:46.568074615Z\", \"@version\": \"1\", \"tags\": [\"_grokparsefailure\"], \"event\": {\"origina
l\": \"192.168.1.10 - - [17/Mar/2026:10:00:00] \"GET /index.html HTTP/1.1\" 200 1043\"}}
{\"message\": \"192.168.1.20 - - [17/Mar/2026:10:01:00] \"POST /login HTTP/1.1\" 401 560
\", \"log\": {\"file\": {\"path\": \"/home/sudeep/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Kn
owledge/Task3-Log-Management-Fundamentals/apache.log\"}}, \"host\": {\"name\": \"sudeep-Virtu
alBox\"}, \"@timestamp\": \"2026-03-17T09:47:46.577298247Z\", \"@version\": \"1\", \"tags\": [\"_grokp
arsefailure\"], \"event\": {\"original\": \"192.168.1.20 - - [17/Mar/2026:10:01:00] \"POST /l
ogin HTTP/1.1\" 401 560\"}} {\"message\": \"192.168.1.10 - - [17/Mar/2026:10:02:00] \"GET
/\"
```

Figure 23: Normalized JSON output generated by Logstash



12.7. SOC Workflow

1. Logs are generated from systems (Apache server)
2. Logstash collects logs
3. Grok filter parses unstructured logs
4. Logs are converted into structured JSON format
5. Normalized logs are stored for analysis in SIEM tools

12.8 Challenges Faced

- Grok parsing failure due to incorrect patterns
- Incorrect file path configuration
- Logstash stdin confusion during testing

12.9. Conclusion

In this lab, log normalization was successfully implemented using Logstash.

Apache logs were converted into structured JSON format, enabling efficient log analysis.

This process is essential in SOC environments for detecting threats and performing

TASK 4. Security Tools Overview

Introduction

Security Operations Centers (SOCs) rely on various security tools to monitor, detect, and respond to cyber threats. These tools work together to provide visibility across networks, endpoints, and systems.

1. SIEM (Security Information and Event Management)



SIEM tools collect and analyze log data from multiple sources in real time.

Key Functions:

- Log collection and aggregation
- Real-time alerting
- Event correlation
- Incident investigation

Examples:

- Splunk
- IBM QRadar

SOC Use:

SOC analysts use SIEM to:

- Detect suspicious activity
- Investigate incidents
- Correlate logs from different sources

2. EDR (Endpoint Detection and Response)

EDR tools monitor endpoint activities such as processes, file changes, and system behavior.

Key Functions:

- Detect malware and suspicious processes
- Monitor endpoint behavior
- Provide response capabilities

Example:

- CrowdStrike Falcon

SOC Use:

- Detect advanced threats
- Perform endpoint investigation
- Isolate compromised systems

3. IDS/IPS (Intrusion Detection/Prevention Systems)

IDS/IPS tools monitor network traffic to detect or block malicious activity.

Key Functions:

- Packet inspection
- Rule-based detection



- Alert generation

Example:

- Snort

SOC Use:

- Detect network-based attacks
- Identify suspicious traffic patterns
- Generate alerts for investigation

4. Vulnerability Scanners

Vulnerability scanners identify security weaknesses in systems and applications.

Key Functions:

- Scan for known vulnerabilities
- Assign CVSS scores
- Provide remediation suggestions

Example:

- Nessus

SOC Use:

- Identify weak systems
- Prioritize patching
- Reduce attack surface

5. Endpoint Monitoring Tools

Endpoint monitoring tools allow querying and monitoring system-level data.

Key Functions:

- Query system processes
- Monitor system state
- Detect suspicious activity

Example:

- Osquery

SOC Use:

- Investigate endpoints
- Detect unauthorized processes
- Perform threat hunting



6.TASK A: SNORT RULE TESTING (Ubuntu)

1.Introduction:

Snort is an **Intrusion Detection System (IDS)**.

Snort is an open-source Intrusion Detection System (IDS) used to monitor network traffic and detect suspicious or malicious activities. It uses rule-based detection to analyze packets and generate alerts. In a SOC environment, Snort helps analysts identify network-based threats in real time.

SOC use-case:

- Detect malicious traffic
- Alert on suspicious activity
- Write custom rules (like detecting malicious domains)

2. Tools Required

- Ubuntu VM
- Snort
- curl

3.Installation on ubuntu

```
sudo apt update
```

```
sudo apt install snort -y
```




```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Mar 18 00:44
sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Theory-Task4-Security-tools-overview$ sudo apt update
sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Theory-Task4-Security-tools-overview$ sudo apt install snort -y
[sudo] password for sudeep:
Hit:1 http://in.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:4 https://artifacts.elastic.co/packages/8.x/apt stable InRelease
Hit:5 https://packages.treasuredata.com/5/ubuntu/jammy jammy InRelease
Get:6 http://in.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [178 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,540 kB]
Get:9 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.7 kB]
Get:10 http://in.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:11 http://in.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:12 http://in.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [33.0 kB]
Get:13 http://in.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:14 http://in.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,344 B]
Get:15 http://in.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:16 http://in.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [13.2 kB]
```

Figure 1:Snort installation on ubuntu

Check version:

snort -V



```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Mar 18 00:47
sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-...
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Theory-Task4-Security-tools-overview$ snort -V
,,_      -*> Snort! <*-
o" )~    Version 2.9.20 GRE (Build 82)
' '      By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
          Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
          Copyright (C) 1998-2013 Sourcefire, Inc., et al.
          Using libpcap version 1.10.4 (with TPACKET_V3)
          Using PCRE version: 8.39 2016-06-14
          Using ZLIB version: 1.3

sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Theory-Task4-Security-tools-overview$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:fa:d9:d5 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 85608sec preferred_lft 85608sec
    inet6 fd17:625c:f037:2:d9ba:40b9:4346:69e3/64 scope global temporary dynamic
        valid_lft 86069sec preferred_lft 14069sec
    inet6 fd17:625c:f037:2:a00:27ff:fefa:d9d5/64 scope global dynamic mngtmpaddr
        valid_lft 86069sec preferred_lft 14069sec
    inet6 fe80::a00:27ff:fefa:d9d5/64 scope link
        valid_lft forever preferred_lft forever
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Theory-Task4-Security-tools-overview$
```

Figure 2 : snort version checking and IP checking

4. Configure Snort

Find config file:

```
sudo nano /etc/snort/snort.conf
```

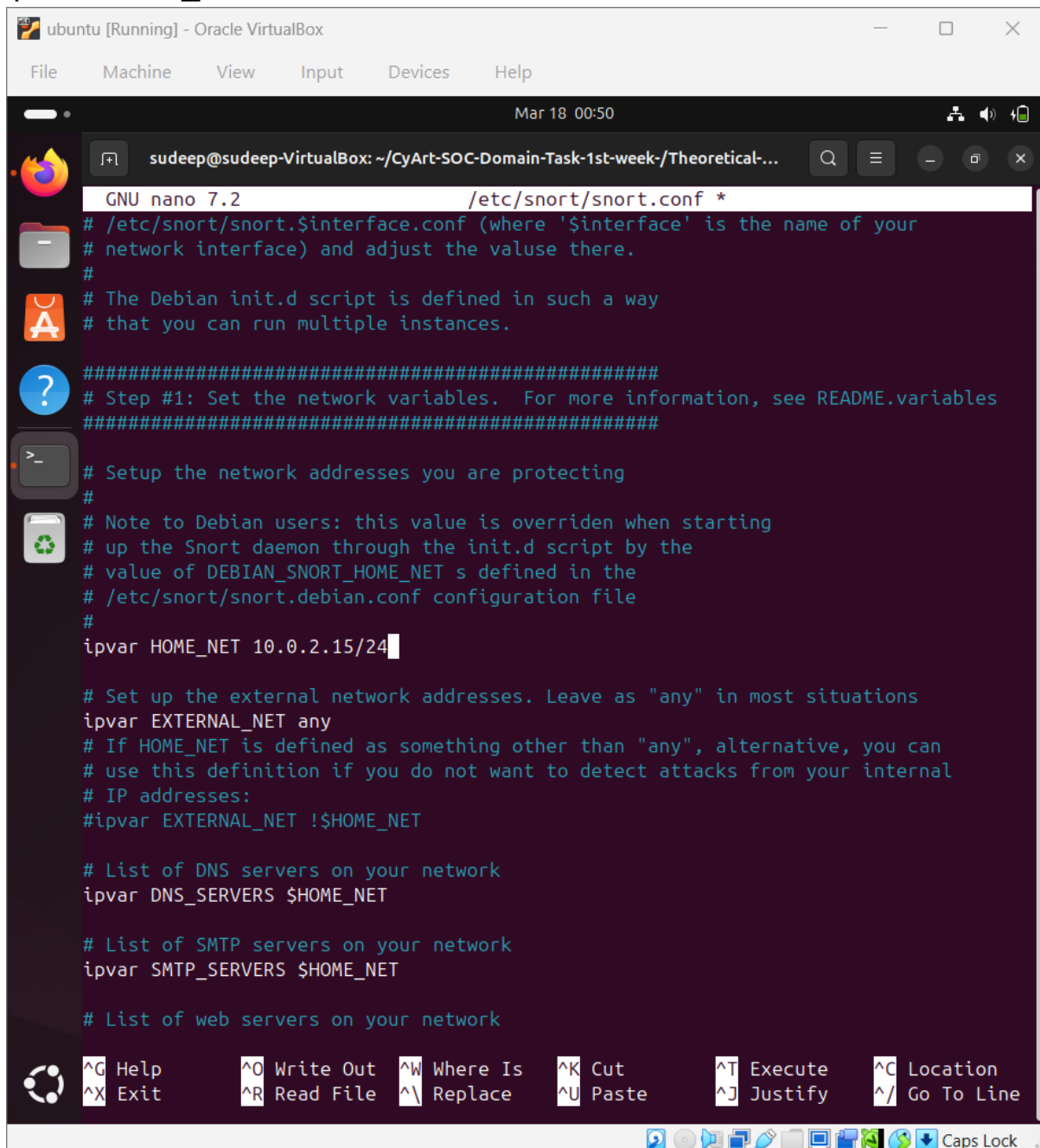
Find:

```
ipvar HOME_NET any
```

Change to:



ipvar HOME_NET 10.0.2.15/24



```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Mar 18 00:50
sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-...
GNU nano 7.2 /etc/snort/snort.conf *
# /etc/snort/snort.$interface.conf (where '$interface' is the name of your
# network interface) and adjust the value there.
#
# The Debian init.d script is defined in such a way
# that you can run multiple instances.
#####
# Step #1: Set the network variables. For more information, see README.variables
#####
# Setup the network addresses you are protecting
#
# Note to Debian users: this value is overridden when starting
# up the Snort daemon through the init.d script by the
# value of DEBIAN_SNORT_HOME_NET s defined in the
# /etc/snort/snort.debian.conf configuration file
#
ipvar HOME_NET 10.0.2.15/24

# Set up the external network addresses. Leave as "any" in most situations
ipvar EXTERNAL_NET any
# If HOME_NET is defined as something other than "any", alternative, you can
# use this definition if you do not want to detect attacks from your internal
# IP addresses:
#ipvar EXTERNAL_NET !$HOME_NET

# List of DNS servers on your network
ipvar DNS_SERVERS $HOME_NET

# List of SMTP servers on your network
ipvar SMTP_SERVERS $HOME_NET

# List of web servers on your network

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line
Caps Lock
```

Figure 3: configuring snort with IP address

5. 🛠️ Add Rule

Open local rules:

`sudo nano /etc/snort/rules/local.rules`



alert tcp any any -> any 80 (msg:"Malicious Domain";
content:"malicious.com"; http_uri; sid:1000001;)

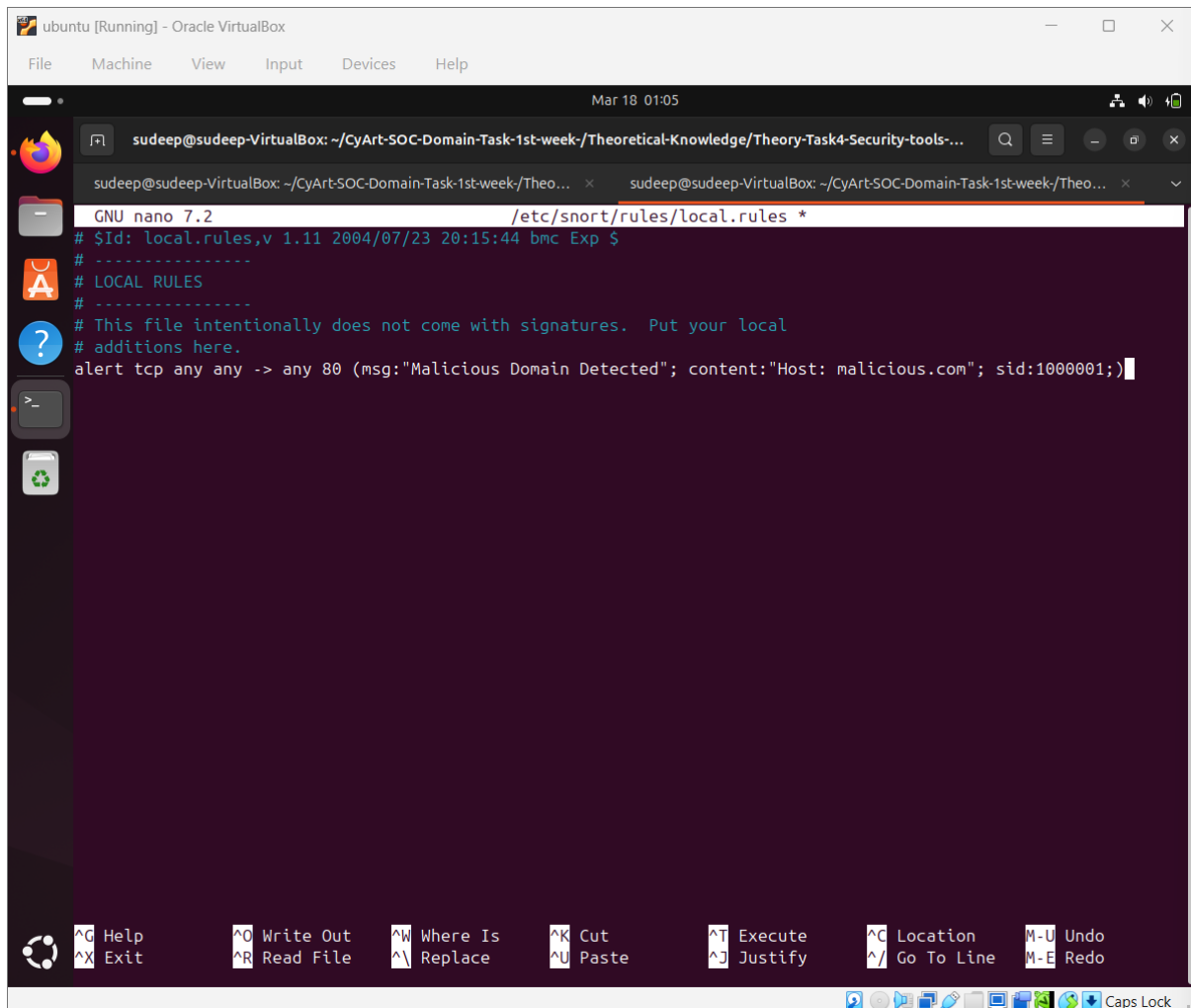


Figure 4: Adding rules on snort

6. Run Snort

sudo snort -A console -q -c /etc/snort/snort.conf -i eth0



```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Mar 18 01:06

sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-Task-1st-week/Theoretical-Knowledge/Theory-Task4-Security-tools-...
sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-Task-1st-week/Theo... x sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-Task-1st-week/Theo... x v

Preprocessor Object: SF_SMTP Version 1.1 <Build 9>
Preprocessor Object: SF_POP Version 1.0 <Build 1>
Preprocessor Object: SF_SIP Version 1.1 <Build 1>
Preprocessor Object: SF_S7COMPLUS Version 1.0 <Build 1>
Preprocessor Object: SF_DNS Version 1.1 <Build 4>
Preprocessor Object: SF_SSH Version 1.1 <Build 3>
Preprocessor Object: SF_DNP3 Version 1.1 <Build 1>
Preprocessor Object: SF_DCERPC2 Version 1.0 <Build 3>
Preprocessor Object: SF_IMAP Version 1.0 <Build 1>
Preprocessor Object: appid Version 1.1 <Build 5>
Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>
Preprocessor Object: SF_SDF Version 1.1 <Build 1>

Total snort Fixed Memory Cost - MaxRss:104676
Snort successfully validated the configuration!
Snort exiting
sudeep@sudeep-VirtualBox:~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Theory-Task4-Security-tools-ove
rview$ sudo snort -A console -c /etc/snort/snort.conf -i enp0s3
Running in IDS mode

--== Initializing Snort ==--
Initializing Output Plugins!
Initializing Preprocessors!
Initializing Plug-ins!
Parsing Rules file "/etc/snort/snort.conf"
PortVar 'HTTP_PORTS' defined : [ 80:81 311 383 591 593 901 1220 1414 1741 1830 2301 2381 2809 3037 3128 3702 43
43 4848 5250 6988 7000:7001 7144:7145 7510 7777 7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181
8243 8280 8300 8800 8888 8899 9000 9060 9080 9090:9091 9443 9999 11371 34443:34444 41080 50002 55555 ]
PortVar 'SHELLCODE_PORTS' defined : [ 0:79 81:65535 ]
PortVar 'ORACLE_PORTS' defined : [ 1024:65535 ]
PortVar 'SSH_PORTS' defined : [ 22 ]
PortVar 'FTP_PORTS' defined : [ 21 2100 3535 ]
PortVar 'SIP_PORTS' defined : [ 5060:5061 5600 ]
PortVar 'FILE_DATA_PORTS' defined : [ 80:81 110 143 311 383 591 593 901 1220 1414 1741 1830 2301 2381 2809 3037
3128 3702 4343 4848 5250 6988 7000:7001 7144:7145 7510 7777 7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8
```

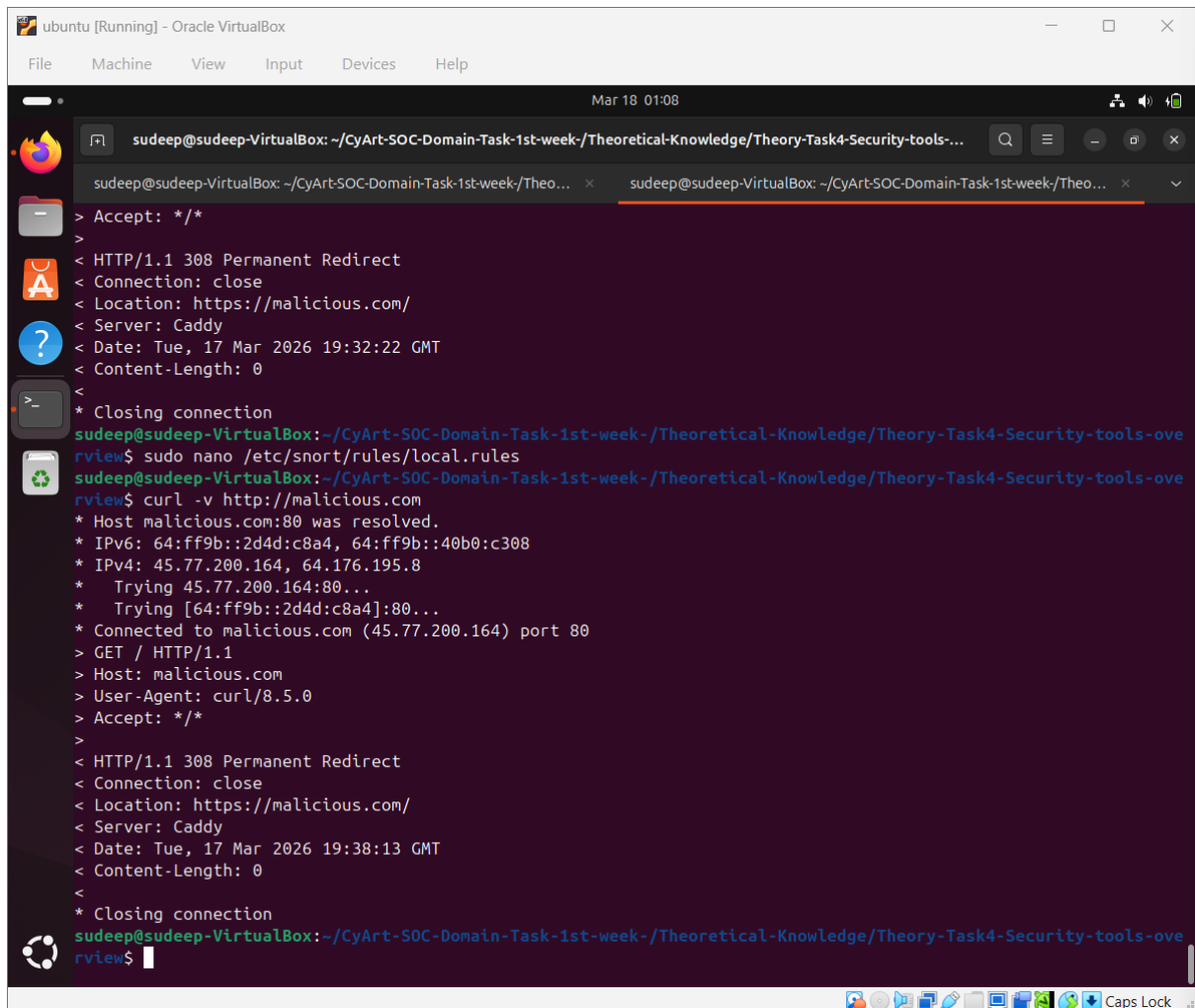
Figure 5 : Running snort

7. Test Attack



curl -v <http://malicious.com>

This command generates HTTP traffic to test the Snort rule.



```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Mar 18 01:08
sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-Task-1st-week/Theoretical-Knowledge/Theory-Task4-Security-tools-...
sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-Task-1st-week/Theo...
> Accept: */*
>
< HTTP/1.1 308 Permanent Redirect
< Connection: close
< Location: https://malicious.com/
< Server: Caddy
< Date: Tue, 17 Mar 2026 19:32:22 GMT
< Content-Length: 0
<
* Closing connection
sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Theory-Task4-Security-tools-ove
rview$ sudo nano /etc/snort/rules/local.rules
sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Theory-Task4-Security-tools-ove
rview$ curl -v http://malicious.com
* Host malicious.com:80 was resolved.
* IPv6: 64:ff9b::2d4d:c8a4, 64:ff9b::40b0:c308
* IPv4: 45.77.200.164, 64.176.195.8
* Trying 45.77.200.164:80...
* Trying [64:ff9b::2d4d:c8a4]:80...
* Connected to malicious.com (45.77.200.164) port 80
> GET / HTTP/1.1
> Host: malicious.com
> User-Agent: curl/8.5.0
> Accept: */*
>
< HTTP/1.1 308 Permanent Redirect
< Connection: close
< Location: https://malicious.com/
< Server: Caddy
< Date: Tue, 17 Mar 2026 19:38:13 GMT
< Content-Length: 0
<
* Closing connection
sudeep@sudeep-VirtualBox: ~/CyArt-SOC-Domain-Task-1st-week-/Theoretical-Knowledge/Theory-Task4-Security-tools-ove
rview$
```

Figure 6: Test attack

Snort successfully generated an alert indicating detection of the malicious domain

8.SOC Workflow Insight

1. Monitor network traffic
2. Apply detection rules
3. Identify suspicious activity
4. Generate alerts for investigation



9.Challenges Faced

- Incorrect network interface (eth0 vs enp0s3)
- Rule not triggering due to HTTP parsing issue

10.Learning Outcome

- Understanding IDS functionality
- Writing and testing Snort rules
- Troubleshooting network monitoring issues

11. Conclusion

This lab demonstrated how Snort IDS can be used to detect malicious network activity using custom rules. It highlights the importance of proper configuration and rule tuning in real-world SOC environments.



7.TASK B: NESSUS SCAN

1.Introduction

Nessus is a widely used vulnerability scanning tool that helps identify security weaknesses in systems. It assigns CVSS scores to vulnerabilities, allowing prioritization based on severity. In SOC environments, Nessus helps analysts detect misconfigurations and vulnerabilities before attackers exploit them.

2.Tools Used

- Kali Linux
- Nessus Essentials
- Metasploitable2

3.Lab Setup

- Both Kali and Metasploitable were connected to the same network
- Connectivity verified using ping

```
Metasploit [Running] - Oracle VirtualBox
File Machine View Input Devices Help

No mail.
msfadmin@metasploitable:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 08:00:27:09:6b:95 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global eth0
    inet6 fd17:625c:f037:2:a00:27ff:fe09:6b95/64 scope global dynamic
        valid_lft 86358sec preferred_lft 14358sec
    inet6 fe80::a00:27ff:fe09:6b95/64 scope link
        valid_lft forever preferred_lft forever
msfadmin@metasploitable:~$ ping 10.0.2.15
PING 10.0.2.15 (10.0.2.15) 56(84) bytes of data.
64 bytes from 10.0.2.15: icmp_seq=1 ttl=64 time=0.034 ms
64 bytes from 10.0.2.15: icmp_seq=2 ttl=64 time=0.064 ms
64 bytes from 10.0.2.15: icmp_seq=3 ttl=64 time=0.028 ms
64 bytes from 10.0.2.15: icmp_seq=4 ttl=64 time=0.031 ms

--- 10.0.2.15 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3001ms
rtt min/avg/max/mdev = 0.028/0.039/0.064/0.015 ms
msfadmin@metasploitable:~$
```



Figure 1:Metasploite lab setup

```
kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
Session Actions Edit View Help

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:1f:b7:23 brd ff:ff:ff:ff:ff:ff
   inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 86001sec preferred_lft 86001sec
   inet6 fd17:625c:f037:2:d996:50fd:f2b3:2965/64 scope global dynamic noprefixroute
       valid_lft 86002sec preferred_lft 14002sec
   inet6 fe80::4253:7840:ce77:5376/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: br-a3c443015808: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
   link/ether 02:42:53:3f:87:ea brd ff:ff:ff:ff:ff:ff
   inet 172.18.0.1/16 brd 172.18.255.255 scope global br-a3c443015808
       valid_lft forever preferred_lft forever
   inet6 fe80::42:53ff:fe3f:87ea/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
4: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
   link/ether 02:42:25:d5:8e:d4 brd ff:ff:ff:ff:ff:ff
   inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
6: vethb5e1a22a1f5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-a3c443015808 state UP group
   default
   link/ether 0a:69:34:87:56:a8 brd ff:ff:ff:ff:ff:ff link-netnsid 1
   inet6 fe80::869:34ff:fe87:56a8/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
8: vethb0f7b22a1f7: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-a3c443015808 state UP group
   default
   link/ether 6a:da:00:df:f8:a9 brd ff:ff:ff:ff:ff:ff link-netnsid 0
   inet6 fe80::68da:ff:fedf:f8a9/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever

(kali@kali)-[~]
$ ping 10.0.2.15
PING 10.0.2.15 (10.0.2.15) 56(84) bytes of data.
64 bytes from 10.0.2.15: icmp_seq=1 ttl=64 time=0.079 ms
64 bytes from 10.0.2.15: icmp_seq=2 ttl=64 time=0.042 ms
64 bytes from 10.0.2.15: icmp_seq=3 ttl=64 time=0.061 ms
64 bytes from 10.0.2.15: icmp_seq=4 ttl=64 time=0.037 ms
64 bytes from 10.0.2.15: icmp_seq=5 ttl=64 time=0.058 ms
^C
-- 10.0.2.15 ping statistics --
5 packets transmitted, 5 received, 0% packet loss, time 4087ms
rtt min/avg/max/mdev = 0.037/0.055/0.079/0.014 ms

(kali@kali)-[~]
$
```

Figure 2 : Kali linux lab setup



4.Installation Nessus

```
kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~/Downloads

Session Actions Edit View Help

cert Mac-spoofing-nmap-readme.pdf task01_network_diagram..png
'Cyber task 9.pdf' 'microsoft - Events.csv' 'TASK01 SUB.pdf'
document1.pdf 'microsoft - Results.csv' 'Udemy EH PT.pdf'
document2.pdf Nessus-10.11.3-ubuntu1604_amd64.deb

(kali@kali)-[~/Downloads]
$ sudo dpkg -i Nessus-10.11.3-ubuntu1604_amd64.deb
(Reading database ... 597617 files and directories currently installed.)
Preparing to unpack Nessus-10.11.3-ubuntu1604_amd64.deb ...
Unpacking nessus (10.11.3) over (10.10.1) ...
Setting up nessus (10.11.3) ...
HMAC : (Module_Integrity) : Pass
SHA1 : (KAT_Digest) : Pass
SHA2 : (KAT_Digest) : Pass
SHA3 : (KAT_Digest) : Pass
TDES : (KAT_Cipher) : Pass
AES_GCM : (KAT_Cipher) : Pass
AES_ECB_Decrypt : (KAT_Cipher) : Pass
RSA : (KAT_Signature) : RNG : (Continuous_RNG_Test) : Pass
Pass
ECDSA : (PCT_Signature) : Pass
ECDSA : (PCT_Signature) : Pass
DSA : (PCT_Signature) : Pass
TLS13_KDF_EXTRACT : (KAT_KDF) : Pass
TLS13_KDF_EXPAND : (KAT_KDF) : Pass
TLS12_PRF : (KAT_KDF) : Pass
PBKDF2 : (KAT_KDF) : Pass
SSHKDF : (KAT_KDF) : Pass
KDKDF : (KAT_KDF) : Pass
HKDF : (KAT_KDF) : Pass
SSKDF : (KAT_KDF) : Pass
X963KDF : (KAT_KDF) : Pass
X942KDF : (KAT_KDF) : Pass
HASH : (DRBG) : Pass
CTR : (DRBG) : Pass
HMAC : (DRBG) : Pass
DH : (KAT_KA) : Pass
ECDH : (KAT_KA) : Pass
RSA_Encrypt : (KAT_AsymmetricCipher) : Pass
RSA_Decrypt : (KAT_AsymmetricCipher) : Pass
RSA_Decrypt : (KAT_AsymmetricCipher) : Pass
INSTALL PASSED
Unpacking Nessus Scanner Core Components...

- You can start Nessus Scanner by typing /bin/systemctl start nessusd.service
- Then go to https://NESSUS_HOSTNAME_OR_IP:8834/ to configure your scanner

(kali@kali)-[~/Downloads]
$
```

Figure 4 : Nessus Installation



5.Service Configuration

Command:

```
sudo systemctl start nessusd
```

```
kali@kali: ~/Downloads
Session Actions Edit View Help
ECDSA : (PCT_Signature) : Pass
ECDSA : (PCT_Signature) : Pass
DSA : (PCT_Signature) : Pass
TLS13_KDF_EXTRACT : (KAT_KDF) : Pass
TLS13_KDF_EXPAND : (KAT_KDF) : Pass
TLS12_PRF : (KAT_KDF) : Pass
PBKDF2 : (KAT_KDF) : Pass
SSHKDF : (KAT_KDF) : Pass
KBKDF : (KAT_KDF) : Pass
HKDF : (KAT_KDF) : Pass
SSKDF : (KAT_KDF) : Pass
X963KDF : (KAT_KDF) : Pass
X942KDF : (KAT_KDF) : Pass
HASH : (DRBG) : Pass
CTR : (DRBG) : Pass
HMAC : (DRBG) : Pass
DH : (KAT_KA) : Pass
ECDH : (KAT_KA) : Pass
RSA_Encrypt : (KAT_AsymmetricCipher) : Pass
RSA_Decrypt : (KAT_AsymmetricCipher) : Pass
RSA_Decrypt : (KAT_AsymmetricCipher) : Pass
INSTALL PASSED
Unpacking Nessus Scanner Core Components...

- You can start Nessus Scanner by typing /bin/systemctl start nessusd.service
- Then go to https://NESSUS_HOSTNAME_OR_IP:8834/ to configure your scanner

(kali@kali)-[~/Downloads]
$ sudo systemctl start nessusd

(kali@kali)-[~/Downloads]
$ sudo systemctl status nessusd
● nessusd.service - The Nessus Vulnerability Scanner
   Loaded: loaded (/usr/lib/systemd/system/nessusd.service; disabled; preset: disabled)
   Active: active (running) since Wed 2026-03-18 00:49:12 EDT; 5s ago
 Invocation: a746b4790cf64bf5915ec73801753247
   Main PID: 13053 (nessus-service)
    Tasks: 23 (limit: 12706)
  Memory: 244.4M (peak: 244.4M)
     CPU: 5.275s
  CGroup: /system.slice/nessusd.service
          └─13053 /opt/nessus/sbin/nessus-service -q
            └─13055 nessusd -q

Mar 18 00:49:12 kali systemd[1]: Started nessusd.service - The Nessus Vulnerability Scanner.
Mar 18 00:49:12 kali nessus-service[13053]: nessus-service [13053][INFO] : Nessus 19.16.3 [build 20047] Started

(kali@kali)-[~/Downloads]
$
```

Figure 5 : Nessus service enabling



Challenges Faced

- Activation issues
- Network connectivity problems
- Incorrect package format (.rpm vs .deb)

Learning Outcome

- Understanding vulnerability scanning
- Interpreting CVSS scores
- Using Nessus in real-world scenarios

Conclusion

This lab demonstrated how Nessus can be used to identify and assess vulnerabilities in a system, which is a critical task in SOC operations to reduce security risks.



8.Task C: Osquery Monitoring (Endpoint Detection Lab)

1.Introduction

Osquery is an open-source endpoint monitoring tool that allows querying system information using SQL-like syntax. It treats the operating system as a relational database, enabling security analysts to investigate system activity efficiently.

In SOC environments, Osquery is used for:

- Endpoint monitoring
- Threat hunting
- Detecting suspicious processes

2.Tools Used

- Windows Operating System
- Osquery

3.Installation

Osquery was downloaded from the official website and installed using the Windows installer (.msi)

4.Running Osquery

Command:

"C:\Program Files\osquery\osqueryi.exe"

5.Query Execution

Query Used:

```
SELECT * FROM processes;
```

This query lists all running processes in the system.

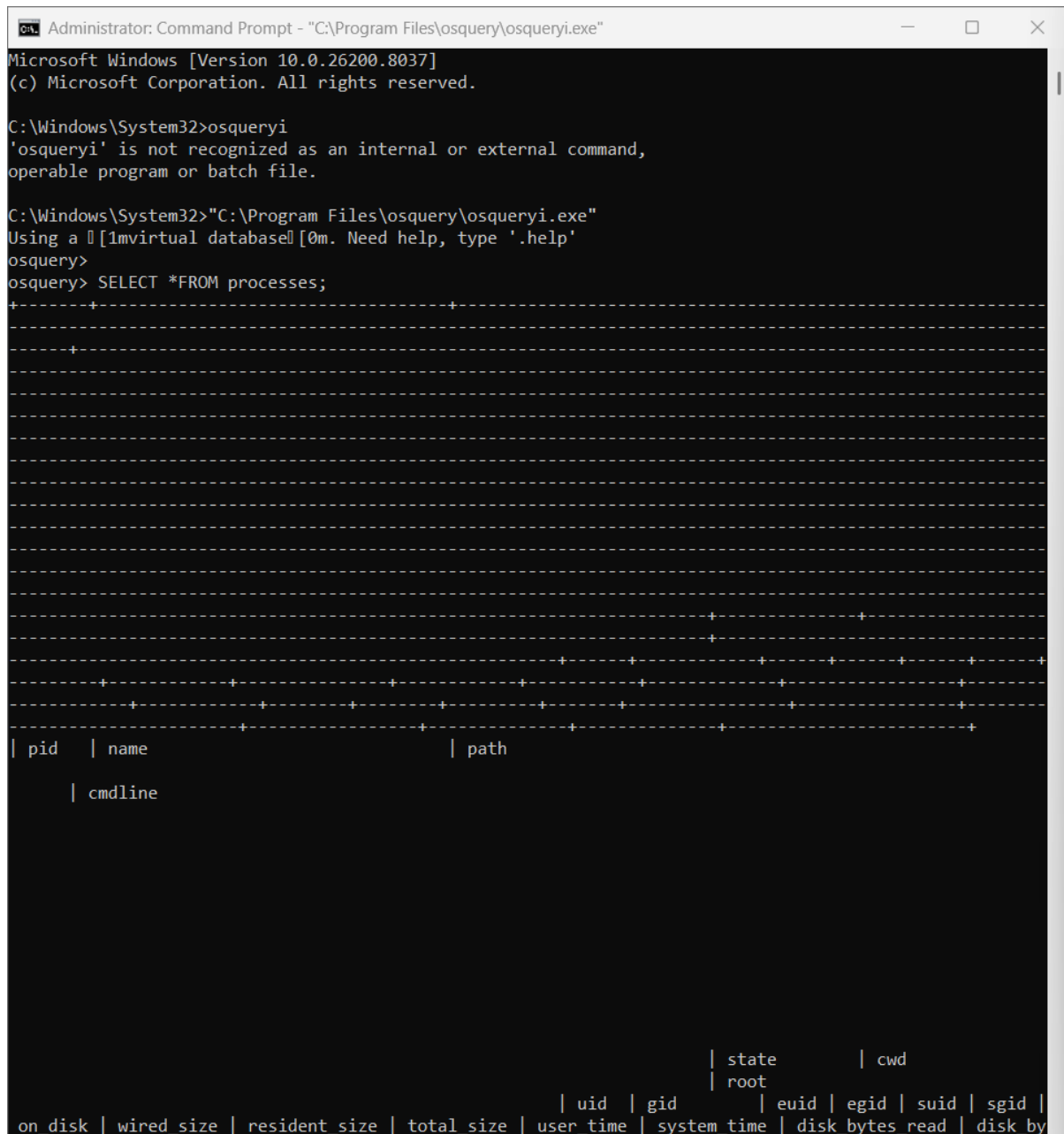


Figure 1 : Query os installation with all process



6.Simulated Malicious Activity

A fake malware process was created using a batch file.

File Name:

fake_malware.bat

Code:

```
@echo off
```

```
ping 127.0.0.1 -t
```

This script continuously runs, simulating a suspicious process.

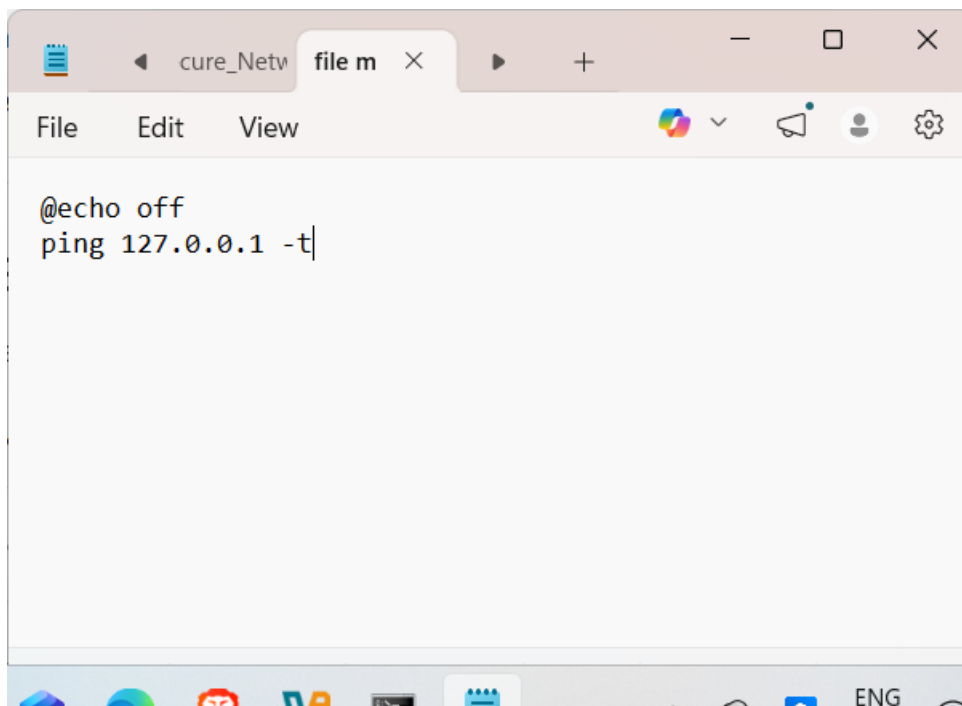


Figure 2 : Fake malware.bat file



This query helps identify running processes and their locations.



```
Administrator: Command Prompt - "C:\Program Files\osquery\osqueryi.exe"
osquery> SELECT name, pid, path FROM processes;
```

name	pid	path
[System Process]	0	
System	4	
Secure System	188	
Registry	232	Registry
smss.exe	724	C:\Windows\System32\smss.exe
csrss.exe	1060	
wininit.exe	1148	C:\Windows\System32\wininit.exe
services.exe	1280	C:\Windows\System32\services.exe
LsaIso.exe	1308	C:\Windows\System32\LsaIso.exe
lsass.exe	1320	C:\Windows\System32\lsass.exe
svchost.exe	1456	C:\Windows\System32\svchost.exe
fontdrvhost.exe	1496	
WUDFHost.exe	1516	

Figure 4 :Detected process



```
Administrator: Command Prompt - "C:\Program Files\osquery\osqueryi.exe"
| svchost.exe | 1816 | C:\Windows\System32\svchost.exe
|
| svchost.exe | 1844 | C:\Windows\System32\svchost.exe
|
| svchost.exe | 1880 | C:\Windows\System32\svchost.exe
|
| svchost.exe | 704 | C:\Windows\System32\svchost.exe
|
| svchost.exe | 696 | C:\Windows\System32\svchost.exe
|
| svchost.exe | 1388 | C:\Windows\System32\svchost.exe
|
| svchost.exe | 2000 | C:\Windows\System32\svchost.exe
|
| svchost.exe | 2096 | C:\Windows\System32\svchost.exe
|
| IntelCpHDCPSvc.exe | 2124 | C:\Windows\System32\DriverStore\FileRepository\iigd_dch.inf_amd64_d4c7d28a7954c5a1\IntelCpHDCPSvc.exe
|
| svchost.exe | 2180 | C:\Windows\System32\svchost.exe
|
| svchost.exe | 2196 | C:\Windows\System32\svchost.exe
|
| WUDFHost.exe | 2220 |
|
| svchost.exe | 2228 | C:\Windows\System32\svchost.exe
|
| svchost.exe | 2284 | C:\Windows\System32\svchost.exe
|
| svchost.exe | 2360 | C:\Windows\System32\svchost.exe
|
| svchost.exe | 2524 | C:\Windows\System32\svchost.exe
|
| svchost.exe | 2648 | C:\Windows\System32\svchost.exe
```

Figure 5 :Detected process

The simulated malicious process (fake_malware.bat) was successfully detected using Osquery.

8.Challenges Faced

- Osquery not recognized in CMD (PATH issue)
- Required full path to execute

9.Learning Outcome

- Understanding endpoint monitoring



- Using SQL queries for investigation
- Detecting suspicious processes

10. Conclusion

This lab demonstrated how Osquery can be used for endpoint monitoring and detection of suspicious processes. It highlights the importance of endpoint visibility in SOC operations.

9. Learning Approach

To gain hands-on experience with these tools:

- Use **free versions** like:
 - Splunk Free
 - Wazuh (open-source XDR)
- Practice in a **sandbox environment**:
 - Configure Snort rules
 - Simulate attacks
 - Perform vulnerability scans

10. Practical Relevance

These tools form the core of modern SOC operations:

Tool Type	Role in SOC
SIEM	Centralized monitoring & alerting
EDR	Endpoint visibility & response
IDS/IPS	Network threat detection
Vulnerability Scanner	Risk assessment
Osquery	Endpoint investigation

11. Conclusion

Understanding and using these tools enables SOC analysts to:

- Detect threats early
- Investigate incidents efficiently
- Strengthen overall security posture

This knowledge is essential for real-world cybersecurity operations.



TASK 5: BASIC SECURITY CONCEPTS

1. Objective (Write this in your report)

Understand fundamental cybersecurity principles used in Security Operations Centers (SOC), including the CIA triad, risk components, and modern security models like Defense-in-Depth and Zero Trust.

2. Concept Explanation (SOC-Friendly)

CIA TRIAD

This is the **foundation of cybersecurity**. Every SOC alert relates to one of these.

1. Confidentiality

- Ensures **data is accessed only by authorized users**
- SOC monitors:
 - Unauthorized login attempts
 - Data exfiltration

Example:

If someone accesses admin data without permission → **Confidentiality breach**

2. Integrity

- Ensures **data is not altered or tampered**
- SOC monitors:
 - File changes
 - Log manipulation

Example:

If attacker modifies logs → **Integrity violation**

3. Availability

- Ensures **systems are accessible when needed**
- SOC monitors:
 - Server downtime
 - DDoS attacks



Example:

Website down due to attack → **Availability issue**

3. Threat vs Vulnerability vs Risk

- Threat: Potential danger (e.g., attacker)
- Vulnerability: Weakness in system (e.g., unpatched software)
- Risk: Likelihood of damage when threat exploits vulnerability

Term	SOC Meaning	Example
Threat	Potential attacker/event	Hacker
Vulnerability	Weakness	Outdated software
Risk	Damage possibility	Data breach

SOC Example:

- Threat → Attacker
- Vulnerability → Weak password
- Risk → Account takeover

5. Defense-in-Depth

A layered security approach where multiple defensive mechanisms are implemented to protect systems.

Multiple layers of security instead of one

Layers:

- Firewall
- Antivirus
- IDS/IPS
- SIEM
- MFA

SOC Insight:

Even if attacker bypasses one layer → others stop them

Real-world:



- Firewall fails → IDS detects → SIEM alerts

5. Zero Trust Model

Principle: **Never trust, always verify**

A security model based on the principle "Never trust, always verify," requiring continuous authentication and validation.

Key Rules:

- Verify every user
- Verify every device
- Continuous authentication

SOC Example:

- Even internal employee → must re-authenticate
- Login from new device → trigger alert

6. Case Study

Equifax Breach (2017)

What happened:

- Vulnerability in web application (**Apache Struts**) not patched
- Attackers exploited it
- The Equifax breach occurred due to an unpatched vulnerability in Apache Struts, leading to the exposure of sensitive data. This highlights the importance of vulnerability management and patching in cybersecurity.

Mapping to Concepts:

Concept	What Failed
Confidentiality	Sensitive data leaked
Integrity	Data compromised
Availability	Systems impacted



Concept	What Failed
Vulnerability	Unpatched software
Risk	Massive data breach

7. Conclusion

Understanding these concepts is essential for SOC analysts to detect, analyze, and respond to security incidents effectively.



TASK 6: Security Operations Workflow

Tools Used

- TheHive (Incident Response Platform)
- Docker (Containerization)
- Elasticsearch (Backend service)

Installation & Setup

Step 1: Install Docker

Docker was installed on Kali Linux to deploy TheHive using containers.

Command used:

```
sudo apt install docker.io -y
```



```
kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~/thehive-lab
Session Actions Edit View Help
Error: Unable to locate package docker-compose-plugin

(kali@kali)-[~]
$ sudo apt install docker.io -y
The following packages were automatically installed and are no longer required:
bloodhound.py libmplex2-2.1-0t64 librte-meter25 linux-image-6.12.38+kali-amd64
dtv-scan-tables libobjc-14-dev librte-net-bond25 linux-image-6.16.8+kali-amd64
gir1.2-girepository-2.0 libpocketsphinx3 librte-net25 mesa-vaapi-drivers
libarmadillo14 libpostproc58 librte-pci25 node-fs.realpath
libavfilter10 libradare2-5.0.0t64 librte-rcu25 node-inflight
libavformat61 librte-bus-pci25 librte-ring25 node-uri-js
libconfig-inifiles-perl librte-bus-vdev25 librte-sched25 pocketsphinx-en-us
libdisplay-info2 librte-eal25 librte-telemetry25 postgresql-17
libgav1-1 librte-ethdev25 libspinxbase3t64 postgresql-17-pg-gvm
libgdal37 librte-hash25 libsqlcipher1 postgresql-client-17
libgirepository-1.0-1 librte-ip-frag25 libswscale8 python3-xlrd
libpgme11t64 librte-kvargs25 libvdpau-va-gl1 tini
libpgmepp6t64 librte-log25 libwireshark18 vdpau-driver-all
libmjpegutils-2.1-0t64 librte-mbuf25 libwiretap15
libmjpeg2encpp-2.1-0t64 librte-mempool25 libwsutil16
Use 'sudo apt autoremove' to remove them.

Upgrading:
docker.io

Summary:
Upgrading: 1, Installing: 0, Removing: 0, Not Upgrading: 1493
Download size: 24.0 MB
Space needed: 3,113 kB / 11.7 GB available

Get:1 http://http.kali.org/kali kali-rolling/main amd64 docker.io amd64 27.5.1+dfsg4-2 [24.0 MB]
Fetched 24.0 MB in 23s (1,024 kB/s)
(Reading database ... 597618 files and directories currently installed.)
Preparing to unpack .../docker.io_27.5.1+dfsg4-2_amd64.deb ...
Unpacking docker.io (27.5.1+dfsg4-2) over (27.5.1+dfsg4-1) ...
Setting up docker.io (27.5.1+dfsg4-2) ...
Processing triggers for kali-menu (2025.4.3) ...
Processing triggers for man-db (2.13.1-1) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...
Service restarts being deferred:
systemctl restart docker.service

No containers need to be restarted.
```

Figure 1: Docker Installation & Setup



The screenshot shows a Kali Linux virtual machine running Oracle VM VirtualBox. The terminal window is titled 'kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox'. The terminal shows the nano 8.7 editor editing a file named 'docker-compose.yml'. The configuration file contains the following content:

```
services:
  elasticsearch:
    image: docker.elastic.co/elasticsearch/elasticsearch:7.17.9
    environment:
      - discovery.type=single-node
      - xpack.security.enabled=false
    ports:
      - "9200:9200"

  thehive:
    image: strangebee/thehive:5
    ports:
      - "9000:9000"
    depends_on:
      - elasticsearch
```

The terminal window also shows a menu bar with options: Session, Actions, Edit, View, Help. The bottom status bar shows various keyboard shortcuts and a 'Caps Lock' indicator.

Figure 2: Docker Compose Configuration



```
kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~/thehive-lab

Session Actions Edit View Help

(kali@kali)-[~/thehive-lab]
$ sudo docker compose up -d
[+] Running 20/20
  ✓ thehive Pulled 270.2s
    ✓ 84a2afebaf4d Pull complete 32.8s
    ✓ f373dafbe488 Pull complete 32.9s
    ✓ 58364b442082 Pull complete 205.7s
    ✓ 1c6833c310cd Pull complete 205.7s
    ✓ 6fc715df1c6a Pull complete 205.8s
    ✓ 1ad2b988e9f2 Pull complete 265.3s
    ✓ 4f4fb700ef54 Pull complete 265.3s
    ✓ 78a08c3c9751 Pull complete 265.3s
    ✓ cb23e5a85f61 Pull complete 265.4s
  ✓ elasticsearch Pulled 332.1s
    ✓ 36a9c60c46d0 Pull complete 80.4s
    ✓ e702cbf68995 Pull complete 93.3s
    ✓ d42ba0f6aa39 Pull complete 94.3s
    ✓ 13c59ecc70cc Pull complete 321.8s
    ✓ 12d112623fed Pull complete 321.8s
    ✓ 3e95eee02a15 Pull complete 321.8s
    ✓ e8819c48f163 Pull complete 321.9s
    ✓ ea0623c40fc9 Pull complete 321.9s
    ✓ a621ebe36959 Pull complete 321.9s
[+] Running 3/3
  ✓ Network thehive-lab_default Created 0.2s
  ✓ Container thehive-lab-elasticsearch-1 Started 10.4s
  ✓ Container thehive-lab-thehive-1 Started 1.0s

(kali@kali)-[~/thehive-lab]
$ sudo docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED   STA
TUS           PORTS
496ab5383ae1   strangebee/thehive:5               "/opt/thehive/entryp..." 16 seconds ago Up
15 seconds    0.0.0.0:9000→9000/tcp, :::9000→9000/tcp thehive-lab-thehive-1
901738338b8a   docker.elastic.co/elasticsearch/elasticsearch:7.17.9 "/bin/tini -- /usr/l..." 26 seconds ago Up
15 seconds    0.0.0.0:9200→9200/tcp, :::9200→9200/tcp, 9300/tcp thehive-lab-elasticsearch-1
e254b1dc5766   ghcr.io/digininja/dvwa:latest      "docker-php-entrypoi..." 3 months ago Up
31 minutes    127.0.0.1:4280→80/tcp dvwa-dvwa-1
1832a58de027   mariadb:10                         "docker-entrypoint.s..." 3 months ago Up
31 minutes    3306/tcp dvwa-db-1

(kali@kali)-[~/thehive-lab]
$ sudo docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED
STATUS        IMAGE      PORTS
496ab5383ae1   strangebee/thehive:5               "/opt/thehive/entryp..." About a minute ago
Up About a minute 0.0.0.0:9000→9000/tcp, :::9000→9000/tcp thehive-lab-thehive-1
901738338b8a   docker.elastic.co/elasticsearch/elasticsearch:7.17.9 "/bin/tini -- /usr/L..." About a minute ago
Up About a minute 0.0.0.0:9200→9200/tcp, :::9200→9200/tcp, 9300/tcp thehive-lab-elasticsearch-1
e254b1dc5766   ghcr.io/digininja/dvwa:latest      "docker-php-entrypoi..." 3 months ago
```

Figure 3: Containers Running



Step 2: TheHive Login

TheHive web interface was accessed and user login was performed.

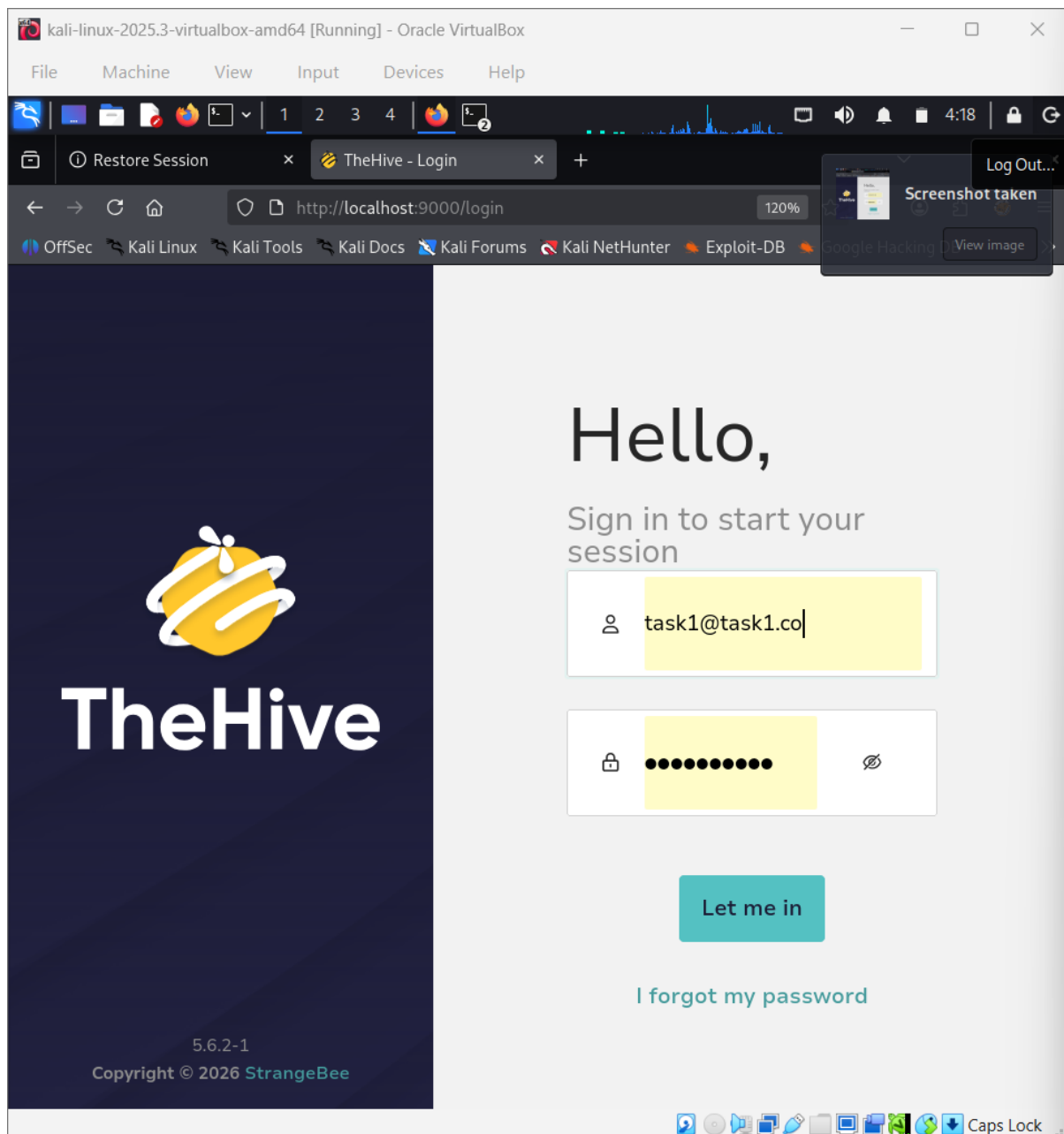


Figure 4: TheHive Login Page

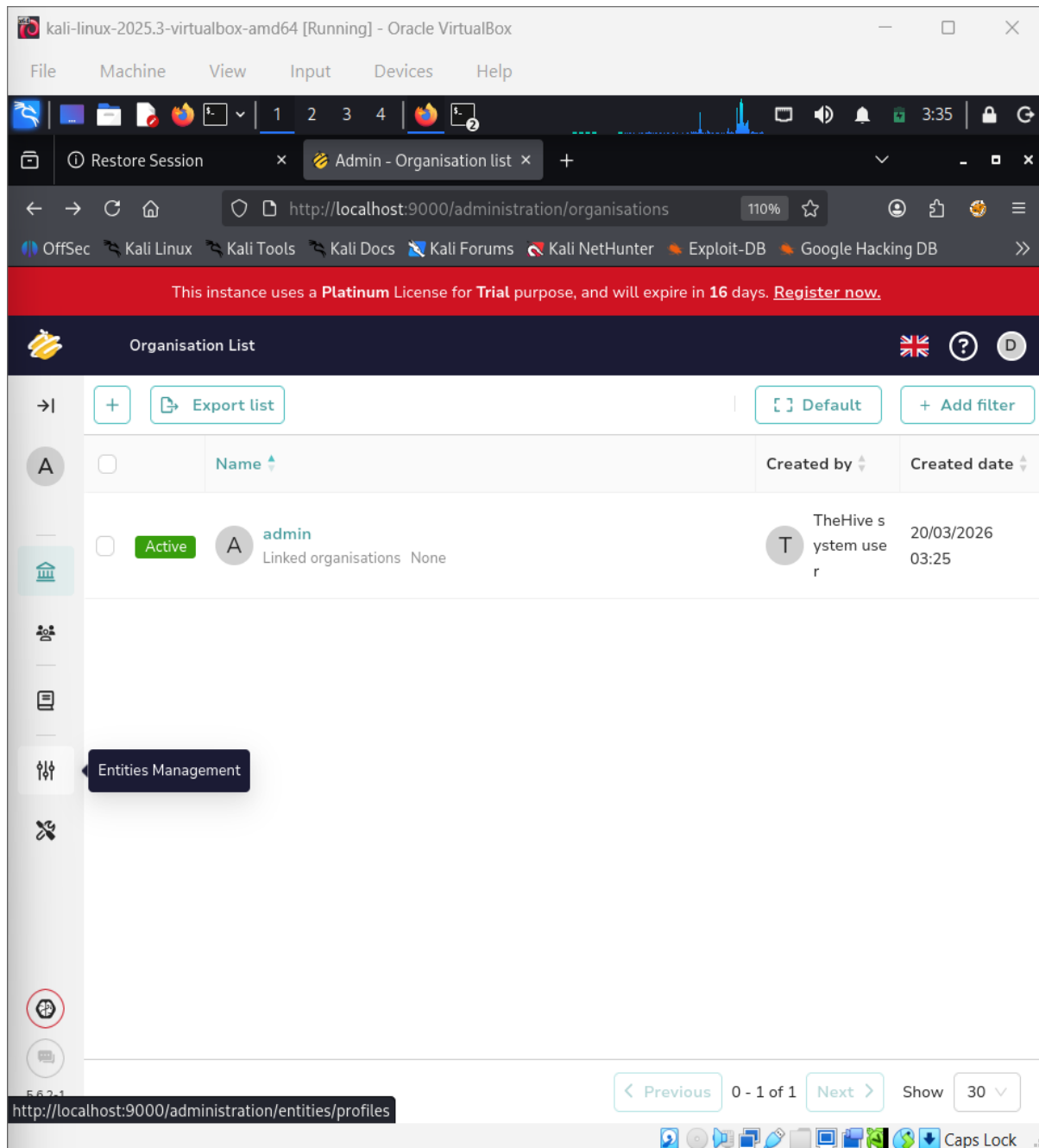


Figure 5: Admin Dashboard



Step 3: Case Creation (Detection Stage)

A phishing email incident was simulated by creating a new case.

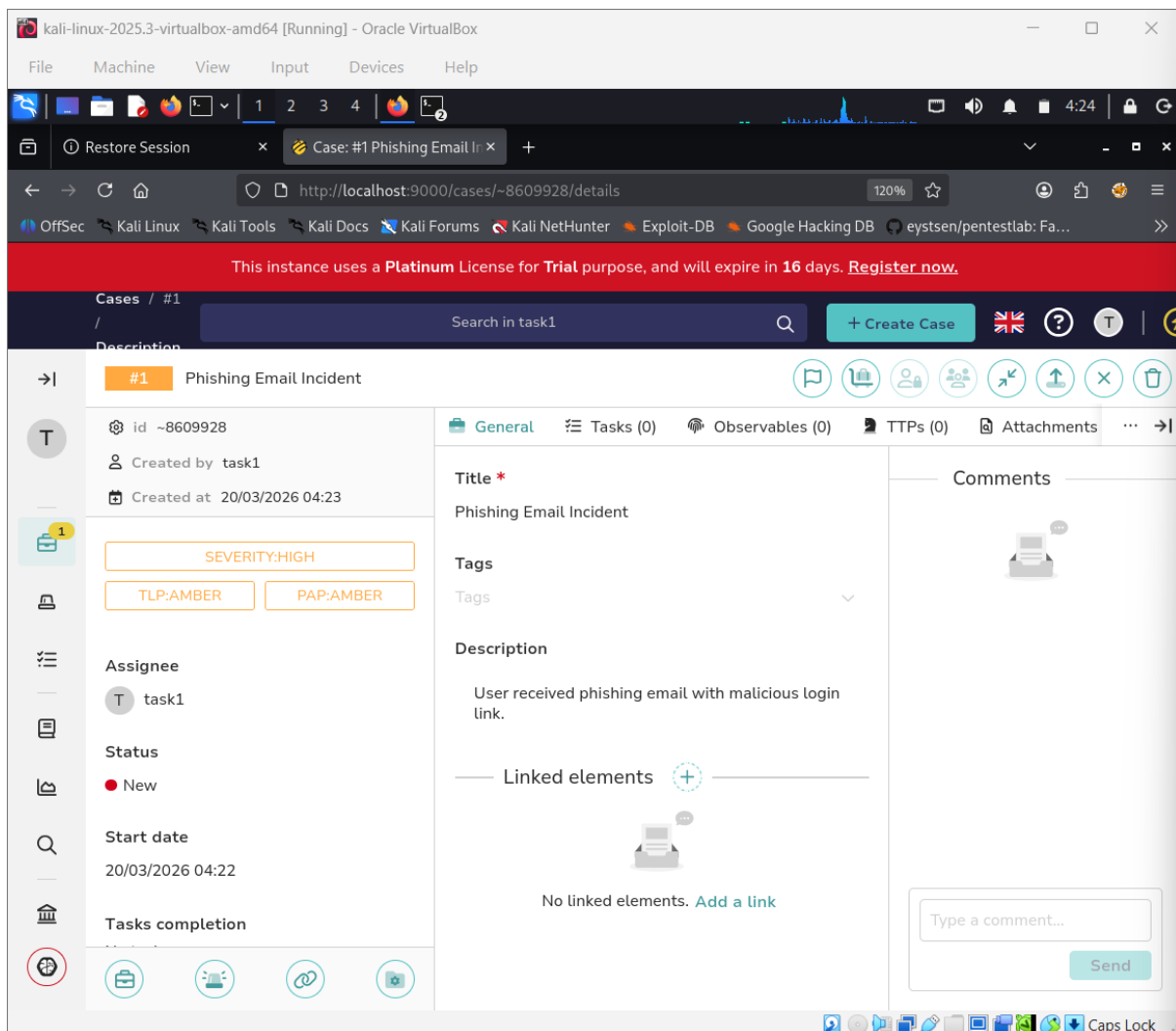


Figure 6: Case Created



Step 4: Investigation (Observables)

Indicators of compromise (IOCs) were added:

- Malicious URL
- Suspicious IP address
- Attacker email

The screenshot shows the CYART web interface in a browser window. The URL is `http://localhost:9000/cases/~8609928/observables`. A red banner at the top indicates a trial license. The interface shows a sidebar with navigation options and a main panel displaying the details of a case titled "#1 Phishing Email Incident". The "Observables" tab is active, showing a table of three added indicators of compromise (IOCs).

Flags	Data type	Value/Filename	Dates
☐ TLP:AMBER ☐ PAP:AMBER	mail	attacker@mail[.]com	S. 20/03/2026 04:28 C. 20/03/2026 04:28
☐ TLP:AMBER ☐ PAP:AMBER	ip	192[.]168[.]1[.]100	S. 20/03/2026 04:28 C. 20/03/2026 04:28
☐ TLP:AMBER ☐ PAP:AMBER	url	hxxp://fake-login[.]com	S. 20/03/2026 04:27 C. 20/03/2026 04:27

Figure 7: Observables Added



Step 5: Response Actions

Response tasks were created:

- Block malicious domain
- Reset user password
- Scan affected system

The screenshot shows the CYART web interface running in a browser. The browser address bar shows the URL `http://localhost:9000/cases/~8609928/tasks`. The interface displays a case titled "#1 Phishing Email Incident" with a sidebar on the left containing various icons and a main panel on the right showing a list of tasks. The tasks are:

Status	Task	Order	Due date	Details	Assignee	Dates	S.	C.	U.
Waiting	Block malicious domain	default		Activity 0		C. 20/03/2026 04:29			
Waiting	Reset user password	default		Activity 0		C. 20/03/2026 04:29			
Waiting	Scan affected system	default		Activity 0		C. 20/03/2026 04:30			

Figure 8: Tasks Added



Step 6: Case Closure

The case was analyzed and classified as:

- True Positive
- Impact: Yes

The case was successfully closed.

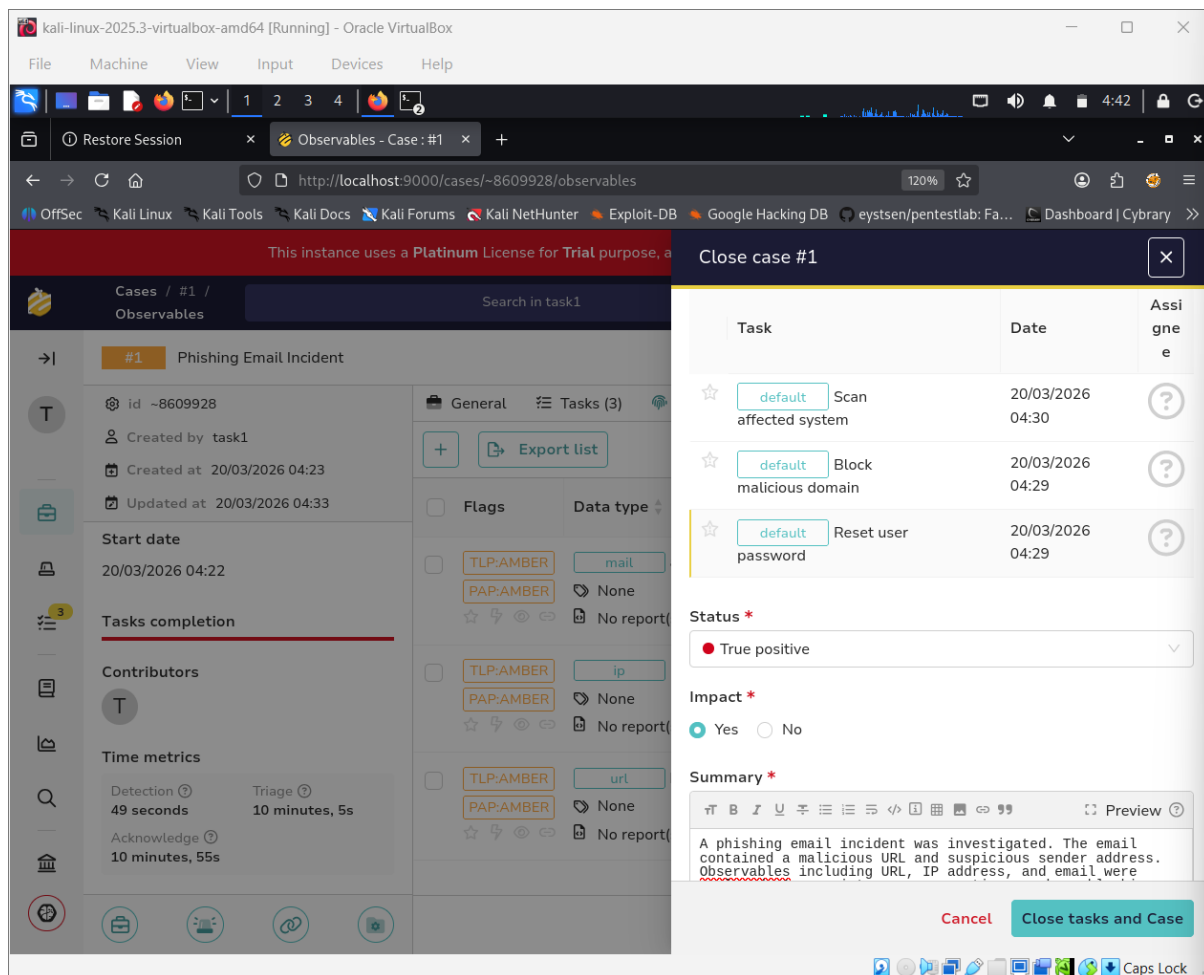


Figure 9: Case Closure

SOC Workflow Implemented



Detection → Triage → Investigation → Response → Closure

Conclusion

This task demonstrated the real-world SOC workflow using TheHive. The phishing incident was investigated using observables, appropriate response actions were taken, and the case was successfully resolved as a true positive.



TASK 7. Incident Response Basics

1. What is Incident Response (IR)?

Incident Response is the **structured process used by a SOC team to detect, analyze, and respond to cybersecurity incidents** like malware, phishing, ransomware, etc.

Goal:

- Minimize damage
- Recover quickly
- Prevent future attacks

2. IR Lifecycle (NIST SP 800-61)

Full Lifecycle Overview





Incident Response Planning



Phases Explained (SOC Style)

1. Preparation

- Set up tools, alerts, policies
- Train SOC team



Examples:

- Configure SIEM (Splunk / Wazuh)
- Create playbooks
- Backup systems

2. Identification

- Detect if an incident occurred

SOC Activities:

- Analyze logs (SIEM alerts)
- Investigate suspicious IPs, files

Example:

- Multiple failed logins → possible brute force

3. Containment

- Stop the attack from spreading

Types:

- Short-term → isolate system
- Long-term → patch vulnerability

Example:

- Disconnect infected machine from network

4. Eradication

- Remove root cause

Example:

- Delete malware
- Remove malicious users
- Patch vulnerabilities

5. Recovery

- Restore systems safely

Example:

- Restore backups
- Monitor system for reinfection

6. Lessons Learned



- Improve future response

SOC Work:

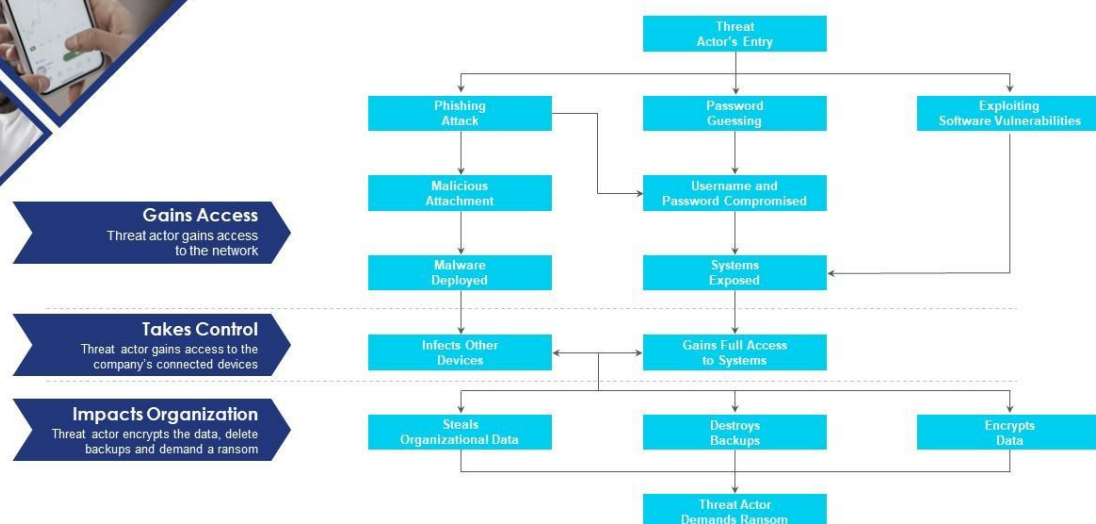
- Write incident report
- Update detection rules
- Improve playbooks

3. SOC Incident Example (Ransomware Scenario) Ransomware Attack Flow



Process flow diagram depicting ransomware incidents occurrence

Mentioned slide depicts the incident workflow diagram of a ransomware event. It starts with the threat actor's entry and ends with the threat actor demanding ransom.



This slide is 100% editable. Adapt it to your needs and capture your audience's attention.



Reconnaissance

Lorem ipsum dolor sit dim amet, mea regione diamet principes gatk.



Weaponization

Lorem ipsum dolor sit dim amet, mea regione diamet principes gatk.



Exploitation

Lorem ipsum dolor sit dim amet, mea regione diamet principes gatk.



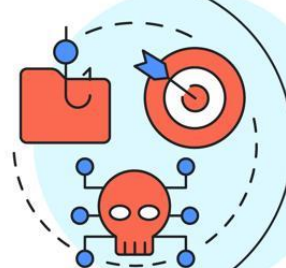
Command & Control

Lorem ipsum dolor sit dim amet, mea regione diamet principes gatk.



Actions on Objectives

Lorem ipsum dolor sit dim amet, mea regione diamet principes gatk.





Scenario Walkthrough

1. Preparation

- EDR + SIEM configured

2. Identification

- Alert: suspicious file execution

3. Containment

- Isolate infected system

4. Eradication

- Remove ransomware + kill process

5. Recovery

- Restore files from backup

6. Lessons Learned

- Add email filtering rules
- Train users on phishing

5. Tools Used in Incident Response

Phase	Tools
Preparation	SIEM (Splunk, Wazuh), EDR
Identification	Logs, IDS (Snort), Threat Intel
Containment	Firewall, EDR isolation
Eradication	Antivirus, forensic tools
Recovery	Backup systems
Lessons Learned	Reporting tools

6. How to Learn Incident Response

1. Study Framework (NIST SP 800-61)

To understand Incident Response properly, I referred to the **NIST SP 800-61**, which is the industry-standard guideline used by SOC teams.

Key Learnings:

- Incident Response follows a structured lifecycle:



- Preparation
- Identification
- Containment
- Eradication
- Recovery
- Lessons Learned
- Emphasis on **documentation and communication**
- Importance of **quick detection and response**

2. Practical Simulation (Attack Scenario)

To gain hands-on understanding, I simulated a basic cyber attack scenario.

Scenario:

- A phishing email delivers a malicious attachment
- User downloads and executes the file

Steps Performed:

1. Generated suspicious activity (login/file execution)
2. Monitored logs using SIEM (e.g., Splunk/Wazuh)
3. Identified abnormal behavior:
 - Unknown file execution
 - Suspicious IP communication

Learning Outcome:

- Understood how alerts are triggered
- Learned how SOC analysts detect threats in logs

3. Tabletop Exercise

A tabletop exercise was performed to simulate a real-world SOC response.

Scenario:

“User clicked a malicious email attachment leading to malware infection.”

Response Plan:

Phase	Action
Identification	Detected alert from SIEM
Containment	Isolated infected system



Phase	Action
Eradication	Removed malware
Recovery	Restored system from backup
Lessons Learned	Improved email filtering rules

Learning Outcome:

- Improved decision-making skills
- Understood incident handling workflow

4. Incident Report Creation

A structured incident report was created to document the findings.

Report Included:

1. Incident Timeline

- Time of detection
- Actions taken at each stage

2. Actions Taken

- System isolation
- Malware removal
- Log analysis

3. Tools Used

- SIEM (Splunk / Wazuh)
- Antivirus / EDR
- Log monitoring tools

7. Conclusion

This task provided a clear understanding of:

- The Incident Response lifecycle
- Real-world SOC workflows
- Importance of quick detection and response
- Continuous improvement through lessons learned

Incident Response is a critical function in cybersecurity, enabling organizations to effectively handle and recover from security incidents.



TASK 8: Documentation Standards

1. Task overview

In a SOC, documentation = **how we record, share, and improve security operations.**

Think of it like this:

- Detection → Logs
- Response → Actions
- Documentation → Evidence + Knowledge

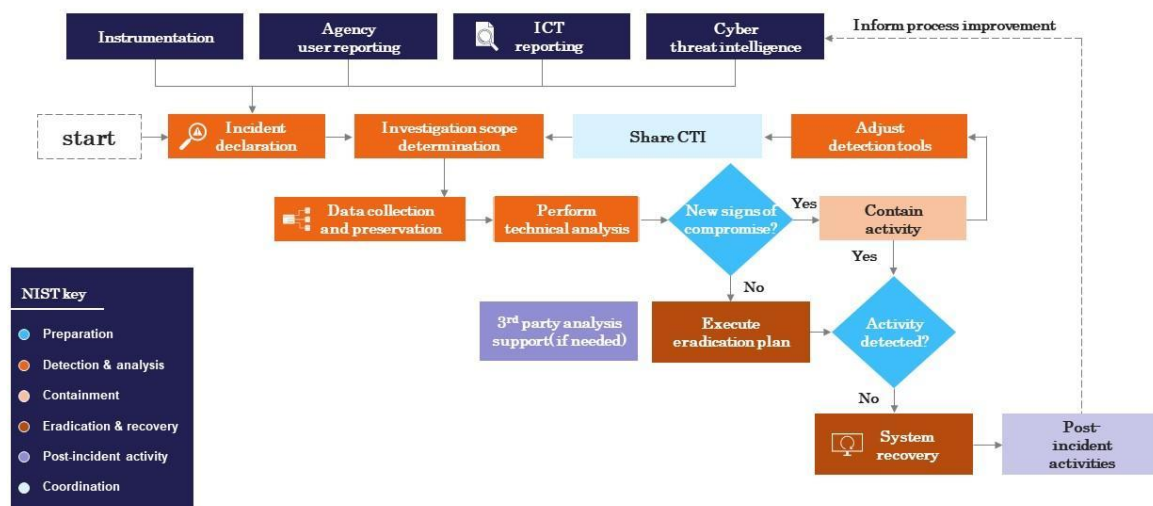
Key Types:

- **Incident Report** → What happened?
- **Runbook** → What to do when it happens again?
- **SOP (Standard Operating Procedure)** → Standard steps for analysts
- **Post-mortem** → What went wrong & how to improve

SOC Documentation Workflow

Cyber security incident response process flow chart

This slide represents the flow chart representing the detection and reaction to cyber security incidents, determination of their scope and risk and reduction of likelihood of incident from reoccurring. It starts with incident declaration and ends with system recovery.



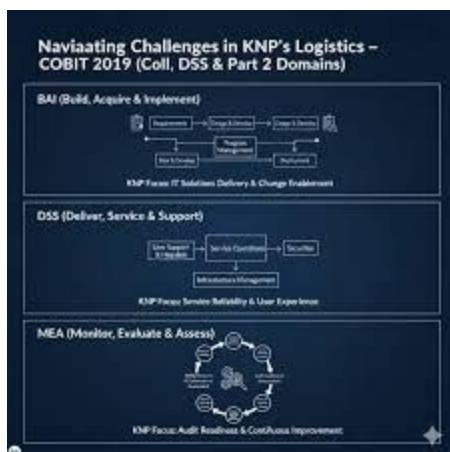
This slide is 100% editable. Adapt it to your needs and capture your audience's attention.



Cyber Incident Response Life Cycle



image: delinea.com/resources/free-incident-response-plan-template





Best Practices	SOPs (Standard Operating Procedures)	Runbooks
Design	Break down tasks into logical steps and use clear, concise language	Document potential incidents, causes, and solutions
Format	Include visuals like diagrams and flowcharts for better comprehension	Use a checklist format for quick reference during incidents
Collaboration	Involve key stakeholders to ensure comprehensive coverage of tasks	Collaborate with teams to address diverse scenarios and validate accuracy
Implementation	Train employees on SOP usage and provide context for their importance	Ensure accessibility via knowledge bases or incident management systems
Continuous Improvement	Update SOPs regularly to reflect changes in processes and incorporate feedback	Test and refine Runbooks based on incident response outcomes and evolving environments
Automation	Consider incorporating automated workflows where applicable	Use automation tools to streamline execution and reduce human error during incidents

GRAPH

2. Tools Required

Basic Tools

- Markdown (.md) files (GitHub)
- Notion / Confluence
- Google Docs / Word

SOC-related Tools

- Splunk (logs reference)
- Wazuh (alerts reference)
- Kibana (visual evidence)

3.Real SOC Workflow

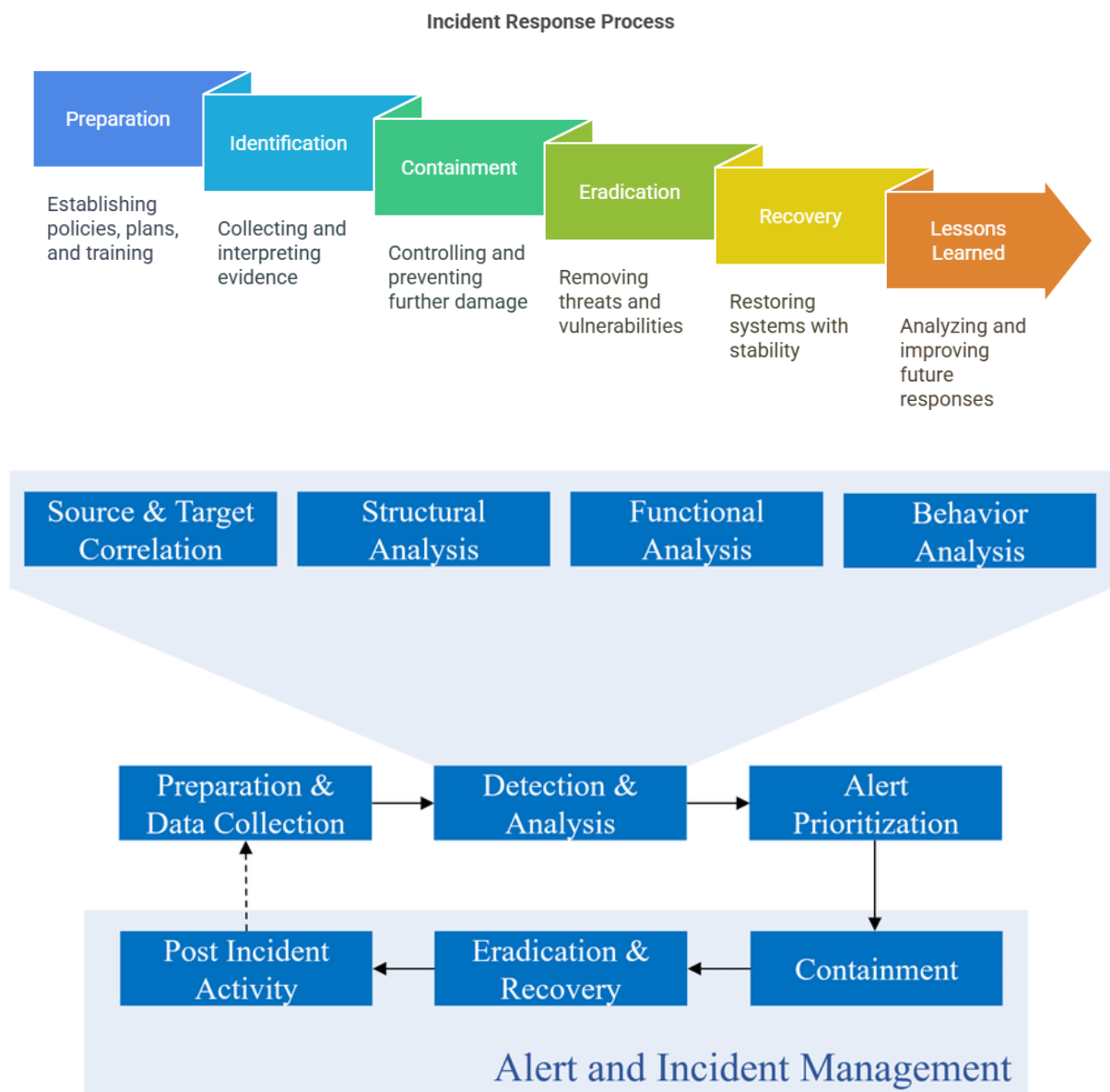
Step-by-step SOC Process

1. Alert Triggered (SIEM)
2. Analyst investigates logs
3. Incident confirmed



4. **Containment actions taken**
5. **Documentation created**
6. **Runbook updated**
7. **Lessons learned shared**

Incident Response + Documentation Flow





4.Installation & Setup

1.Install Git

```
sudo apt update
sudo apt install git -y
```

2.Create documentation file

```
cd ~/CyArt-SOC-Domain-Task-1st-week-
mkdir Task8-Documentation
cd Task8-Documentation
touch incident-report.md
```



```
kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~/CyArt-SOC-Domain-Task-1st-week-/Task8-Documentation
Session Actions Edit View Help

(kali@kali)~$ sudo apt update
Get:1 http://deb.debian.org/debian bookworm InRelease
Get:2 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Packages [20.9 MB]
Get:4 https://packages.microsoft.com/repos/code stable InRelease [3,590 B]
Get:5 https://packages.microsoft.com/repos/code stable/main amd64 Packages [23.2 kB]
Get:6 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [52.8 MB]
Get:7 http://kali.download/kali kali-rolling/contrib amd64 Packages [118 kB]
Get:8 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [274 kB]
Get:9 http://kali.download/kali kali-rolling/non-free amd64 Packages [185 kB]
Get:10 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [880 kB]
Get:11 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [14.3 kB]
Fetched 75.2 MB in 36s (2,078 kB/s)
1492 packages can be upgraded. Run 'apt list --upgradable' to see them.
git is already the newest version (1:2.51.0-1).
The following packages were automatically installed and are no longer required:
bloodhound.py libplex2-2.1-0t64 librtmeter25 linux-image-6.12.38+kali-amd64
dtv-scan-tables libobjc-14-dev librtmeter25 linux-image-6.16.8+kali-amd64
gir1.2-girepository-2.0 libpocketsphinx3 librtmeter25 mesa-vaapi-drivers
libarmadillo14 libpostproc58 librtmeter25 node-fs.realpath
libavfilter10 libradare2-5.0.0t64 librtmeter25 node-inflight
libavformat61 librtmeter25 librtmeter25 node-uri-js
libconfig-inifiles-perl librtmeter25 librtmeter25 pocketsphinx-en-us
libdisplay-info2 librtmeter25 librtmeter25 postgresql-17
libgav1-1 librtmeter25 librtmeter25 postgresql-17-pg-gvm
libgdal37 librtmeter25 librtmeter25 postgresql-client-17
libgirepository-1.0-1 librtmeter25 librtmeter25 python3-xldr
libgpgme11t64 librtmeter25 librtmeter25 tini
libgpgmepp6t64 librtmeter25 librtmeter25 vdpau-driver-all
libmjpegutils-2.1-0t64 librtmeter25 librtmeter25
libmpeg2encpp-2.1-0t64 librtmeter25 librtmeter25
Use 'sudo apt autoremove' to remove them.

Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1492

(kali@kali)~$ cd ~/CyArt-SOC-Domain-Task-1st-week-
(kali@kali)~/CyArt-SOC-Domain-Task-1st-week-$
$ mkdir Task8-Documentation
(kali@kali)~/CyArt-SOC-Domain-Task-1st-week-$
$ cd Task8-Documentation
(kali@kali)~/CyArt-SOC-Domain-Task-1st-week-/Task8-Documentation$
$ touch incident-report.md
```

Figure 1 : git installation and project structure



3. Practical Task: Write a Mock DDoS Incident Report

INCIDENT REPORT TEMPLATE

Incident Report – DDoS Attack

1. Incident Summary

- Incident ID: SOC-2026-001
- Date: [18.3.2026]
- Analyst: [sudeep]
- Severity: High
- Status: Resolved

2. Description

A Distributed Denial of Service (DDoS) attack was detected targeting the web server, causing service disruption.

3. Detection

- Tool: Splunk / Wazuh
- Alert: High traffic spike detected
- Logs:
 - Multiple IP requests
 - Traffic > threshold

4. Affected Systems

- Web Server (192.168.1.10)

5. Indicators of Compromise (IOCs)

- Suspicious IPs: 45.67.x.x, 89.23.x.x
- Traffic pattern: SYN flood

6. Actions Taken

- Blocked IPs via firewall
- Enabled rate limiting
- Restarted services

7. Root Cause

Lack of rate limiting and firewall rules.

8. Lessons Learned

- Need better traffic monitoring
- Implement DDoS protection

9. Recommendations



- Use WAF (Web Application Firewall)
- Enable auto-scaling
- Improve alert thresholds

```
kali@kali: ~/CyArt-SOC-Domain-Task-1st-week/Task8-Documentation
GNU nano 8.7 incident-report.md
Incident Report - DDoS Attack

## 1. Incident Summary
- Incident ID: SOC-2026-001
- Date: [Add Today Date]
- Analyst: Sudeep
- Severity: High
- Status: Resolved

## 2. Description
A Distributed Denial of Service (DDoS) attack caused high traffic on the web server leading to downtime.

## 3. Detection
- Tool: Simulated Logs / SIEM
- Alert: Traffic spike detected

## 4. Affected System
- Web Server (192.168.1.10)

## 5. IOCs
- Multiple IP requests
- SYN flood traffic

## 6. Actions Taken
- Blocked IPs
- Enabled rate limiting

## 7. Root Cause
No traffic filtering rules were configured.

## 8. Lessons Learned
Need proper firewall and monitoring setup.

## 9. Recommendation
Implement WAF and alert thresholds.
```

Figure 2 Incident report file creation



4.Runbook Example

Runbook – DDoS Attack Response

Step 1: Identify Attack

- Check traffic logs
- Verify spike in requests

Step 2: Validate

- Confirm abnormal traffic patterns
- Identify IP sources

Step 3: Containment

- Block IPs
- Enable rate limiting

Step 4: Mitigation

- Use firewall rules
- Activate CDN protection

Step 5: Recovery

- Restore services
- Monitor system stability

Step 6: Documentation

- Update incident report
- Record lessons learned



```
kali@kali: ~/CyArt-SOC-Domain-Task-1st-week-/Task8-Documentation
GNU nano 8.7 runbook.md
Runbook - DDoS Response

## Step 1: Identify
Check logs for traffic spike

## Step 2: Validate
Confirm abnormal behavior

## Step 3: Containment
Block suspicious IPs

## Step 4: Mitigation
Enable firewall rules

## Step 5: Recovery
Restart services and monitor

## Step 6: Documentation
Update incident report
```

Figure 3: Runbook file creation

5.SOP , Postmortem and all list of files



kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

kali@kali: ~/CyArt-SOC-Domain-Task-1st-week/Task8-Documentation

Session Actions Edit View Help

GNU nano 8.7 postmortem.md

Postmortem - DDoS Incident

What Happened
DDoS attack caused downtime.

Why It Happened
No rate limiting configured.

Impact
Service disruption.

Fix
Firewall rules applied.

Improvement
Better monitoring and alerting needed.

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line

kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

kali@kali: ~/CyArt-SOC-Domain-Task-1st-week/Task8-Documentation

Session Actions Edit View Help

GNU nano 8.7 sop.md

SOP - Incident Handling

1. Monitor alerts
2. Investigate logs
3. Confirm incident
4. Escalate if needed
5. Take action
6. Document everything

Help Exit Write Out Read File Where Is Replace Read 8 lines Cut Paste Execute Justify Location Go To Line Undo Redo



```
kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~/CyArt-SOC-Domain-Task-1st-week-/Task8-Documentation
Session Actions Edit View Help

(kali@kali)-[~/CyArt-SOC-Domain-Task-1st-week-/Task8-Documentation]
$ touch runbook.md sop.md postmortem.md

(kali@kali)-[~/CyArt-SOC-Domain-Task-1st-week-/Task8-Documentation]
$ ls
incident-report.md postmortem.md runbook.md sop.md

(kali@kali)-[~/CyArt-SOC-Domain-Task-1st-week-/Task8-Documentation]
$ nano incident-report.md

(kali@kali)-[~/CyArt-SOC-Domain-Task-1st-week-/Task8-Documentation]
$ nano runbook.md

(kali@kali)-[~/CyArt-SOC-Domain-Task-1st-week-/Task8-Documentation]
$ nano sop.md

(kali@kali)-[~/CyArt-SOC-Domain-Task-1st-week-/Task8-Documentation]
$ nano postmortem.md

(kali@kali)-[~/CyArt-SOC-Domain-Task-1st-week-/Task8-Documentation]
$ ls -l
total 16
-rw-rw-r-- 1 kali kali 719 Mar 18 03:55 incident-report.md
-rw-rw-r-- 1 kali kali 245 Mar 18 03:56 postmortem.md
-rw-rw-r-- 1 kali kali 314 Mar 18 03:55 runbook.md
-rw-rw-r-- 1 kali kali 147 Mar 18 03:56 sop.md

(kali@kali)-[~/CyArt-SOC-Domain-Task-1st-week-/Task8-Documentation]
$ ls -l
total 16
-rw-rw-r-- 1 kali kali 719 Mar 18 03:59 incident-report.md
-rw-rw-r-- 1 kali kali 245 Mar 18 04:02 postmortem.md
-rw-rw-r-- 1 kali kali 314 Mar 18 04:00 runbook.md
-rw-rw-r-- 1 kali kali 149 Mar 18 04:01 sop.md

(kali@kali)-[~/CyArt-SOC-Domain-Task-1st-week-/Task8-Documentation]
$
```

Figure 4,5,6 : created SOP, Postmortem and list of the all files

6.Output

- Properly structured documentation files:
 - incident-report.md
 - runbook.md
 - sop.md
 - postmortem.md
- Files successfully uploaded to GitHub repository.



7.Outcome

Successfully created SOC-standard documentation including:

- Incident report
- Runbook
- SOP
- Postmortem

This task improved understanding of real-world SOC documentation practices.

8. Conclusion

Documentation is a critical part of SOC operations. It ensures proper incident tracking, improves response efficiency, and helps teams learn from past incidents. This task provided hands-on experience in creating professional SOC documentation.

For more information about task Documentation Basics check out my github repo link: <https://github.com/sudeepri-pks/CyArt-SOC-Domain-Task-1st-week-/tree/main/Theoretical-Knowledge/Task8-Documentation>



PRACTICAL APPLICATION

TASK 1: Log Analysis and Browser Forensics

Task A: Brute-Force Detection

Tools Used:

- Windows Event Viewer
- PowerShell (Get-WinEvent)
- Command Prompt (wevtutil)

Step 1: Generate Failed Logins

- Multiple incorrect login attempts were performed on the system to simulate a brute-force attack.

Step 2: Analyze Logs using PowerShell

Command used:

```
Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4625}
```



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Windows\system32> Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4625}

ProviderName: Microsoft-Windows-Security-Auditing

TimeCreated          Id LevelDisplayName Message
-----
19-03-2026 09:58:45    4625 Information    An account failed to log on....
19-03-2026 09:58:12    4625 Information    An account failed to log on....
19-03-2026 09:58:08    4625 Information    An account failed to log on....
19-03-2026 09:58:05    4625 Information    An account failed to log on....
19-03-2026 09:57:58    4625 Information    An account failed to log on....

PS C:\Windows\system32>
```

Figure 1:Failed login logs

PowerShell Output

- Shows multiple failed login attempts
- Confirms Event ID 4625 detection



Step 3: Analyze Logs using CLI

Command used:

```
wevtutil qe Security /q:"*[System[(EventID=4625)]]" /f:text /c:50
```

```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>wevtutil qe Security /q:"*[System[(EventID=4625)]]" /f:text /c:50
Event[0]
  Log Name: Security
  Source: Microsoft-Windows-Security-Auditing
  Date: 2026-03-19T09:57:58.822000Z
  Event ID: 4625
  Task: Logon
  Level: Information
  Opcode: Info
  Keyword: Audit Failure,
  User: N/A
  User Name: N/A
  Computer: BLACK-HAT-HACKER
  Description:
An account failed to log on.

Subject:
  Security ID: S-1-5-18
  Account Name: BLACK-HAT-HACKER$
  Account Domain: WORKGROUP
  Logon ID: 0x3E7

Logon Type: 2

Account For Which Logon Failed:
  Security ID: S-1-0-0
  Account Name: -
  Account Domain: -

Failure Information:
  Failure Reason: An Error occurred during Logon.
  Status: 0xC000006D
  Sub Status: 0xC0000380

Process Information:
  Caller Process ID: 0x9dc
  Caller Process Name: C:\Windows\System32\svchost.exe

Network Information:
  Workstation Name: -
  Source Network Address: 127.0.0.1
  Source Port: 0

Detailed Authentication Information:
  Logon Process: User32
  Authentication Package: Negotiate
```

Figure 2 : Failed login limit 50 results

wevtutil Output



- Detailed log including:
 - Failure reason
 - Logon type
 - Source IP

Step 4: Open Event Viewer

Win + R → eventvwr.msc

Navigate:

Windows Logs → Security

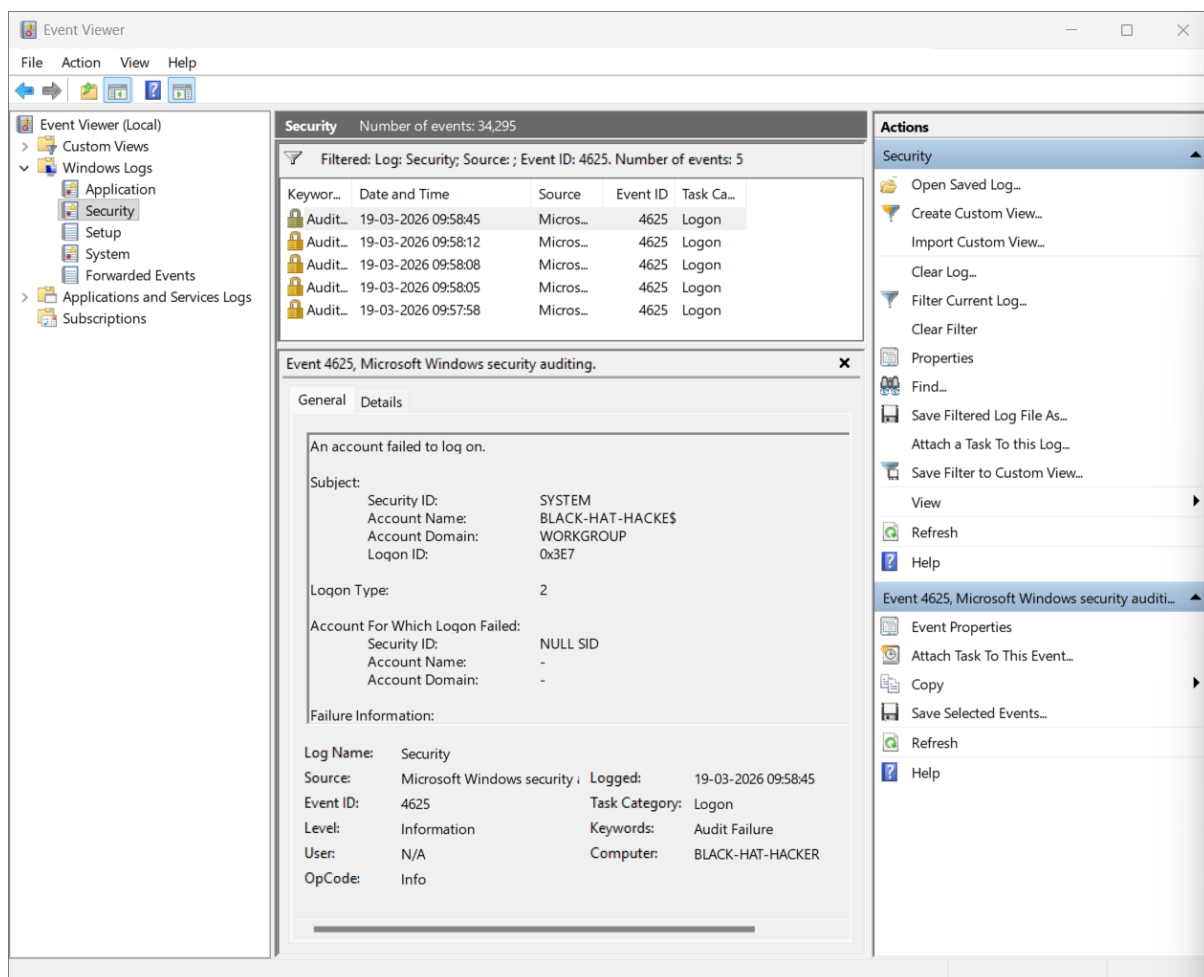


Figure 3: Event viewer failed login attempt logs

Event Viewer

- Filtered view of Event ID 4625
- Timeline of repeated login failures



SOC Analysis

- Repeated failed attempts in short time = **Brute-force behavior**
- Localhost (127.0.0.1) indicates **internal simulation**
- In real SOC:
 - Would correlate with IP reputation
 - Block attacker IP
 - Alert incident response team

Conclusion

The system successfully detected multiple failed login attempts, demonstrating how SOC analysts identify brute-force attacks using Windows logs. .

TASK 2. Browser History Analysis

Tools Used

- BrowsingHistoryView (Nirsoft)

Step 1: Generate Browser Activity

- Visited multiple websites including test URLs

Step 2: Extract Browser History

- Ran BrowsingHistoryView tool
- Automatically parsed Chrome history

Step 3: Analyze URLs

- Reviewed visited URLs
- Checked timestamps and visit counts

Findings

- Multiple URLs extracted from Chrome history
- Includes:
 - Google login pages
 - YouTube
- Visit timestamps and counts recorded



URL	Title	Visit Time	Visit Count	Visited From
https://accounts.googl...	Sign in - Google Accounts	10-03-2026 13:39:53	1	
https://accounts.googl...	Sign in - Google Accounts	12-03-2026 12:47:07	1	https://accounts.google....
https://accounts.googl...	Sign in - Google Accounts	10-03-2026 13:39:53	1	https://accounts.google....
https://accounts.googl...	Sign in - Google Accounts	14-03-2026 18:45:35	1	https://accounts.google....
https://accounts.googl...	Sign in - Google Accounts	18-03-2026 16:46:25	1	https://accounts.google....
https://accounts.googl...	YouTube	17-03-2026 10:07:14	1	https://www.youtube.co...
https://accounts.googl...		10-03-2026 13:40:57	1	
https://accounts.googl...		18-03-2026 16:46:29	1	https://accounts.google....
https://accounts.googl...		14-03-2026 18:45:39	1	
https://accounts.googl...	Sign in - Google Accounts	18-03-2026 16:46:25	1	https://accounts.google....
https://accounts.googl...	Sign in - Google Accounts	18-03-2026 16:46:25	1	https://accounts.google....
https://accounts.googl...	Sign in - Google Accounts	12-03-2026 12:47:07	1	https://accounts.google....
https://accounts.googl...	Haxcamp - Cybersecurity...	12-03-2026 12:47:29	1	https://accounts.google....
https://accounts.googl...	Sign in - Google Accounts	10-03-2026 13:39:53	1	https://accounts.google....
https://accounts.googl...	Sign in - Google Accounts	10-03-2026 13:39:53	1	https://accounts.google....
https://accounts.googl...	Sign in - Google Accounts	14-03-2026 18:45:35	1	https://accounts.google....
https://accounts.googl...	Sign in - Google Accounts	14-03-2026 18:45:35	1	https://accounts.google....
https://accounts.googl...	Sign in - Google Accounts	10-03-2026 13:39:53	1	
https://accounts.googl...	Sign in - Google Accounts	10-03-2026 13:39:53	1	https://accounts.google....
https://accounts.googl...	Sign in - Google Accounts	18-03-2026 16:46:25	1	https://accounts.google....
https://accounts.googl...	Sign in - Google Accounts	14-03-2026 18:45:35	2	https://accounts.google....
https://accounts.googl...	Sign in - Google Accounts	14-03-2026 18:45:56	2	
https://accounts.googl...	Sign in - Google Accounts	14-03-2026 18:45:56	2	
https://accounts.googl...	Sign in - Google Accounts	18-03-2026 16:46:18	1	https://accounts.google....
https://accounts.googl...	Sign in - Google Accounts	11-03-2026 18:39:53	1	https://accounts.google....
https://accounts.googl...	Sign in - Google Accounts	10-03-2026 13:39:47	1	
https://accounts.googl...	Sign in - Google Accounts	10-03-2026 13:39:47	1	https://accounts.google....
https://accounts.googl...	Sign in - Google Accounts	12-03-2026 12:47:00	1	https://accounts.google....
https://accounts.googl...	Sign in - Google Accounts	12-03-2026 12:47:00	1	
https://accounts.googl...	Sign in - Google Accounts	16-03-2026 14:31:03	2	
https://accounts.googl...	Sign in - Google Accounts	16-03-2026 14:31:03	2	
https://accounts.googl...	Sign in - Google Accounts	15-03-2026 14:36:22	2	
https://accounts.googl...	Sign in - Google Accounts	16-03-2026 19:58:48	2	
https://accounts.googl...	Sign in - Google Accounts	15-03-2026 14:36:22	2	
https://accounts.googl...	Sign in - Google Accounts	14-03-2026 18:45:58	2	

Figure 1 : BrowsingHistoryView (Nirsoft) shows chrome history

SOC Analysis

- Analysts use browser history to:
 - Detect phishing attempts
 - Identify malicious domains
 - Track user activity



- In real SOC:
 - URLs are checked against threat intelligence feeds
 - Suspicious domains are flagged

Tool Learning Note

- Initially attempted with **LECcmd**
- Identified that it is used for .lnk files
- Switched to **BrowsingHistoryView** for correct analysis

Conclusion

Browser history analysis successfully revealed user activity and demonstrates how SOC analysts investigate potential threats using forensic tools.

Overall Outcome

- ✓ Successfully performed log analysis
- ✓ Detected brute-force attack patterns
- ✓ Extracted and analyzed browser history
- ✓ Used multiple tools and troubleshooting methods



TASK 2. Document Security Events

Tools Used

- Kali Linux Terminal
- Nano Text Editor
- Linux Authentication Logs (/var/log/auth.log)

Event Log Structure

The following fields were used to document security events:

- **Date/Time** – Timestamp of the event
- **Source IP** – Origin of the activity
- **Event ID** – Unique identifier for the event
- **Description** – Details of the event
- **Action Taken** – Response performed

Sample Event Log

Create a table in Word:

Date/Time	Source IP	Event ID	Description	Action Taken
2026-03-19 11:00	192.168.1.10	EVT1001	Multiple failed login attempts	Monitoring initiated
2026-03-19 11:05	192.168.1.10	EVT1002	Possible brute force attack	IP blocked using firewall
2026-03-19 11:10	127.0.0.1	EVT1003	Failed SSH login attempts detected	Investigated and monitored



The screenshot shows a terminal window titled "kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox". The terminal is running the nano text editor, editing a file named "security_events.log". The file content is as follows:

```
GNU nano 8.7 security_events.log *
Date/Time | Source IP | Event ID | Description | Action Taken
```

The bottom of the terminal shows the nano editor's command palette with options: ^G Help, ^X Exit, ^O Write Out, ^R Read File, ^F Where Is, ^I Replace, ^K Cut, ^U Paste, ^T Execute, and ^J Justify.

Figure 1: Shows initial log structure

The screenshot shows the same terminal window as Figure 1, but now the "security_events.log" file contains two documented events:

```
GNU nano 8.7 security_events.log *
Date/Time | Source IP | Event ID | Description | Action Taken
2026-03-19 11:00 | 192.168.1.10 | EVT1001 | Multiple failed login attempts | Monitoring initiated
2026-03-19 11:05 | 192.168.1.10 | EVT1002 | Possible brute force attack | IP blocked using firewall
```

The bottom of the terminal shows the nano editor's command palette with options: ^G Help, ^X Exit, ^O Write Out, ^R Read File, ^F Where Is, ^I Replace, ^K Cut, ^U Paste, ^T Execute, ^J Justify, ^C Location, ^M Go To Line, ^U Undo, and ^E Redo.

Figure 2: Shows documented events



```
kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~/CyArt-SOC-Domain-Task-1st-week-/Practical-Application/Document-security-Events

Session Actions Edit View Help

(kali@kali)~$ cd CyArt-SOC-Domain-Task-1st-week-
(kali@kali)~$ cd Practical-Application
(kali@kali)~$ cd Document-security-Events
(kali@kali)~$ ls
screenshots security_events.log
(kali@kali)~$ sudo grep -a "Failed password" /var/log/auth.log
[sudo] password for kali:
2026-03-19T03:04:29.284324-04:00 kali sshd-session[5903]: Failed password for invalid user fakeuser from ::1 port 5
8430 ssh2
2026-03-19T03:04:35.276303-04:00 kali sshd-session[5903]: Failed password for invalid user fakeuser from ::1 port 5
8430 ssh2
2026-03-19T03:04:39.551196-04:00 kali sshd-session[5903]: Failed password for invalid user fakeuser from ::1 port 5
8430 ssh2
2026-03-19T03:04:49.770011-04:00 kali sudo:      kali : TTY=pts/1 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/gr
ep 'Failed password' /var/log/auth.log
2026-03-19T03:05:53.465901-04:00 kali sudo:      kali : TTY=pts/0 ; PWD=/home/kali/CyArt-SOC-Domain-Task-1st-week-/P
ractical-Application/Document-security-Events ; USER=root ; COMMAND=/usr/bin/grep -a 'Failed password' /var/log/aut
h.log
(kali@kali)~$
```

Figure 3: Shows failed login logs from terminal

```
kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~/CyArt-SOC-Domain-Task-1st-week-/Practical-Application/Document-security-Events

Session Actions Edit View Help

GNU nano 8.7 security_events.log *
Date/Time | Source IP | Event ID | Description | Action Taken
2026-03-19 11:00 | 192.168.1.10 | EVT1001 | Multiple failed login attempts | Monitoring initiated
2026-03-19 11:05 | 192.168.1.10 | EVT1002 | Possible brute force attack | IP blocked using firewall
2026-03-19 03:05 | 127.0.0.1 | EVT1003 | Failed SSH login attempts detected | Investigated and monitored

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo
```

Figure 4: Final completed log file



Analysis

The logs indicate multiple failed login attempts from a single IP address, which suggests a potential brute-force attack. Such behavior is considered suspicious and requires monitoring or blocking to prevent unauthorized access.

Outcome

- Successfully created a structured security event log
- Simulated real-world attack scenario
- Practiced log extraction and analysis using Linux

10. Conclusion

This task highlights the importance of proper documentation in SOC operations. Structured event logging helps analysts track incidents, respond effectively, and maintain security visibility across systems.



TASK 3. Set Up Monitoring Dashboards

TOOLS USED

The following tools were used in this project:

1. Elasticsearch – for storing logs
2. Logstash – for log ingestion and processing
3. Kibana – for visualization and dashboards

LOG GENERATION

Sample security logs were manually created to simulate login activity.

Each log contains:

- EventID (4624 = success, 4625 = failed login)
- SourceIP
- Username

Example logs:

EventID=4625 SourceIP=192.168.1.10 User=admin

EventID=4624 SourceIP=192.168.1.15 User=user

LOG INGESTION USING LOGSTASH

Logstash was configured to read the log file and send data to Elasticsearch.

Pipeline includes:

- Input: file (log file path)
- Filter: grok pattern to extract fields
- Output: Elasticsearch index (security-logs)

Logs are continuously monitored and ingested into Elasticsearch.



DATA VERIFICATION (KIBANA DISCOVER)

Logs were verified in Kibana Discover section.

Fields extracted:

- EventID
- SourceIP
- User

This confirms successful ingestion and parsing.

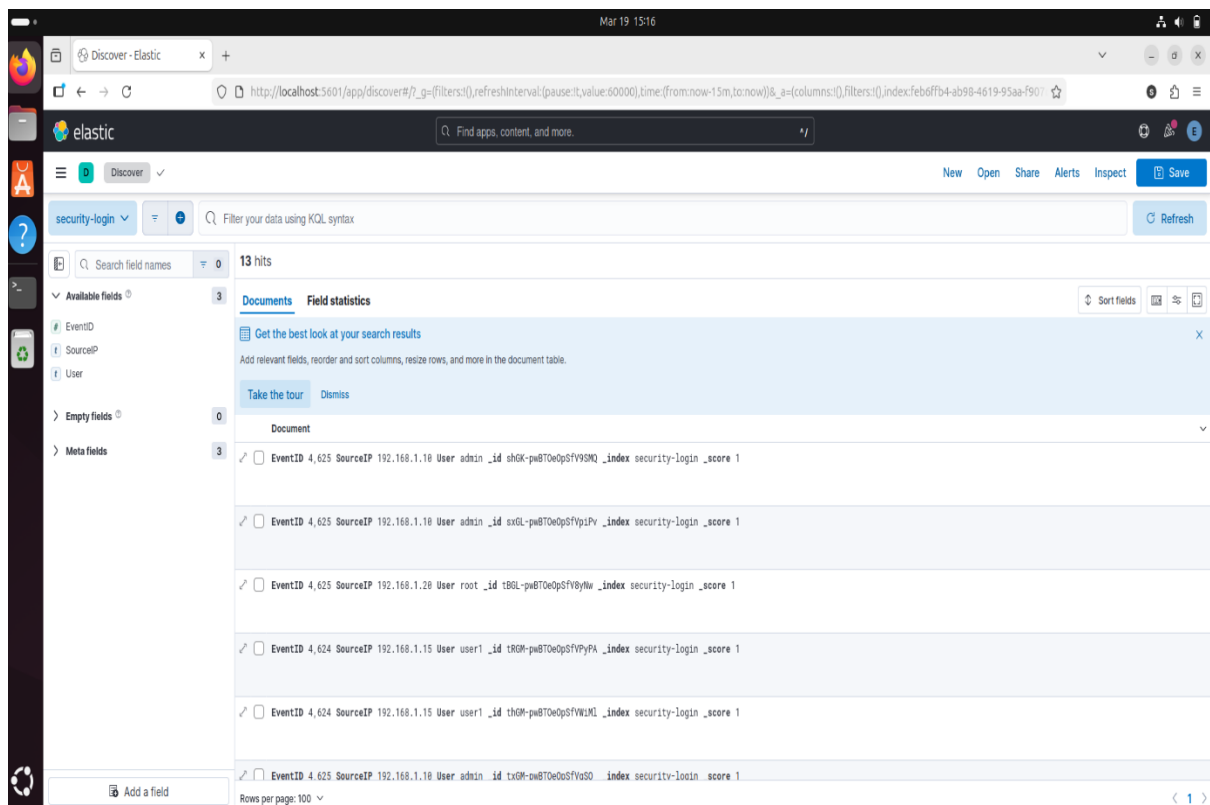


Figure 1: Logs visible in Kibana Discover



VISUALIZATION 1 – TOP SOURCE IPs

A bar chart was created to identify top source IP addresses generating events.

Configuration:

- X-axis: SourceIP.keyword
- Y-axis: Count of records

This helps identify suspicious or frequent IP activity.

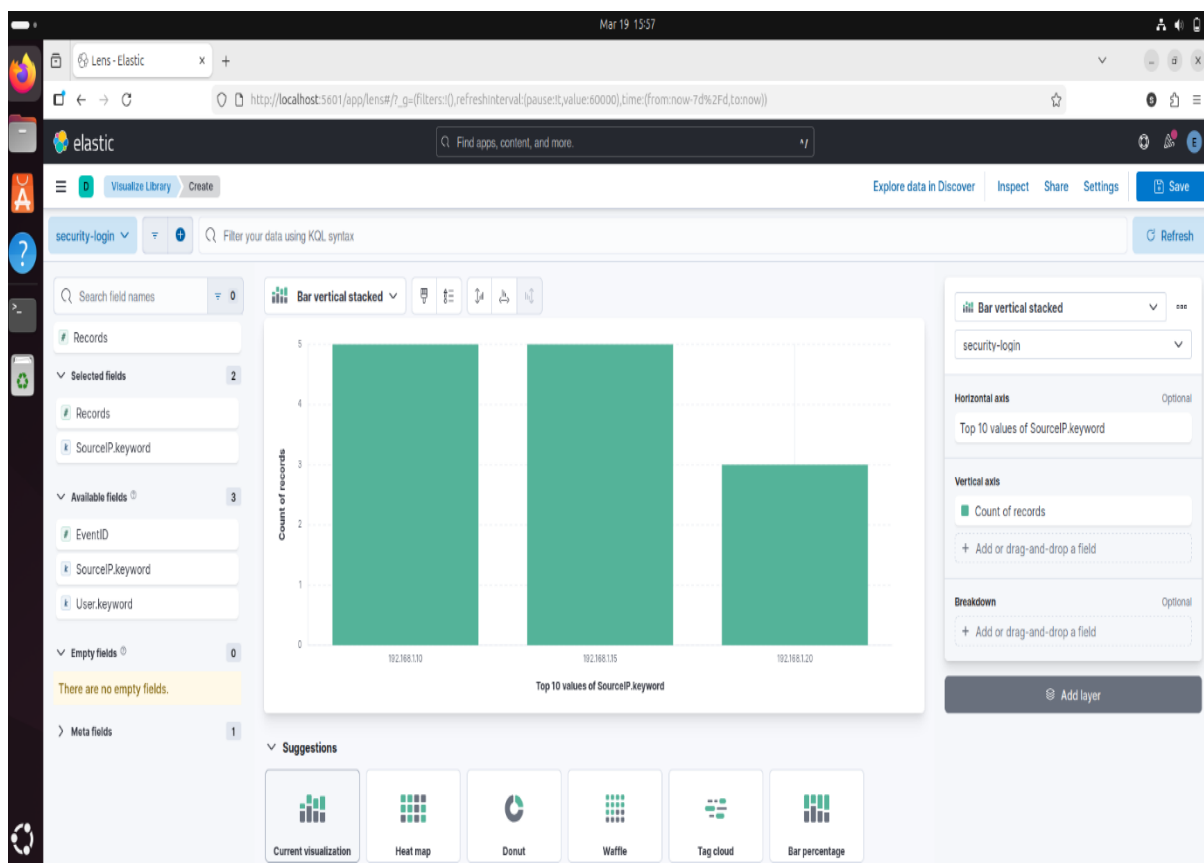


Figure 2: Top Source Ips



VISUALIZATION 2 – EVENT FREQUENCY

A bar chart was created to analyze frequency of Event IDs.

Configuration:

- X-axis: EventID
- Y-axis: Count

This helps differentiate successful vs failed login attempts.

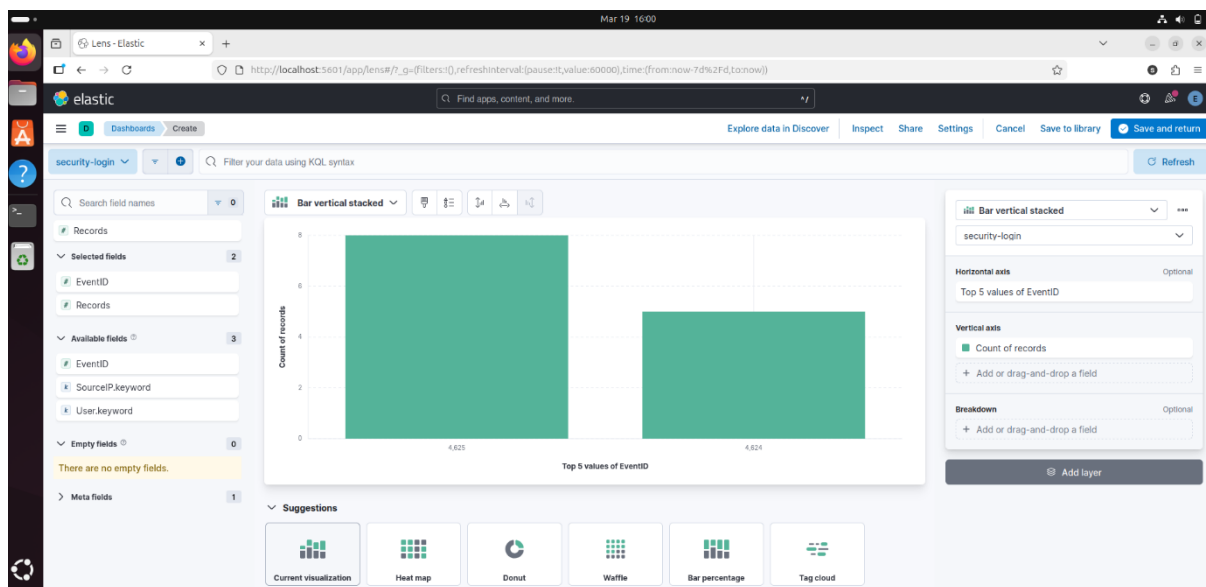


Figure 3: Event Frequency (4624 vs 4625)

DASHBOARD CREATION

Both visualizations were added into a single dashboard.

The dashboard provides:

- Centralized monitoring
- Quick visibility of suspicious activity
- SOC-level overview of login events

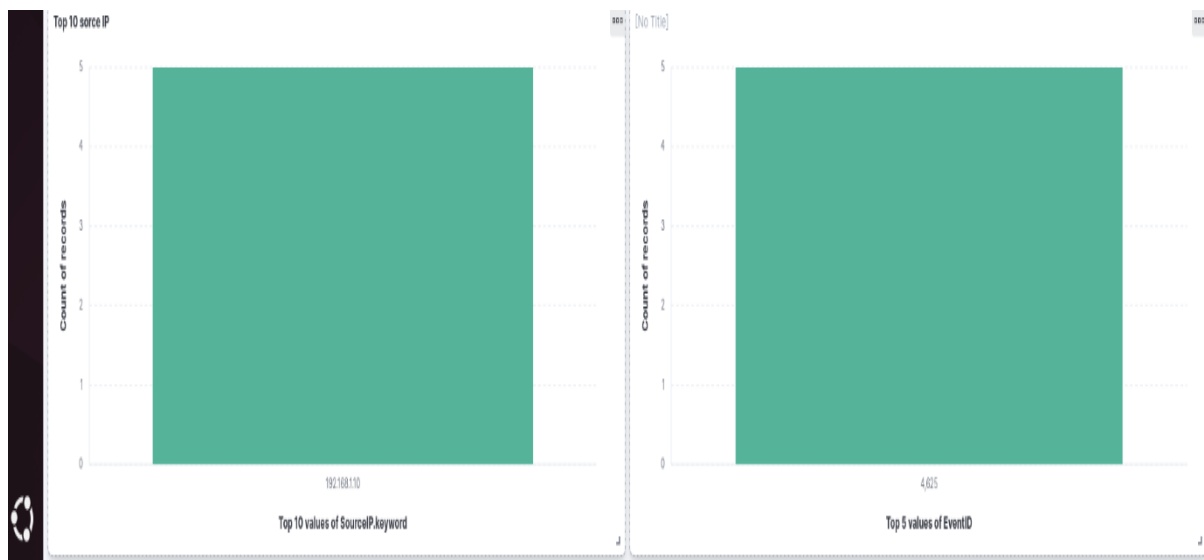


Figure 4: Security Monitoring Dashboard

ANALYSIS

- EventID 4625 (failed login) appears more frequently → indicates potential brute-force attempts
- Multiple IPs are generating login events
- Certain IPs show repeated activity → may require investigation

This simulates real SOC alert scenarios.

CONCLUSION

This task demonstrates how security logs can be collected, processed, and visualized using ELK Stack.

Key learning:

- Log ingestion pipeline setup
- Data parsing using Logstash
- Visualization using Kibana
- Basic SOC monitoring workflow

This setup can be extended for real-time threat detection



TASK 4 : Configure Alert Rules

Tools Used

- Wazuh Manager
- Ubuntu (VirtualBox)
- SSH (OpenSSH)

Step 1: Wazuh Installation

Command used:

```
sudo apt install wazuh-manager -y
```

```
sudeep@sudeep-VirtualBox:~$ sudo apt install wazuh-manager -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  expect
The following NEW packages will be installed:
  wazuh-manager
0 upgraded, 1 newly installed, 0 to remove and 10 not upgraded.
Need to get 490 MB of archives.
After this operation, 1,081 MB of additional disk space will be used.
Get:1 https://packages.wazuh.com/4.x/apt stable/main amd64 wazuh-manager amd64 4.14.4-1 [490 MB]
Ign:1 https://packages.wazuh.com/4.x/apt stable/main amd64 wazuh-manager amd64 4.14.4-1
Get:1 https://packages.wazuh.com/4.x/apt stable/main amd64 wazuh-manager amd64 4.14.4-1 [490 MB]
Fetched 343 MB in 2min 47s (2,051 kB/s)
Selecting previously unselected package wazuh-manager.
(Reading database ... 273295 files and directories currently installed.)
Preparing to unpack .../wazuh-manager_4.14.4-1_amd64.deb ...
Unpacking wazuh-manager (4.14.4-1) ...
Setting up wazuh-manager (4.14.4-1) ...
```

Figure 1 : Wazuh Manager Installation



Step 2: Start and Enable Service

```
Setting up wazuh-manager (4.14.4-1) ...
sudeep@sudeep-VirtualBox: ~$ sudo systemctl daemon-reexec
sudo systemctl daemon-reload
sudo systemctl enable wazuh-manager
sudo systemctl start wazuh-manager
Created symlink /etc/systemd/system/multi-user.target.wants/wazuh-manager.service → /usr/lib/systemd/system/wazuh-manager.service.
sudeep@sudeep-VirtualBox: ~$ sudo systemctl status wazuh-manager
● wazuh-manager.service - Wazuh manager
   Loaded: loaded (/usr/lib/systemd/system/wazuh-manager.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-03-19 20:31:23 IST; 6s ago
     Process: 62163 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
    Tasks: 278 (limit: 12323)
   Memory: 2.0G (peak: 2.0G)
      CPU: 38.835s
   CGroup: /system.slice/wazuh-manager.service
           └─62241 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             └─62280 /var/ossec/bin/wazuh-authd
               └─62296 /var/ossec/bin/wazuh-db
                 └─62322 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                   └─62323 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                     └─62326 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                       └─62329 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                         └─62341 /var/ossec/bin/wazuh-execd
                           └─62356 /var/ossec/bin/wazuh-analysisd
                             └─62369 /var/ossec/bin/wazuh-syscheckd
                               └─62383 /var/ossec/bin/wazuh-remoted
                                 └─62508 /var/ossec/bin/wazuh-logcollector
                                   └─62527 /var/ossec/bin/wazuh-monitord
                                     └─62550 /var/ossec/bin/wazuh-modulesd

Mar 19 20:31:17 sudeep-VirtualBox env[62163]: Started wazuh-syscheckd...
Mar 19 20:31:18 sudeep-VirtualBox env[62163]: Started wazuh-remoted...
Mar 19 20:31:19 sudeep-VirtualBox env[62163]: Started wazuh-logcollector...
Mar 19 20:31:20 sudeep-VirtualBox env[62163]: Started wazuh-monitord...
Mar 19 20:31:20 sudeep-VirtualBox env[62548]: 2026/03/19 20:31:20 wazuh-modulesd:router: INFO: Loaded router module.
Mar 19 20:31:20 sudeep-VirtualBox env[62548]: 2026/03/19 20:31:20 wazuh-modulesd:content_manager: INFO: Loaded content_manager module.
Mar 19 20:31:20 sudeep-VirtualBox env[62548]: 2026/03/19 20:31:20 wazuh-modulesd:inventory-harvester: INFO: Loaded Inventory harvester module.
```

Figure2 :Wazuh Manager Service Running

Enabled and started Wazuh Manager
Verified status using systemctl

Step 3: Custom Rule Creation

/var/ossec/etc/rules/local_rules.xml

Custom Rule:

```
<group name="local,ssh">
  <rule id="100002" level="10">
    <match>Failed password|authentication failure|PAM</match>
    <description>SSH Brute Force Attempt Detected</description>
    <mitre>
      <id>T1110</id>
```



</mitre>

</rule>

</group>

✓ Purpose:

- Detect multiple SSH login failures
- Map attack to MITRE ATT&CK

```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Mar 20 16:21
sudeep@sudeep-VirtualBox: ~
GNU nano 7.2 /var/ossec/etc/rules/local_rules.xml
<group name="local,ssh">
  <rule id="100002" level="10">
    <match>Failed password|authentication failure|PAM</match>
    <description>SSH Brute Force Attempt Detected</description>
    <mitre>
      <id>T1110</id>
    </mitre>
  </rule>
</group>
```

Figure 3: Custom Wazuh Rule Configuration



Step 4: Restart Wazuh

`sudo systemctl restart wazuh-manager`

Step 5: Attack Simulation

- Simulated brute-force attack:

`ssh fakeuser@localhost`

- Entered wrong password multiple times

Result:

- Authentication failures generated

```
Mar 19 20:36:44 sudeep-VirtualBox systemd[1]: Started wazuh-manager.service - Wazuh manager.
sudeep@sudeep-VirtualBox:~$ ssh fakeuser@localhost
fakeuser@localhost's password:
Permission denied, please try again.
fakeuser@localhost's password:
Permission denied, please try again.
fakeuser@localhost's password:
fakeuser@localhost: Permission denied (publickey,password).
sudeep@sudeep-VirtualBox:~$
```

Figure 4 - SSH Failed Login Attempt Simulation

Step 6: Alert Monitoring

Command:

`sudo tail -f /var/ossec/logs/alerts/alerts.log`

Observed logs:

- PAM authentication events
- AppArmor alerts
- sudo activity logs



```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Mar 20 16:19
sudeep@sudeep-VirtualBox: ~
sudeep@sudeep-VirtualBox:~$ sudo nano /var/ossec/etc/rules/local_rules.xml
[sudo] password for sudeep:
sudeep@sudeep-VirtualBox:~$ sudo systemctl restart wazuh-manager
sudeep@sudeep-VirtualBox:~$ ssh fakeuser@localhost
fakeuser@localhost's password:
Permission denied, please try again.
fakeuser@localhost's password:
Permission denied, please try again.
fakeuser@localhost's password:
fakeuser@localhost: Permission denied (publickey,password).
sudeep@sudeep-VirtualBox:~$ sudo tail -f /var/ossec/logs/alerts/alerts.log
sca.invalid: 34
sca.total_checks: 279
sca.score: 48
sca.file: cis_ubuntu24-04.yml

** Alert 1774003724.180307: - local,syslog,apparmor,
2026 Mar 20 16:18:44 sudeep-VirtualBox->journald
Rule: 52002 (level 3) -> 'Apparmor DENIED'
Mar 20 10:48:44 sudeep-VirtualBox kernel: audit: type=1400 audit(1774003724.653:226
): apparmor="DENIED" operation="open" class="file" profile="snap.firefox.firefox" n
ame="/proc/pressure/memory" pid=5242 comm="MemoryPoller" requested_mask="r" denied_
mask="r" fsuid=1000 ouid=0

** Alert 1774003726.180729: - pam,syslog,authentication_success,pci_dss_10.2.5,gpg1
3_7.8,gpg13_7.9,gdpr_IV_32.2,hipaa_164.312.b,nist_800_53_AU.14,nist_800_53_AC.7,tsc
_CC6.8,tsc_CC7.2,tsc_CC7.3,
2026 Mar 20 16:18:46 sudeep-VirtualBox->/var/log/auth.log
Rule: 5501 (level 3) -> 'PAM: Login session opened.'
User: root
2026-03-20T16:18:46.627988+05:30 sudeep-VirtualBox sudo: pam_unix(sudo:session): se
ssion opened for user root(uid=0) by sudeep(uid=1000)
uid: 1000

** Alert 1774003726.181193: - syslog,sudo,pci_dss_10.2.5,pci_dss_10.2.2,gpg13_7.6,g
pg13_7.8,gpg13_7.13,gdpr_IV_32.2,hipaa_164.312.b,nist_800_53_AU.14,nist_800_53_AC.7
,nist_800_53_AC.6,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,
```

Figure 5 - Wazuh Alert Logs Output



```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Mar 20 16:20
sudeep@sudeep-VirtualBox: ~
,nist_800_53_AC.6,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,
2026 Mar 20 16:18:46 sudeep-VirtualBox->journald
Rule: 5402 (level 3) -> 'Successful sudo to ROOT executed.'
User: root
Mar 20 10:48:46 sudeep-VirtualBox sudo[12243]: sudeep : TTY=pts/0 ; PWD=/home/sudeep ; USER=root ; COMMAND=/usr/bin/tail -f /var/ossec/logs/alerts/alerts.log
tty: pts/0
pwd: /home/sudeep
command: /usr/bin/tail -f /var/ossec/logs/alerts/alerts.log

** Alert 1774003726.182382: - pam,syslog,authentication_success,pci_dss_10.2.5,gpg13_7.8,gpg13_7.9,gdpr_IV_32.2,hipaa_164.312.b,nist_800_53_AU.14,nist_800_53_AC.7,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,
2026 Mar 20 16:18:46 sudeep-VirtualBox->journald
Rule: 5501 (level 3) -> 'PAM: Login session opened.'
User: root
Mar 20 10:48:46 sudeep-VirtualBox sudo[12243]: pam_unix(sudo:session): session opened for user root(uid=0) by sudeep(uid=1000)
uid: 1000

** Alert 1774003730.182827: - local,syslog,apparmor,
2026 Mar 20 16:18:50 sudeep-VirtualBox->journald
Rule: 52002 (level 3) -> 'Apparmor DENIED'
Mar 20 10:48:49 sudeep-VirtualBox kernel: audit: type=1400 audit(1774003729.656:227): apparmor="DENIED" operation="open" class="file" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=5242 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0

** Alert 1774003734.183249: - local,syslog,apparmor,
2026 Mar 20 16:18:54 sudeep-VirtualBox->journald
Rule: 52002 (level 3) -> 'Apparmor DENIED'
Mar 20 10:48:54 sudeep-VirtualBox kernel: audit: type=1400 audit(1774003734.657:228): apparmor="DENIED" operation="open" class="file" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=5242 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0

** Alert 1774003745.183671: - local,syslog,apparmor,
```

Figure 6 - Wazuh Alert Logs Output-a



```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Mar 20 16:20
sudeep@sudeep-VirtualBox: ~
): apparmor="DENIED" operation="open" class="file" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=5242 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
** Alert 1774003734.183249: - local,syslog,apparmor,
2026 Mar 20 16:18:54 sudeep-VirtualBox->journald
Rule: 52002 (level 3) -> 'Apparmor DENIED'
Mar 20 10:48:54 sudeep-VirtualBox kernel: audit: type=1400 audit(1774003734.657:228): apparmor="DENIED" operation="open" class="file" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=5242 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
** Alert 1774003745.183671: - local,syslog,apparmor,
2026 Mar 20 16:19:05 sudeep-VirtualBox->journald
Rule: 52002 (level 3) -> 'Apparmor DENIED'
Mar 20 10:49:04 sudeep-VirtualBox kernel: audit: type=1400 audit(1774003744.666:229): apparmor="DENIED" operation="open" class="file" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=5242 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
** Alert 1774003751.184093: - local,syslog,apparmor,
2026 Mar 20 16:19:11 sudeep-VirtualBox->journald
Rule: 52002 (level 3) -> 'Apparmor DENIED'
Mar 20 10:49:09 sudeep-VirtualBox kernel: audit: type=1400 audit(1774003749.669:230): apparmor="DENIED" operation="open" class="file" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=5242 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
** Alert 1774003755.184515: - local,syslog,apparmor,
2026 Mar 20 16:19:15 sudeep-VirtualBox->journald
Rule: 52002 (level 3) -> 'Apparmor DENIED'
Mar 20 10:49:14 sudeep-VirtualBox kernel: audit: type=1400 audit(1774003754.722:231): apparmor="DENIED" operation="open" class="file" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=5242 comm="MemoryPoller" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
```

Figure 7 - Wazuh Alert Logs Output-b

MITRE ATT&CK Mapping

- Technique ID: **T1110**
- Technique Name: **Brute Force Attack**



Conclusion

- Wazuh Manager successfully installed and configured
- Custom rule created for SSH brute-force detection
- Attack simulation performed using invalid SSH login
- Alerts successfully generated and monitored

This demonstrates:

- Real-time log monitoring
- Threat detection using SIEM
- Mapping alerts to MITRE ATT&CK



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